

Freie Universität Berlin
FB Mathematik und Informatik

Numerics II

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Numerics 2

$$\frac{dy}{dt} =$$

Ordinary
Differential
Equations

Differential
Algebraic
Equations

Symplectic
Methods

$$\begin{bmatrix} a_{1,2} & a_{1,4} \\ a_{3,1} & a_{3,3} \\ a_{4,2} & a_{4,4} \end{bmatrix}$$

Iterative Solvers
of Linear Systems

(c) R. Gruhlke

Preface

This script accompanies the lecture Numerics II, taught by the author at Freie Universität Berlin during the winter term 2025/2026. It draws on several sources and follows presentation styles inspired by my former Numerics II lecture from Prof. Harry Yserentant (TU Berlin) and partially by Prof. Marcus Bachmayr (RWTH Aachen).

This lecture script is intended as an aid for reviewing and following up on the material covered in the course. It does not, and is not meant to, reproduce the lecture verbatim. In the lectures, we will motivate the definitions, theorems, and algorithms, discuss their results, and illustrate them with examples.

Suggestions for improvements of any kind are very welcome: **r.gruhlke@fu-berlin.de**

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1. Numerical Methods for Ordinary Differential Equations

Examples and Motivation

For reasons of limited time, various (derivation) examples for ordinary differential equations are omitted in this lecture and maybe added at a later stage for the interested reader. For some applications, please refer to the introductory information slides or to previous courses such as Computerorientierte Mathematik II (FU Berlin) or Numerics I (FU Berlin).

1.1 Regularity, Existence and Uniqueness

Consider the ordinary differential equation (ODE) of **first order** for $G \subset \mathbb{R}^n$ open for $n \in \mathbb{N}$ and $t \in [a, b]$ and $f: [a, b] \times G \rightarrow \mathbb{R}^n$ given as

$$y'(t) = f(t, y(t)), \quad a \leq t \leq b. \quad (1.1)$$

A solution (if existing) $y: [a, b] \rightarrow \mathbb{R}^n$ of (1.1) is called *classical solution* if it continuous differentiable, i.e. $y \in \mathcal{C}([a, b]; \mathbb{R}^n)$.

Theorem 1.1 (Regularity) If $f: [a, b] \times G \rightarrow \mathbb{R}^n$ is continuous, then y is continuously differentiable. If f is r -times continuously differentiable, then y is $(r + 1)$ -times continuously differentiable.

Proof. By induction. □

In general there are infinite solutions of the ODE 1.1. In order to fix a possible unique solution, additional constraints are required.

Definition 1.2 (Initial value problem (IVP)) An initial value problem is given for $y_0 \in G$ as

$$\begin{cases} y'(t) = f(t, y(t)), & a \leq t \leq b, \\ y(a) = y_0 \end{cases} \quad (\text{IVP})$$

Remark 1.3 (General form and high order ODEs) A general m -th order ordinary differential equation is given of the form

$$F(t, y(t), y'(t), \dots, y^{(m)}(t)) = 0.$$

The equation is called explicit, if it can be written as

$$y^{(m)} = f(t, y(t), y'(t), \dots, y^{(m-1)}(t)). \quad (1.2)$$

Every explicit differential equation of m -th order can be transformed to a system of m differential equations of first order using

$$\begin{aligned} y'_i(t) &= y_{i+1}(t), & i &= 1, \dots, m-1, \\ y'_m(t) &= f(t, y_1(t), \dots, y_m(t)) \end{aligned}$$

or compactly written for $\mathbf{y}(t) = (y_1(t), \dots, y_m(t))^T$ as

$$\mathbf{y}'(t) = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} \mathbf{y}(t) + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ f(t, y_1, \dots, y_m) \end{pmatrix}$$

Then, y_1 solves (1.2). Moreover, together with the conditions

$$y(a) = y_0, \quad y'(a) = y'_0, \quad \dots, y^{(m-1)}(a) = y'_{m-1}$$

the general form becomes an initial value problem. △

Theorem 1.4 (Peano - Existence of solutions) Assume $f: [a, b] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous and bounded. Then, for any $y_0 \in \mathbb{R}^n$ the initial value problem (IVP) has a solution.

Proof. See, e.g., Theorem 1.2. in [?]. □

We call a continuous function $f: [a, b] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, Lipschitz w.r.t. to its second input if there exists $L > 0$ such that

$$\|f(t, y_1) - f(t, y_2)\| \leq L \|y_1 - y_2\|, \quad \forall t \in [a, b], y_1, y_2 \in \mathbb{R}^n. \quad (1.3)$$

Theorem 1.5 (Picard-Lindelöf - Uniqueness of solution) Assume $f: [a, b] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous w.r.t. t and Lipschitz-continuous w.r.t. y . Then, for any $y_0 \in \mathbb{R}^n$ the initial value problem (IVP) has a **unique** solution.

Proof. See, e.g., Theorem 1.1. in [?]. □

Remark 1.6 Theorem 1.4 has a local version for $f: [a, b] \times G$ continuous only. Theorem 1.5 has a local version for a continuous differentiable right hand side f , yielding local existence and uniqueness. Moreover, the boundedness in Theorem 1.4 is needed for the existence of the solution

y on the whole interval $[a, b]$. For this consider the ODE $y(t)' = y(t)^2$ on the interval $[0, 2]$ with general form of solutions $y(t) = \frac{1}{c-t}$ for any $c \in \mathbb{R}$. So let $y(0) = 1$, then $c = 1$ and the solution is not defined at $t = 1$. In particular the right hand side does not satisfy the assumptions of Picard-Lindelöf. $\triangle$

We introduce the space $\mathcal{C}_{\text{PL}}$ of right hand sides for our IVP satisfying the conditions for the Picard-Lindelöf theorem, i.e.

$$\begin{aligned}\mathcal{C}_{\text{PL}} &:= \mathcal{C}_{\text{PL}}([a, b] \times \mathbb{R}^n; \mathbb{R}^n) \\ &:= \{f: [a, b] \times \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ continuous and satisfies (1.3)}\}.\end{aligned}$$

Example 1.7 A linear ODE is of the form

$$y'(t) = A(t)y(t) + g(t), \quad a \leq t \leq b,$$

with continuous matrix-valued function A and continuous function g . The right hand side is Lipschitz w.r.t. the y coordinate with constant

$$L = \max_{a \leq t \leq b} \|A(t)\|_2.$$

Consequently, IVPs based on linear ODEs are uniquely solvable by Theorem 1.5. $\triangle$

As a next result we want to understand the effect of perturbation of the initial value for general IVPs and for the case of dissipative systems. For this we need an auxillary result

Lemma 1.8 (Gronwall Lemma) Let $\Phi: [a, b] \rightarrow \mathbb{R}$ continuous, $\alpha \in \mathbb{R}$ and $\beta > 0$ such that

$$\Phi(t) \leq \alpha + \beta \int_a^t \Phi(s) \, ds, \quad a \leq t \leq b.$$

Then, it holds $\Phi(t) \leq \alpha e^{\beta(t-a)}$ for all $t \in [a, b]$.

Proof. Let $\epsilon > 0$ and $\Psi(t) := (\alpha + \epsilon)e^{\beta(t-a)}$. Then either $\Phi(t) < \Psi(t)$ or the set $\{t \in [a, b] \mid \Phi(t) \geq \Psi(t)\}$ is not empty. Since the latter set is compact subset of $\mathbb{R}$, there exists a minimal t_0 . Since, for this t_0 it holds $\Phi(t_0) \geq \Psi(t_0)$ and $\Phi(a) \leq \alpha < \alpha + \epsilon = \Psi(a)$ it must hold $t_0 > a$. Consequently, $\Phi(t) < \Psi(t)$ for $a \leq t \leq t_0$ and since $\beta > 0$ it holds

$$\Phi(t_0) \leq \alpha + \beta \int_a^{t_0} \Phi(t) \, dt < \alpha + \beta \int_a^{t_0} \Psi(t) \, dt = \alpha + \int_a^{t_0} \Psi'(t) \, dt = \Psi(t_0) - \epsilon < \Psi(t_0).$$

This is a contradiction to the definition of t_0 . It follows that

$$\Phi(t) < \Psi(t) = (\alpha + \epsilon)e^{\beta(t-a)}.$$

Since $\epsilon > 0$ can be chosen arbitrary small, it yields the claim. $\square$

Theorem 1.9 Let $f \in \mathcal{C}_{\text{PL}}$ with Lipschitz constant $L > 0$ and let $y, z: [a, b] \rightarrow \mathbb{R}^n$ be the solution of the IVPs

$$y'(t) = f(t, y(t)), \quad y(a) = y_0 \quad \text{and} \quad z'(t) = f(t, z(t)), \quad z(a) = z_0.$$

Then, it holds

$$\|y(t) - z(t)\| \leq e^{L(t-a)} \|y_0 - z_0\|, \quad a \leq t \leq b. \quad (1.4)$$

Proof. For all $t \in [a, b]$ it holds

$$y(t) = y_0 + \int_a^t f(s, y(s)) \, ds, \quad z(t) = z_0 + \int_a^t f(s, z(s)) \, ds.$$

Hence,

$$\begin{aligned} \|y(t) - z(t)\| &= \left\| (y_0 - z_0) + \int_a^t f(s, y(s)) - f(s, z(s)) \, ds \right\| \\ &\leq \|y_0 - z_0\| + \int_a^t \|f(s, y(s)) - f(s, z(s))\| \, ds \\ &\leq \|y_0 - z_0\| + L \int_a^t \|y(s) - z(s)\| \, ds \end{aligned}$$

Then the claim follows by application of Gronwall Lemma 1.8 with $\Phi(t) = \|y(t) - z(t)\|$, $\alpha = \|y_0 - z_0\|$ and $\beta = L$. $\square$

Theorem 1.10 Let $f \in \mathcal{C}_{\text{PL}}$ and it satisfies

$$\langle y_1 - y_2, f(t, y_1) - f(t, y_2) \rangle \leq \theta \|y_1 - y_2\|^2 \quad \forall t \in [a, b], y_1, y_2 \in \mathbb{R}^n$$

for an inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^n$ with induced norm $\|\cdot\|$ and for $\theta \in \mathbb{R}$. Then, the solutions $t \mapsto y_1(t), t \mapsto y_2(t)$ of the IVPs

$$y'(t) = f(t, y(t)), \quad t \in [a, b], \quad y(a) \in \{y_1(a), y_2(a)\},$$

satisfy

$$\|y_1(t) - y_2(t)\| \leq e^{\theta(t-a)} \|y_1(a) - y_2(a)\|.$$

Proof. Define the function $\Psi(t) = \|y_1(t) - y_2(t)\|^2 > 0$. It satisfies

$$\begin{aligned} \Psi'(t) &= 2\langle y_1(t) - y_2(t), y_1'(t) - y_2'(t) \rangle \\ &= 2\langle y_1(t) - y_2(t), f(t, y_1(t)) - f(t, y_2(t)) \rangle \\ &\leq 2\theta \|y_1(t) - y_2(t)\|^2 \\ &= 2\theta \Psi(t). \end{aligned}$$

Hence,

$$\frac{\Psi'(s)}{\Psi(s)} \leq 2\theta, \quad \forall s \in [a, b].$$

Integration over $[a, t]$, yields

$$\ln(\Psi(t)) - \ln(\Psi(a)) \leq 2\theta(t - a) \quad \Leftrightarrow \quad \Psi(t) \leq e^{2\theta(t-a)} \Psi(a).$$

□

Remark 1.11 If $f \in \mathcal{C}_{PL}$ with Lipschitz constant L then automatically

$$\langle y_1 - y_2, f(t, y_1) - f(t, y_2) \rangle \leq \|y_1 - y_2\| \|f(t, y_1) - f(t, y_2)\| \leq L \|y_1 - y_2\|^2.$$

However, Theorem 1.10 includes the case of arbitrary $\theta \in \mathbb{R}$. △

The relevance of Theorem 1.10 becomes visible in the case of dissipative systems, see Example 1.12. We call a system $y'(t) = f(t, y(t))$ *monotone or dissipative in the sense of Dahlquist* with respect to a inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^n$ with induced norm $\|\cdot\|$ if there exists $\eta > 0$ such that for all $y_1, y_2 \in \mathbb{R}^n$:

$$\langle f(t, y_1) - f(t, y_2), y_1 - y_2 \rangle \leq -\eta \|y_1 - y_2\|^2. \quad (1.5)$$

Example 1.12 Consider the IVPs for $\lambda \in \mathbb{R}$ given as

$$y'(t) = \lambda y(t), \quad y(a) = y_0 \quad \text{and} \quad z'(t) = \lambda z(t), \quad z(a) = z_0$$

with Lipschitz constant $L = |\lambda|$ of the right hand side. In this case

$$y(t) - z(t) = e^{\lambda(t-a)}(y(a) - z(a)).$$

Consequently, following Remark 1.11 the choice $\theta = L = |\lambda|$ is only sharp if $\lambda > 0$ and else represents a huge overestimation for the case of $\lambda < 0$. △

Example 1.13 Consider

$$y'(t) = A(t)y(t), \quad y(0) = y_0$$

with

$$A(t) \equiv A := \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$$

for $\alpha, \beta \in \mathbb{R}$. Then, $L = \sqrt{\alpha^2 + \beta^2}$ and $\theta = \alpha$. [Exercise](#). △

1.2 Explicit One-step methods

We seek for approximate values Y_k for the solution $y(t_k)$ for points t_k of a grid

$$a = t_0 < t_1 < \dots < t_N = b, \quad (1.6)$$

of the interval $[a, b]$. In view of a one-step scheme to generate the sequence $(Y_k)_k$, we denote $h_k = t_{k+1} - t_k$ the step size. For equidistant grids, i.e. $h_0 = h_1 = \dots = h_{N-1}$ the notation of h is used.

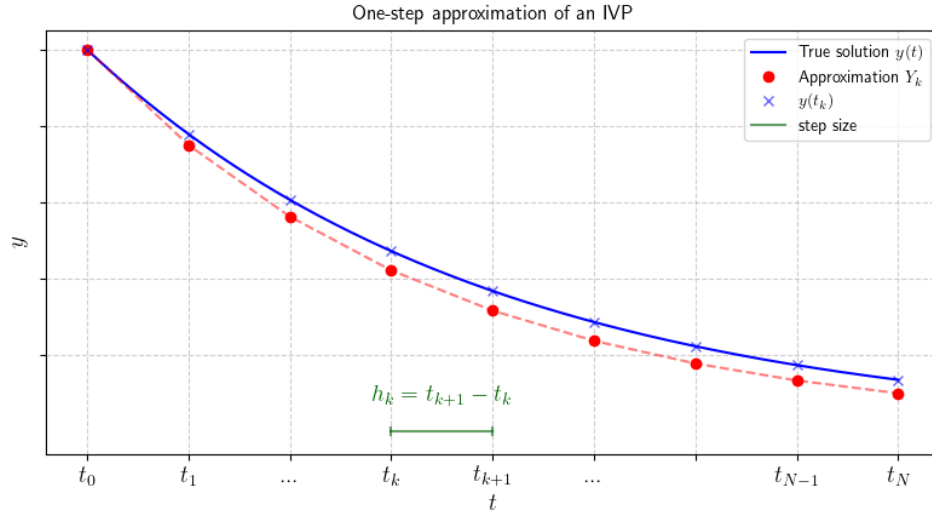


Figure 1.1: Illustration of approximation of $t \mapsto y(t)$ on grid points t_k .

Definition 1.14 (Incremental function) An explicit one-step method is given as

$$Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k), \quad k = 0, \dots, N-1, \quad Y_0 = y(t_0), \quad (1.7)$$

and determined by an *incremental function* $\Phi: t \times \mathbb{R}^n \times \mathbb{R}_+ \rightarrow \mathbb{R}^n$.

Example 1.15 (Explicit Euler method) Motivated by the first order approximation using rectangle rule given as

$$y(t_{k+1}) = y(t_k) + \int_{t_k}^{t_{k+1}} f(t, y(t)) dt \approx y(t_k) + h_k f(t_k, y(t_k))$$

the *explicit Euler method* reads

$$Y_{k+1} = Y_k + h_k f(t_k, Y_k). \quad (1.8)$$

Consequently, $\Phi(t, Y, h) = f(t, Y)$. △

Example 1.16 (Improved Euler method) Motivated by the midpoint rule for integration taking the form

$$y(t_{k+1}) = y(t_k) + \int_{t_k}^{t_{k+1}} f(t, y(t)) dt \approx y(t_k) + h_k f\left(t_k + \frac{h_k}{2}, y\left(t_k + \frac{h_k}{2}\right)\right)$$

one obtains the scheme

$$Y_{k+1} = Y_k + h_k f\left(t_k + \frac{h_k}{2}, Y_k + \frac{h_k}{2} f(t_k, Y_k)\right). \quad (1.9)$$

Consequently, $\Phi(t, Y, h) = f\left(t + \frac{h}{2}, Y + \frac{h}{2} f(t, Y)\right)$. △

Example 1.17 Consider the initial value problem

$$\begin{aligned} y'(t) &= \cos(t)y & -8 \leq t \leq 0 \\ y(-8) &= \exp(\sin(-8)) \end{aligned} \quad (1.10)$$

with solution $y(t) = \exp(\sin(t))$.

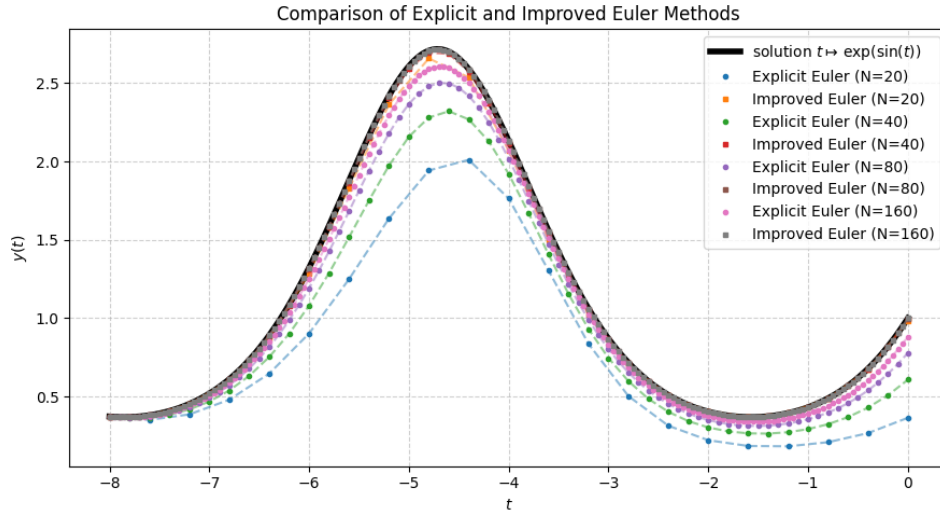


Figure 1.2: Numerical results of explicit and improved Euler methods for various equidistant grids for Example 1.17.

Table 1.1: Error $|Y_N - y(t)|$ at endpoint $t = 0$ for the explicit and improved Euler for Example 1.17

N	Explicit Euler error	Improved Euler error
20	6.362×10^{-1}	1.928×10^{-2}
40	3.929×10^{-1}	6.105×10^{-3}
80	2.218×10^{-1}	1.719×10^{-3}
160	1.184×10^{-1}	4.549×10^{-4}

Observation: Explicit Euler converges as $\mathcal{O}(h)$ and improved Euler method as $\mathcal{O}(h^2)$. △

There are two main sources of errors for the discretization of IVP:

1. **Local error:** Each iteration introduces a new error.
2. **Error propagation:** Errors introduced once are propagated and possible amplified by the recursion.

1.2.1 Consistency, Stability and Convergence

This section is devoted to the general theory leading to convergence of one-step methods.

As a first step we formalize the introduced error by a single step of the method.

Definition 1.18 (Local truncation error) The quantity

$$\tau(t, h) := \frac{1}{h}(y(t+h) - y(t)) - \Phi(t, y(t), h) \quad (1.11)$$

is called the local truncation error (*dt. Abbruchfehler*) of the one-step method at the solution point $y(t)$.

In other words, it holds

$$y(t_{k+1}) = y(t_k) + h_k \Phi(t_k, y(t_k), h_k) + h_k \tau(t_k, h_k).$$

Then, the one-step method at k -th step yields a value

$$\hat{y}_{k+1} = y(t_k) + h_k \Phi(t_k, y(t_k), h_k)$$

leading to the local deviation $\hat{d}_{k+1} = y(t_{k+1}) - \hat{y}_{k+1}$ with error

$$\|\hat{d}_{k+1}\| = \|y(t_{k+1}) - \hat{y}_{k+1}\| = \|y(t_{k+1}) - (y(t_k) + h_k \Phi(t_k, y(t_k), h_k))\|.$$

Hence, the design of Φ should be necessarily meaningful in terms of controlling the local truncation error, ideally $\tau(t, h) \in \mathcal{O}(h^q)$ for some $q > 0$.

Definition 1.19 (Consistency of one-step methods) Let $h = \max_k h_k$. Then, the one-step method (1.7) is called consistent if for any $f \in \mathcal{C}_{\text{PL}}$ and corresponding unique solution y of (IVP), it holds

$$\lim_{h \rightarrow 0} \left(\max_{k=0, \dots, N-1} \|\tau(t_k, h_k)\| \right) = 0. \quad (1.12)$$

The condition (1.12) is equivalent to

$$\lim_{h \rightarrow 0} \left(\max_{k=0, \dots, N-1} \|f(t_k, y(t_k)) - \Phi(t_k, y(t_k), h_k)\| \right) = 0.$$

In particular, consistency requires that the incremental function approximates the derivative y' of the function y sufficiently well. This allows a comparison of different one-step methods.

Let us for simplicity assume that $h_k \equiv h$ in the following.

Definition 1.20 (Order of consistency) An explicit one-step method has consistency order $q \in \mathbb{N}$ if, q is the largest natural number such that for any $f \in \mathcal{C}_{\text{PL}}$ and related incremental function $\Phi = \Phi(f)$ it holds

$$\|\tau(t_k, h)\| \leq Ch^q, \quad \forall k = 0, \dots, N-1$$

for a constant $C = C(y, f)$ and all $h \in (0, \bar{h})$ for some $\bar{h} > 0$.

Remark 1.21 The constant $C = C(y, f)$ may depend on (partial) derivatives of y and f . $\triangle$

Example 1.22 The explicit Euler method has consistency order $q = 1$. Assume $y \in \mathcal{C}^2([a, b]; \mathbb{R}^n)$. By Taylor expansion with integral remainder it holds

$$y(t_k + h) = y(t_k) + h_k y'(t_k) + \int_{t_k}^{t_k + h_k} (t_k + h_k - s) y''(s) ds.$$

Consequently, since for the explicit Euler method $\Phi(t_k, y(t_k), h_k) = f(t, y(t_k)) = y'(t_k)$ it holds

$$\begin{aligned} h_k \|\tau(t_k, h_k)\| &= \|y(t_k + h) - \hat{y}_{k+1}\| \\ &= \left\| y(t_k) + h_k y'(t_k) + \int_{t_k}^{t_k + h} (t_k + h_k - s) y''(s) ds - y(t_k) - h_k f(t_k, y(t_k)) \right\| \\ &= \left\| \int_{t_k}^{t_k + h_k} (t_k + h_k - s) y''(s) ds \right\| \\ &\leq \left(\int_{t_k}^{t_k + h_k} (t_k + h_k - s) y''(s) ds \right) \max_{s \in [t_k, t_k + h_k]} \|y''(s)\| \\ &\leq \frac{h_k^2}{2} \|y''\|_\infty \end{aligned}$$

. Hence $\|\tau(t_k, h_k)\| \in \mathcal{O}(h_k)$. △

Definition 1.23 (Global error) The global error of the one-step method is defined as

$$\max_{k=0, \dots, N} \|Y_k - y(t_k)\|.$$

Now the goal is to estimate the global error with the local truncation error.

Theorem 1.24 (Stability) Let y be solution of (IVP) defined on the whole interval $[a, b]$ and for some $\bar{h} > 0$ assume that the incremental function $\Phi: [a, b] \times \mathbb{R}^n \times [0, \bar{h}]$ is uniformly L -Lipschitz w.r.t the second component, i.e.

$$\|\Phi(t, y_1, h) - \Phi(t, y_2, h)\| \leq L \|y_1 - y_2\|, \quad y_1, y_2 \in \mathbb{R}^n,$$

for $t \in [a, b], h \in [0, \bar{h}]$. Then, it holds the stability estimate

$$\|y(t_k) - Y_k\| \leq e^{L(t_k - a)} \sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\|, \quad k = 1, \dots, N.$$

Proof. For $k = 0, \dots, N - 1$ it holds that

$$\begin{aligned} \|y(t_{k+1}) - Y_{k+1}\| &= \|y(t_k) + h_k \Phi(t_k, y(t_k), h_k) + h_k \tau(t_k, h_k) - [Y_k + h_k \Phi(t_k, Y_k, h_k)]\| \\ &\leq \|y(t_k) - Y_k\| + h_k \|\Phi(t_k, y(t_k), h_k) - \Phi(t_k, Y_k, h_k)\| + h_k \|\tau(t_k, h_k)\| \\ &\leq \|y(t_k) - Y_k\| + h_k L \|y(t_k) - Y_k\| + h_k \|\tau(t_k, h_k)\|. \end{aligned}$$

Since $1 + t \leq e^t$ for $t \geq 0$, it follows

$$\|y(t_{k+1}) - Y_{k+1}\| \leq e^{Lh_k} \|y(t_k) - Y_k\| + h_k \|\tau(t_k, h_k)\|.$$

Hence, by induction

$$\begin{aligned} \|y(t_{k+1}) - Y_{k+1}\| &\leq e^{Lh_k} \left(e^{L(t_k - a)} \sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\| \right) + h_k \|\tau(t_k, h_k)\| \\ &\leq e^{L(t_{k+1} - a)} \sum_{j=0}^k h_j \|\tau(t_j, h_j)\| \end{aligned}$$

□

Example 1.25 Let $f \in \mathcal{C}_{\text{PL}}$ with Lipschitz constant L , then the improved Euler-Method with increment function $\Phi(t, y, h) = f(t + \frac{h}{2}, y + \frac{h}{2}f(t, y))$ satisfies

$$\|\Phi(t, y_1, h) - \Phi(t, y_2, h)\| \leq (1 + \frac{h}{2}L)L\|y_1 - y_2\|$$

for $0 \leq h \leq b - a$, $a \leq t \leq b - h$ and all $y_1, y_2 \in \mathbb{R}^n$.

△

Remark 1.26 The constant $e^{L(t_k - a)}$ in Theorem 1.24 can become very large due to the dependence of L from the incremental function Φ , that in turn is dependent on the Lipschitz constant of f . An improvement is obtained with the next Theorem. △

Theorem 1.27 (improved Stability) Let y be solution of (IVP) defined on the whole $[a, b]$. Let $\|\cdot\| = \langle \cdot, \cdot \rangle^{\frac{1}{2}}$ and for some $\bar{h} > 0$ assume that the incremental function $\Phi: [a, b] \times \mathbb{R}^n \times [0, \bar{h}]$ is uniformly L -Lipschitz w.r.t the second component, i.e.

$$\|\Phi(t, y_1, h) - \Phi(t, y_2, h)\| \leq L\|y_1 - y_2\|, \quad y_1, y_2 \in \mathbb{R}^n,$$

for $t \in [a, b]$, $h \in [0, \bar{h}]$ and satisfies

$$\langle y_1 - y_2, \Phi(t, y_1, h) - \Phi(t, y_2, h) \rangle \leq \theta \|y_1 - y_2\|^2$$

for $\theta \in \mathbb{R}$. Then for $h = \max_k h_k$ it holds the estimate

$$\|y(t_k) - Y_k\| \leq e^{\max\{\theta + \frac{h}{2}L^2, 0\}(t_k - a)} \sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\|, \quad k = 1, \dots, N.$$

Proof. Exercise :).

□

Now we define the concept of convergence for the one-step methods in mind.

Definition 1.28 (Convergence) A one-step method converges for our (IVP) on the interval $[a, b]$ if for each sequence of step sizes (h_k) such that $h = \max_k h_k \rightarrow 0$ it holds that the global error goes to zero. In particular

$$\max_{k=0, \dots, N-1} \|Y_k - y(t_k)\| \rightarrow 0 \text{ as } h \rightarrow 0.$$

The method has order of convergence q if q is the largest natural number such that there exists $C > 0$ and for some $\bar{h} > 0$ and all $h \in (0, \bar{h})$ it holds

$$\max_{k=0, \dots, N-1} \|Y_k - y(t_k)\| \leq Ch^q.$$

It holds

$$\sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\| \leq (x_k - a) \max_{j=0, \dots, k-1} \|\tau(t_j, h_j)\| \leq (b - a) \max_{j=0, \dots, N-1} \|\tau(t_j, h_j)\|.$$

So assume that we have a consistency estimate of the form

$$\|\tau(t, h)\| \leq Kh^q,$$

for some $K > 0$ and $q \in \mathbb{N}$ and $h \in (0, \bar{h}]$. Then for $h = \max_j h_j$, it holds

$$\sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\| \leq (b - a)Kh^q.$$

So under the setup of Theorem 1.24 it follows that

$$\|y(t_k) - Y_k\| \leq e^{L(t_k - a)} \sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\| \leq e^{L(t_k - a)} (b - a)Kh^q.$$

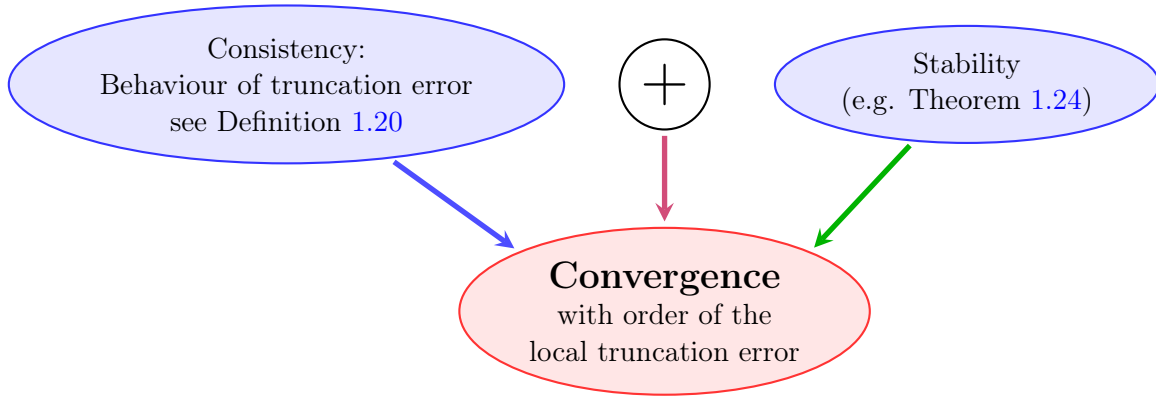
Taking the maximum on both sides it follows for $C = e^{L(b-a)}(b-a)K$

$$\max_k \|y(t_k) - Y_k\| \leq Ch^q.$$

For many ODEs, the right hand side f is **not defined** on the whole strip $[a, b] \times \mathbb{R}^n$ and in particular **not Lipschitz**.

Theorem 1.29 Consider our (IVP) with right hand side $f: [a, b] \times G \rightarrow \mathbb{R}^n$ and solution $y: [a, b] \rightarrow G$ for $G \subset \mathbb{R}^n$ open. Assume the increment function Φ is defined on

$$S_\delta := \{(t, Y, h) \mid 0 \leq h \leq \bar{h}, t \in [a, b - h], \|Y - y(t)\| \leq \delta\}$$



for some $\delta > 0$ and for all $(t, Y, u) \in S_\delta$ it holds

$$\|\Phi(t, y(t), h) - \Phi(t, Y, h)\| \leq L\|y(t) - Y\|.$$

Now additional assume that

$$e^{L(b-a)} \sum_{j=0}^{N-1} h_j \|\tau(t_j, h_j)\| \leq \delta.$$

Then, the iterates $Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k)$ for $k = 0, \dots, N-1$ with $Y_0 = y_0$ are welldefined and satisfy

$$\|y(t_k) - Y_k\| \leq e^{L(t_k-a)} \sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\|.$$

Proof. Similar to Theorem 1.24 using induction. [Exercise](#). □

At the end of this section we discuss the effect of perturbation errors.

Theorem 1.30 (Effect of perturbation errors) Let the assumptions of Theorem 1.29 hold. Moreover, let σ_k , for $k = 0, \dots, N-1$ be a sequence of perturbations satisfying

$$e^{L(b-a)} \sum_{j=0}^{N-1} \|\sigma_j\| \leq \frac{\delta}{2}$$

and assume

$$e^{L(b-a)} \sum_{j=0}^{N-1} h_j \|\tau(t_j, h_j)\| \leq \frac{\delta}{2}.$$

Then, the iteration

$$Y_0 = y_0, \quad Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k) + \sigma_k$$

is well-defined and it holds

$$\|y(t_k) - Y_k\| \leq e^{L(t_k-a)} \left[\sum_{j=0}^{k-1} h_j \|\tau(t_j, h_j)\| + \sum_{j=0}^{k-1} \|\sigma_j\| \right]$$

Proof. As in Theorem 1.29. □

So assume

$$\|\sigma_j\| \sim \epsilon > 0 \quad \Rightarrow \quad \sum_{j=0}^{k-1} \|\sigma_j\| \sim k\epsilon.$$

The influence of the perturbations grows proportional to the number of grid points and at some points dominates the influence of the local truncation errors.

Example 1.31 Consider the IVP

$$y'(t) = -200ty(t)^2, \quad -1 \leq t \leq 0, \quad y(-1) = \frac{1}{101}.$$

with exact solution $y(t) = \frac{1}{100t^2+1}$. Compute approximations of $y(0) = 1$ with the Euler scheme and the improved Euler scheme with constant step size using float and double floating point accuracy. [Exercise](#). △

Remark 1.32 One-step methods yield approximation Y_k to $y(t_k)$ at each grid point t_k for $k = 1, \dots, N$. In order to make the approximation a function on $[a, b]$ a typical approach is piecewise linear interpolation, i.e. connecting (t_k, Y_k) linearly with (t_{k+1}, Y_{k+1}) . △

1.2.2 Explicit Runge-Kutta schemes

Idea: Construct an incremental function Φ , that is a linear combination of values of $f(t, Y)$ in different points to obtain methods of high order at the cost of evaluating f more often per iteration step.

For simplicity assume $y'(t) = f(t)$, hence

$$y(t_{k+1}) = y(t_k) + \int_{t_k}^{t_{k+1}} f(t) dt.$$

Then, we can apply an arbitrary quadrature rule of level $s \in \mathbb{N}$ of the form

$$\int_{t_k}^{t_{k+1}} f(t) dt \approx h_k \sum_{j=1}^s b_j f(t_k + c_j h_k)$$

for suitable $b_j, c_j \in \mathbb{R}$. The local error at starting value $Y_k = y(t_k)$ is given as

$$y(t_{k+1}) - Y_{k+1} = y(t_{k+1}) - y(t_k) - h_k \sum_{j=1}^s b_j f(t_k + c_j h_k) = \int_{t_k}^{t_{k+1}} f(t) dt - h_k \sum_{j=1}^s b_j f(t_k + c_j h_k),$$

that is exactly the quadrature error. Basic concept for a Runge-Kutta scheme for our problem (IVP) with right hand side $f(t, y)$ is a similar ansatz

$$\int_{t_i}^{t_{k+1}} f(t, y(t)) dt \approx h_k \sum_{j=1}^s b_j f(t_k + c_j h_k, \boldsymbol{\eta}_j), \quad (1.13)$$

where now additional approximations $\boldsymbol{\eta}_j$ for intermediate values $y(t_k + c_j h_k)$ have to be considered for $j = 1, \dots, s$.

Example 1.33 (Improved Euler method) Making the ansatz with midpoint rule we first obtain

$$Y_{k+1} = Y_k + h_k f(t_k + \frac{h_k}{2}, \boldsymbol{\eta}_1)$$

for an approximation $\boldsymbol{\eta}_1 \approx y(t_k + \frac{h_k}{2})$ to be chosen. Using the explicit Euler scheme with step size $\frac{h_k}{2}$ yields

$$\boldsymbol{\eta}_1 = Y_k + \frac{h_k}{2} f(t_k, Y_k),$$

and in total

$$Y_{k+1} = Y_k + h_k f(t_k + \frac{h_k}{2}, Y_k + \frac{h_k}{2} f(t_k, Y_k))$$

which is the improved Euler scheme from (1.16) known as *Runge scheme* (1895, aka *explicit midpoint rule*). Let f be sufficiently smooth, then for $h_k \equiv h$ by Taylor expansion:

$$\begin{aligned} Y_{k+1} &= Y_k + h f(t_k + \frac{h}{2}, Y_k + \frac{h}{2} f(t_k, Y_k)) \\ &= Y_k + h [f(t_k, Y_k) + \frac{h}{2} (\partial_t f)(t_k, Y_k) + \frac{h}{2} (\partial_y f)(t_k, Y_k) + \mathcal{O}(h^2)]. \end{aligned}$$

Differentiating the differential equation leads

$$y''(t) = \frac{\partial}{\partial t} f(t, y(t)) = (\partial_t f)(t, y(t)) + (\partial_y f)(t, y(t)) f(t, y(t)).$$

Hence, for the local error starting at $Y_k = y(t_k)$ we conclude by Taylor expansion that

$$\begin{aligned} y(t_{k+1}) &= y(t_k) + h y'(t_k) + \frac{h^2}{2} y''(t_k) + \mathcal{O}(h^3) \\ &= Y_k + h f(t_k, Y_k) + \frac{h^2}{2} [(\partial_t f)(t_k, Y_k) + (\partial_y f)(t_k, Y_k)] + \mathcal{O}(h^3). \end{aligned}$$

Consequently,

$$\|Y_{k+1} - y(t_{k+1})\| \in \mathcal{O}(h^3).$$

The method has **consistency order** $q = 2$. △

Definition 1.34 (explicit Runge-Kutta methods) Let $s \in \mathbb{N}$. A s -stage explicit Runge-Kutta method has the form

$$Y_0 = y(t_0), \quad Y_{k+1} = Y_k + h \Phi(t, Y_k, h), \quad k = 0, 1, \dots$$

with

$$\Phi(t, Y, h) = \sum_{i=1}^s b_i K_i(t, Y, h),$$

based on increments

$$\begin{aligned} K_1(t, Y, h) &= f(t, Y) \\ K_i(t, Y, h) &= f \left(t + c_i h, Y + h \sum_{j=1}^{i-1} a_{ij} K_j(t, Y, h) \right), \quad i = 2, \dots, s \end{aligned}$$

and constants $c_2, \dots, c_s; b_1, \dots, b_s; a_{ij} \in \mathbb{R}$ for $i = 1, \dots, s$ and $j = 1, \dots, i-1$.

In the spirit of (1.13) let $\boldsymbol{\eta}_{k+1}^{(i)} = Y_k + h \sum_{j=1}^{i-1} a_{ij} f(t_k + c_j h, \boldsymbol{\eta}_{k+1}^{(j)})$, then we can equivalently represent the increment function as

$$\Phi(t, Y, h) = \sum_{i=1}^s b_i f(t_k + c_i h, \boldsymbol{\eta}_{k+1}^{(i)}).$$

This representation is useful, when extending explicit Runge-Kutta methods to the **implicit** case, see also Remark 1.35.

We can summarize an explicit Runge Kutta scheme in a so-called *Butcher tableau*:

$$\begin{array}{c|cccc} 0 & 0 & \dots & \dots & 0 \\ c_2 & a_{21} & & & \\ c_3 & a_{31} & a_{32} & & \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & a_{s2} & \dots & a_{s,s-1} \\ \hline & b_1 & b_2 & \dots & b_{s-1} & b_s \end{array} = \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^\top \end{array}$$

with nodes $\mathbf{c}$, matrix $\mathbf{A}$ of the method and weights $\mathbf{b}$.

Remark 1.35 The matrix $\mathbf{A}$ is strictly lower triangular for explicit Runge-Kutta methods. For so called **implicit** Runge Kutta methods, that we introduce later, $\mathbf{A}$ might be lower triangular (non-zero diagonal) or even dense. Then, $\boldsymbol{\eta}_{k+1}^{(i)} = Y_k + h \sum_{j=1}^s a_{ij} f(t_k + c_j h, \boldsymbol{\eta}_{k+1}^{(j)})$. and in general $\boldsymbol{\eta}_j$ is given implicitly. $\triangle$

Example 1.36 The following Butcher tableaus

$$\begin{array}{c|c} 0 & 0 \\ \hline & 1 \end{array}, \quad \begin{array}{c|cc} 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \hline & 0 & 1 \end{array}, \quad \begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}, \quad \begin{array}{c|cccc} 0 & 0 & \dots & \dots & 0 \\ \frac{1}{2} & \frac{1}{2} & & & \\ \frac{1}{2} & 0 & \frac{1}{2} & & \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array} \quad (1.14)$$

represent the *explicit Euler method*, the *Runge method* (aka *improved Euler method*), the *Heun method* (based on trapezoid + explicit Euler for $\boldsymbol{\eta}_1$) and the *classical Runge-Kutta method* (based on Simpson rule), respectively. $\triangle$

Lemma 1.37 Assume a one-step method of the form

$$Y_{k+1} = Y_k + h \sum_{i=1}^s b_i f(t_k + c_i h, \boldsymbol{\eta}_j)$$

has consistency order $q \in \mathbb{N}$. Then, the quadrature rule

$$\mathcal{Q}[g] := \sum_{i=1}^s b_i g(c_i) \approx \int_0^1 g(t) dt$$

is exact for polynomials up to degree $q - 1$.

Proof. W.l.o.g. let $n = 1$. Let $p = 0, \dots, q-1$ and $y'(t) = (t-a)^p$, $y(a) = 0$, with unique solution $y(t) = \frac{(t-a)^{p+1}}{p+1}$ on $[a, b]$. By consistency assumption

$$|y(t_1) - Y_1| = |y(a+h) - Y_1| = \left| \frac{1}{p+1} h^{p+1} - h \sum_{j=1}^s b_j (c_j h)^p \right| \in \mathcal{O}(h^{p+1}), \quad \text{for } h \rightarrow 0.$$

Hence,

$$\left| \frac{1}{p+1} - \sum_{j=1}^s b_j c_j^p \right| \in \mathcal{O}(h^{q-p}), \quad \text{for } h \rightarrow 0.$$

Consequently, in the limit $h \rightarrow 0$

$$\sum_{j=1}^s b_j c_j^p = \frac{1}{p+1} = \int_0^1 t^p dt,$$

which means exactness for all monomials t^p for $p = 0, \dots, q-1$. $\square$

Remark 1.38 As a consequence, a Runge-Kutta method of stage s can have **maximum consistency order** $2s$ due to the limit of exactness order $2s-1$ for quadrature formulas with s abscissas. However for Runge-Kutta methods also η_j is designed by quadrature and this will limit the consistency order further for fixed s (in the explicit case). $\triangle$

Our next goal is the discussion of the degrees of freedoms of a Runge-Kutta scheme, namely $\mathbf{c}$, $\mathbf{b}$ and $\mathbf{A}$.

For this we will consider test IVPs to discuss preliminary constraints on the degrees of freedom. Let $y'(t) = 1$, i.e. $f(t, y) \equiv 1$. Then, $y(t) = y_0 + (t-a)$ and $y(t_k + h) = y(t_k) + h$. Then, stage computation leads (using $c_1 = 0$):

$$K_i(t_k, Y_k, h) = f(t_k + c_i h, Y_k + h \sum_{j=1}^{i-1} a_{ij} K_j(t_k, Y_k, h)) \equiv 1.$$

Hence, for the iteration

$$Y_{k+1} = Y_k + h \sum_{i=1}^s b_i K_i(t_k, Y_k, h) = Y_k + h \sum_{i=1}^s b_i = y_0 + (t_k - a) \sum_{i=1}^s b_i.$$

Comparing with $y(t) = y_0 + (t-a)$ consistency is only possible for

$$\boxed{\sum_{i=1}^s b_i = 1.} \tag{1.15}$$

Now recall for each $i = 1, \dots, s$ the ansatz of the Runge-Kutta scheme (choice of η_i) generalized to the **implicit regime** as in Remark 1.35, s.t.

$$\begin{aligned} y(t_k + c_i h) &= y(t_k) + \int_{t_k}^{t_k + c_i h} f(t, y(t)) dt \\ &\approx \eta_i = Y_k + h \sum_{j=1}^s a_{ij} f(t_k + c_j h, \eta_j). \end{aligned}$$

With the choice of $f \equiv 1$ yields the necessary condition

$$\boxed{\sum_{j=1}^s a_{ij} = \frac{1}{h} \int_{t_k}^{t_k+c_i h} 1 \, dt = c_i.} \quad (1.16)$$

Both necessary conditions have in fact a bigger impact.

Theorem 1.39 Let f be continuous on $[a, b] \times \mathbb{R}^n$. Then, an explicit Runge-Kutta scheme is consistent if and only if

$$\sum_{i=1}^s b_i = 1. \quad (1.17)$$

Proof. By continuity of f and the definition of K_i it holds

$$\lim_{h \rightarrow 0} K_i(t, Y, h) = f(t, Y), \quad \forall (t, Y) \in [a, b] \times \mathbb{R}^n.$$

Then, for $t = t_k, Y = y(t_k)$ it follows

$$\begin{aligned} \lim_{h \rightarrow 0} |f(t_k, y(t_k)) - \Phi(t_k, y(t_k), h)| &= \lim_{h \rightarrow 0} \left| f(t_k, y(t_k)) - \sum_{i=1}^s b_i K_i(t_k, y(t_k), h) \right| \\ &= \left| f(t_k, y(t_k)) \left(1 - \sum_{i=1}^s b_i \right) \right| = 0 \end{aligned}$$

if and only if $\sum_{i=1}^s b_i = 1$. □

Theorem 1.40 Let (IVP) have a solution $y \in \mathcal{C}^2([a, b])$ and let $f \in \mathcal{C}_{\text{PL}}$. Let $Y_k = y(t_k)$ and

$$c_i = \sum_{j=1}^{i-1} a_{ij}, \quad i \geq 2.$$

Then, $K_i(t_k, Y_k, h)$ is an first order approximation of $y'(t_k + c_i h)$, i.e.

$$\|y'(t_k + c_i h) - K_i(t_k, Y_k, h)\| \in \mathcal{O}(h^2).$$

Proof. By induction: Let $i = 2$, by Lipschitz continuity and Taylor it follows

$$\begin{aligned} \|y'(t_k + c_2 h) - K_2(t_k, Y_k, h)\| &= \|f(t_k + c_2 h, y(t_k + c_2 h)) - f(t_k + c_2 h, y(t_k) + h a_{21} f(t_k, y(t_k)))\| \\ &\leq L \|y(t_k + c_2 h) - y(t_k) - h a_{21} f(t_k, y(t_k))\| \\ &= L \|y(t_k) + c_2 h y'(t_k) + \mathcal{O}(h^2) - y(t_k) - h a_{21} y'(t_k)\| \\ &= L \|(c_2 - a_{21}) h y'(t_k) + \mathcal{O}(h^2)\| \end{aligned}$$

Hence for $c_2 = a_{21}$, the approximation error is of order $\mathcal{O}(h^2)$. For $i > 2$, let the statement holds for all indices $2, \dots, i-1$. Then, similarly

$$\begin{aligned}
\|y'(t_k + c_i h) - K_i(t_k, Y_k, h)\| &= \left\| f(t_k + c_i h, y(t_k + c_i h)) - f(t_k + c_i h, y(t_k) + h \sum_{j=1}^{i-1} a_{ij} K_j(t_k, Y_k, h)) \right\| \\
&\leq L \left\| y(t_k + c_i h) - y(t_k) - h \sum_{j=1}^{i-1} a_{ij} K_j(t_k, Y_k, h) \right\| \\
&= L \left\| c_i h y'(t_k) + \mathcal{O}(h^2) - h \sum_{j=1}^{i-1} a_{ij} (y'(t_k + c_j h) + \mathcal{O}(h^2)) \right\| \\
&= L \left\| c_i h y'(t_k) + \mathcal{O}(h^2) - h \sum_{j=1}^{i-1} a_{ij} (y'(t_k + c_j h) + \mathcal{O}(h^2)) \right\| \\
&= L \left\| c_i h y'(t_k) + \mathcal{O}(h^2) - h \sum_{j=1}^{i-1} a_{ij} (y'(t_k) + \mathcal{O}(h)) \right\| \\
&= L \left\| \left(c_i - \sum_{j=1}^{i-1} a_{ij} \right) h y'(t_k) + \mathcal{O}(h^2) \right\|,
\end{aligned}$$

where we used Taylor expansion (using the integral remainder) of $y'(t_k + c_j h) = y'(t_k) + \mathcal{O}(h)$. The approximation error is of order $\mathcal{O}(h^2)$ if $c_i = \sum_{j=1}^{i-1} a_{ij}$. $\square$

Remark 1.41 For $f \in C^1([a, b] \times \mathbb{R}^n)$ it can directly be shown that the explicit Runge-Kutta schema is consistent of order 1 if the condition $\sum_{i=1}^s b_i = 1$ holds. [Exercise](#) $\triangle$

A generalized result to Remark 1.41 is giving without proof.

Theorem 1.42 (See [?], Theorem 4.18) An explicit Runge-Kutta method is consistent for any right hand side $f \in \mathcal{C}^q([a, b] \times \mathbb{R}^n; \mathbb{R}^n)$ with order

1. $q = 1$: if

$$\sum_{i=1}^s b_i = 1$$

2. $q = 2$: if additional

$$\sum_{i=1}^s b_i c_i = 1/2$$

3. $q = 3$: if additional

$$\sum_{i=1}^s b_i c_i^2 = 1/3, \quad \sum_{i,j=1}^s b_i a_{ij} c_j = 1/6$$

4. $q = 4$: if additional

$$\sum_{i=1}^s b_i c_i^3 = 1/4, \quad \sum_{i,j=1}^s b_i c_i a_{ij} c_j = 1/8, \quad \sum_{i,j=1}^s b_i a_{ij} c_j^2 = 1/12, \quad \sum_{i,j,k=1}^s b_i a_{ij} a_{jk} c_k = 1/24$$

Note that the sufficient conditions for $p > 1$ become non-linear.

Remark 1.43 For a requested consistency order p a number N_p of conditions has to be fulfilled. Up to $p = 8$ the original work of J.C. Butcher, see [?], contains those conditions. It turns out that the number N_p can be calculated by combinatorial arguments, but their number grows exponentially in p as illustrated in the following table:

Table 1.2: Number of order conditions for Runge-Kutta-methods

p	1	2	3	4	5	10	20
N_p	1	2	4	8	17	1205	20247374.

△

Corollary 1.44 The explicit Euler method has consistency order $q = 1$, the Runge and Heun method have consistency order $q = 2$ and the classical Runge Kutta scheme has consistency order $q = 4$ for any right hand side $f \in \mathcal{C}^q([a, b] \times \mathbb{R}^n; \mathbb{R}^n)$.

As a next result we will refine Remark 1.38 and show that explicit stage s Runge-Kutta methods are of consistency order at most s

For this we need to derive the **stability function** (including the **implicit case**).

Consider the scalar (in particular $n = 1$) linear test equation

$$y'(t) = \lambda y(t), \quad \lambda \in \mathbb{C},$$

with initial value $y(0) = y_0$. Let an s -stage Runge-Kutta method with Butcher tableau

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^\top \end{array}$$

be applied to this problem.

The internal stages of the Runge-Kutta scheme are defined as

$$K_i = f \left(t_n + c_i h, Y_k + h \sum_{j=1}^s a_{ij} K_j \right), \quad i = 1, \dots, s.$$

For the test equation $f(t, y) = \lambda y$, this becomes

$$K_i = \lambda \left(Y_k + h \sum_{j=1}^s a_{ij} K_j \right), \quad i = 1, \dots, s.$$

In vector form, introducing $\mathbf{K} = (K_1, \dots, K_s)^\top \in \mathbb{C}^s$ and $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^s$, we can write

$$\mathbf{K} = \lambda (Y_k \mathbf{1} + h \mathbf{A} \mathbf{K}).$$

Rearranging terms yields (note that $Y_k \in \mathbb{C}$)

$$(\mathbf{I} - h\lambda \mathbf{A}) \mathbf{K} = \lambda Y_k \mathbf{1}.$$

Assuming $\mathbf{I} - h\lambda \mathbf{A}$ is invertible, we obtain

$$\mathbf{K} = \lambda (\mathbf{I} - h\lambda \mathbf{A})^{-1} Y_k \mathbf{1}.$$

Hence, the Runge–Kutta update reads

$$Y_{k+1} = Y_k + h \mathbf{b}^\top \mathbf{K}.$$

Substituting the expression for $\mathbf{K}$ gives

$$Y_{k+1} = Y_k + h \mathbf{b}^\top (\lambda (\mathbf{I} - h\lambda \mathbf{A})^{-1} Y_k \mathbf{1}) = \left(1 + h\lambda \mathbf{b}^\top (\mathbf{I} - h\lambda \mathbf{A})^{-1} \mathbf{1}\right) Y_k.$$

Defining $z = h\lambda \in \mathbb{C}$, we can write

$$Y_{k+1} = R(z) Y_k, \quad R(z) = 1 + z \mathbf{b}^\top (\mathbf{I} - z \mathbf{A})^{-1} \mathbf{1}.$$

Definition 1.45 (Stability function) Let $\overline{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$. Then, the associated *stability function* $R: \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$ for a (possible implicate) Runge Kutta scheme with coefficients $\mathbf{A}, \mathbf{b}, \mathbf{c}$ is given as

$$R(z) = 1 + z \mathbf{b}^\top (\mathbf{I} - z \mathbf{A})^{-1} \mathbf{1}.$$

Note that R is welldefined for all $z = \lambda h$, which are not eigenvalue of $\mathbf{A}$.

Theorem 1.46 (Consistency of maximum order s) A explicit Runge-Kutta scheme of stage $s \in \mathbb{N}$ has consistency order of maximum s .

Proof. In the case of explicit Runge-Kutta the matrix $\mathbf{A} \in \mathbb{R}^{s,s}$ is strictly lower triangular and hence nilpotent with $\mathbf{A}^s = \mathbf{0}$. It holds that

$$(\mathbf{I} - z \mathbf{A})(\mathbf{I} + z \mathbf{A} + \dots + z^{s-1} \mathbf{A}^{s-1}) = \mathbf{I} - z^s \mathbf{A}^s = \mathbf{I}$$

and thus $(\mathbf{I} - z \mathbf{A})^{-1}$ is invertible for each $z \in \mathbb{C}$ given as a matrix polynomial of degree $s - 1$. Consequently, R is a polynomial of degree s . Assume for some $q \in \mathbb{N}$ that the explicit Runge-Kutta schema is of consistency order q . Then, for the test equation

$$y'(t) = y(t), \quad y(a) = 1,$$

and step size $h > 0$ the local error satisfies

$$|Y_1 - y(h)| = |Y_1 - e^h| = |R(h) - e^h| \in \mathcal{O}(h^{q+1}), \quad \text{as } h \rightarrow 0.$$

Since $R(z)$ is a polynomial in z and e^z can be written as a local convergent Taylor expansion at $z = 0$, their Taylor polynomials must match up to order q . It follows $q \leq s$ and if $q = s$ then

$$R(z) = \sum_{k=0}^q \frac{z^k}{k!}.$$

□

So we have seen with Corollary 1.44, that the explicit Euler-, Runge-, Heun- and the classical Runge-Kutta method, all achieve maximum possible consistency order. The question arises, if the same holds true for higher number of stages $s \geq 5$.

Remark 1.47 (Order limitations, Butcher (1963,1965,1985)) For $s \geq 5$, the order $q = s$ can no longer be achieved by explicit Runge-Kutta methods. A more detailed analysis yields the following values $s_{\min}(q)$ for the minimal number of stages required by an explicit method to attain order q :

q	1	2	3	4	5	6	7	8
$s_{\min}(q)$	1	2	3	4	6	7	9	11

As we shall see later, for implicit Runge-Kutta methods one always has $q = 2s$. $\triangle$

Example 1.48 (stage 2 Runge-Kutta schemes of consistency order 2) For simplicity let $n = 1$. Runge-Kutta schemes of stage $s = 2$ have the incremental function

$$\Phi(t, Y, h) = b_1 f(t, Y) + b_2 f(t + c_2 h, Y + h a_{21} f(t, Y)) \in \mathbb{R}.$$

Taylor expansion around $h = 0$ and requiring consistency order $q = s = 2$, yields the condition for $q = 2$ from Theorem 1.42. Cp. Tutorium. $\triangle$

Theorem 1.49 (Stability of explicit Runge-Kutta methods) Let $f \in \mathcal{C}_{\text{PL}}$ with Lipschitz constant L . Then, the incremental function Φ satisfies

$$\|\Phi(t, y_1, h) - \Phi(t, y_2, h)\| \leq L(h) \|y_1 - y_2\|$$

with $L(h) = \sum_{i=1}^s |b_i| L_i(h)$ and

$$L_1(h) = L, \quad L_i(h) = \left(1 + h \sum_{j=1}^{i-1} |a_{ij}| L_j(h) \right) L, \quad i = 2, \dots, s.$$

For $i = 1, \dots, s$ it holds

$$\|K_i(t, y_1, h) - K_i(t, y_2, h)\| \leq L_i(h) \|y_1 - y_2\|. \quad (1.18)$$

Proof. Show (1.18) via induction and conclude the claim. $\square$

It follows

$$\lim_{h \rightarrow 0^+} L_i(h) = L \quad \text{and} \quad \lim_{h \rightarrow 0^+} L(h) = \left(\sum_{i=1}^s |b_i| \right) L.$$

So if $b_i \geq 0$, by consistency $\sum_{i=1}^s |b_i| = \sum_{i=1}^s b_i = 1$.

Remark 1.50 (Stability in light of dissipative systems) If additional to the assumption of Theorem 1.49 it holds that

$$\langle y_1 - y_2, f(t, y_1) - f(t, y_2) \rangle \leq \theta \|y_1 - y_2\|^2$$

and $b_i \geq 0$ for all $i = 1, \dots, s$. Then,

$$\langle y_1 - y_2, \Phi(t, y_1, h) - \Phi(t, y_2, h) \rangle \leq \theta(h) \|y_1 - y_2\|^2$$

with $\lim_{h \rightarrow 0^+} \theta(h) = \theta \sum_{i=1}^s b_i = \theta$. $\triangle$

Theorem 1.51 (Convergence of explicit Runge-Kutta methods) Explicit Runge-Kutta methods of stage s with consistency order $q \leq s$ are convergent with order q . The constant in the convergence estimate can be improved for dissipative systems.

Proof. Using Theorem 1.49 Runge-Kutta methods are stable by application of Theorem 1.24 and with improved constants due to Remark 1.50 and application of Theorem 1.27. Then, **stability** + **Consistency** yields **convergence**. $\square$

1.2.3 Step length control

When solving an ordinary differential equation numerically, the choice of the step size plays a decisive role in the quality and efficiency of the computed solution. If the step size is too large, the numerical method may lose accuracy or even become unstable, producing results that no longer resemble the true behavior of the system. If the step size is too small, the computation becomes unnecessarily expensive, and rounding errors may begin to accumulate.

In practice, the solution of an ODE often exhibits regions of very different behavior: it may vary slowly for a long time and then change rapidly within a short interval. Using a fixed step size throughout such a computation is inefficient—either one wastes computational effort in the smooth parts or loses accuracy in the rapidly changing ones.

Step size control addresses this problem by adapting the step length dynamically to the local behavior of the solution. The idea is simple yet powerful: estimate the local error made in each step, and adjust the next step size so that the error remains within a prescribed tolerance. In this way, the numerical method automatically takes small steps where the solution demands high resolution and larger steps where the dynamics are smooth.

This adaptive approach not only improves computational efficiency but also enhances robustness, allowing numerical solvers to handle stiff or highly variable problems with minimal user intervention. Step size control is thus a central component of modern ODE solvers and an essential tool in scientific computing.

Theorem 1.52 Consider different incremental functions $\tilde{\Phi}$ and Φ an associated one-step scheme

$$Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k), \quad \tilde{Y}_{k+1} = Y_k + h_k \tilde{\Phi}_{k+1}(t_k, Y_k, h_k)$$

for the solution of (IVP) with step sizes $h_k > 0$ and nodes $t_{k+1} = t_k + h_k$. Assume $\tilde{\Phi}$ is L -Lipschitz w.r.t. to the second input uniformly in t, h , i.e.

$$\|\tilde{\Phi}(t, y_1, h) - \tilde{\Phi}(t, y_2, h)\| \leq L \|y_1 - y_2\|, \quad \forall t \in [a, b], y_1, y_2 \in \mathbb{R}^n, h > 0.$$

Then

$$\|y(t_k) - Y_k\| \leq e^{L(t_k - t_0)} \sum_{j=0}^{k-1} \left(\|Y_{j+1} - \tilde{Y}_{j+1}\| + h_j \|\tilde{\tau}(t_j, h_j)\| \right), \quad k = 1, 2, \dots,$$

where $\tilde{\tau}$ denotes the local truncation error with respect to $\tilde{\Phi}$ as in Definition 1.18.

Proof. Analogue to the proof of Theorem 1.24 based on $1 + t \leq e^t$ and using

$$\begin{aligned} \|y(t_{k+1}) - Y_{k+1}\| &= \|y(t_k) + h_k \tilde{\Phi}(t_k, y(t_k), h_k) + h_k \tilde{\tau}(t_k, h_k) \underbrace{- Y_k - h_k \tilde{\Phi}(t_k, Y_k, h_k) + \tilde{Y}_{k+1}}_{=0} - Y_{k+1}\| \\ &\leq \|y(t_k) - Y_k + h_k(\tilde{\Phi}(t_k, y(t_k), h_k) - \tilde{\Phi}(t_k, Y_k, h_k))\| + \|\tilde{Y}_{k+1} - Y_{k+1}\| + h_k \|\tilde{\tau}(t_k, h_k)\| \\ &\leq e^{h_k L} \|y(t_k) - Y_k\| + \|\tilde{Y}_{k+1} - Y_{k+1}\| + h_k \|\tilde{\tau}(t_k, h_k)\| \end{aligned}$$

Unrolling the recurrence yields

$$\begin{aligned} \|y(t_{k+1}) - Y_{k+1}\| &\leq \sum_{j=0}^k \left((\|Y_{j+1} - \tilde{Y}_{j+1}\| + h_j \|\tilde{\tau}(t_j, h_j)\|) \prod_{i=j+1}^k e^{h_i L} \right) \\ &= \sum_{j=0}^k \left((\|Y_{j+1} - \tilde{Y}_{j+1}\| + h_j \|\tilde{\tau}(t_j, h_j)\|) e^{L(t_{k+1} - t_{j+1})} \right). \end{aligned}$$

Then, the claim follows by noting that $t_{k+1} - t_{j+1} \leq t_{k+1} - t_0$. $\square$

As a consequence of Theorem 1.52 if Φ is of consistency order q and $\tilde{\Phi}$ is of consistency order $\geq q + 1$, then the term $\sum_{j=0}^{k-1} h_j \|\tilde{\tau}(t_k, h_k)\|$ is of higher order and dominated by $\sum_{j=0}^{k-1} \|Y_{j+1} - \tilde{Y}_{j+1}\|$. The goal is now to choose step sizes h_j making the latter controlled, such that for some prescribed $\epsilon > 0$

$$\|Y_{j+1} - \tilde{Y}_{j+1}\| \approx h_j \epsilon \quad \Rightarrow \quad \sum_{j=0}^{k-1} \|Y_{j+1} - \tilde{Y}_{j+1}\| \approx (t_k - t_0) \epsilon.$$

One approach to choose h_k is to analyze the behavior of

$$\|\{Y_k + h_k \Phi(t_k, Y_k, h)\} - \{Y_k + h \tilde{\Phi}(t_k, Y_k, h)\}\|, \quad \text{as } h \rightarrow 0.$$

So let z be solution of

$$z'(t) = f(t, z(t)), \quad z(t_k) = Y_k.$$

Then, by construction for some function $e(t)$ we have

$$\frac{z(t+h) - z(t)}{h} - \Phi(t, z(t), h) = e(t)h^q + \mathcal{O}(h^{q+1}) \quad \text{and} \quad \frac{z(t+h) - z(t)}{h} - \tilde{\Phi}(t, z(t), h) = \mathcal{O}(h^{q+1}).$$

Hence, for a single step of approximation of $z(t+h)$ given as $Y = Y_k + h\Phi(t_k, Y_k, h)$ and $\tilde{Y} = Y_k + h\tilde{\Phi}(t_k, Y_k, h)$ it holds

$$\begin{aligned} Y - \tilde{Y} &= \{z(t_k) + h\Phi(t_k, Y_k, h)\} - \{z(t_k) + h\tilde{\Phi}(t_k, Y_k, h)\} \\ &= h \left\{ \frac{z(t_k+h) - z(t_k)}{h} - \tilde{\Phi}(t_k, Y_k, h) \right\} - h \left\{ \frac{z(t_k+h) - z(t_k)}{h} - \Phi(t_k, Y_k, h) \right\} \\ &= e(t_k)h^{q+1} + \mathcal{O}(h^{q+2}). \end{aligned}$$

As a consequence up to higher order terms for

$$Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k) \quad \text{and} \quad \tilde{Y}_{k+1} = Y_k + h_k \tilde{\Phi}(t_k, Y_k, h_k)$$

with the representation $h_k = sh$ for some unknown $s > 0$, it holds

$$\|Y_{k+1} - \tilde{Y}_{k+1}\| \approx Ch_k^{q+1} = C(sh)^{q+1} = s^{q+1}(Ch^{q+1}) = s^{q+1}\|Y - \tilde{Y}\|.$$

Hence, the requirement $\|Y_{k+1} - \tilde{Y}_{k+1}\| \approx h_k\epsilon = sh\epsilon$ leads for given h to the scaling factor

$$s = \left(\frac{h\epsilon}{\|Y - \tilde{Y}\|} \right)^{1/q}. \quad (1.19)$$

and hence for the recommended step size $h_k = sh \approx h \left(\frac{h\epsilon}{\|Y - \tilde{Y}\|} \right)^{1/q}$.

Algorithmus 1.2.1 Adaptive Step Size Control

Require: Current step size estimate h , current value Y_k at time t_k , increment functions Φ of order q and $\tilde{\Phi}$ of order $q + 1$, tolerance $\epsilon > 0$.

Ensure: Next value Y_{k+1} and next step size estimate h_{next} .

1: Compute trial steps:

$$Y = Y_k + h \Phi(t_k, Y_k, h), \quad \tilde{Y} = Y_k + h \tilde{\Phi}(t_k, Y_k, h).$$

2: Estimate the scaling factor:

$$s = \left(\frac{h\epsilon}{\|Y - \tilde{Y}\|} \right)^{1/q}.$$

3: **if** $s \geq 1$ **then**

4: Accept the step: $Y_{k+1} \leftarrow \tilde{Y}$ (typically).

5: Advance time: $t_{k+1} \leftarrow t_k + h$.

6: Update step size estimate: {Possibly double the step size}

$$h_{\text{next}} \leftarrow \min\{2, s\} h.$$

7: **else**

8: Reject the step.

9: Reduce step size: {possibly half the step size}

$$h \leftarrow \max\{1/2, s\} h.$$

10: Repeat from line 1 with new h .

11: **end if**

Richardson Extrapolation The approach of halving the step size can be used to construct schemes of higher order based on extrapolation.

Let us assume we have a reference scheme of order q with incremental function $\hat{\Phi}$. Let us compute Y_{k+1} based on two half steps using this incremental function, s.t.

$$\begin{aligned} Y_{k+\frac{1}{2}} &= Y_k + \frac{1}{2} \hat{\Phi}(t_k, Y_k, \frac{h_k}{2}) \\ Y_{k+1} &= Y_{k+\frac{1}{2}} + \frac{h_k}{2} \hat{\Phi}(t_k + \frac{h_k}{2}, Y_{k+\frac{1}{2}}, \frac{h_k}{2}). \end{aligned}$$

This corresponds to a one step method

$$Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k)$$

with incremental function

$$\Phi(t, y, h) = \frac{1}{2} \widehat{\Phi}(t, y, \frac{h}{2}) + \frac{1}{2} \widehat{\Phi}(t + \frac{h}{2}, y + \frac{h}{2} \widehat{\Phi}(t, y, \frac{h}{2}), \frac{h}{2}).$$

Now, let

$$\widehat{Y}_{k+1} = Y_k + h_k \widehat{\Phi}(t_k, Y_k, h_k)$$

and by extrapolation define

$$\widetilde{Y}_{k+1} = \frac{1}{2^q - 1} \left(2^q Y_{k+1} - \widehat{Y}_{k+1} \right).$$

This step can be interpreted as a one-step method with incremental function

$$\widetilde{\Phi}(t, y, h) = \frac{1}{2^q - 1} \left(2^q \Phi(t, y, h) - \widehat{\Phi}(t, y, h) \right).$$

Theorem 1.53 Let y be solution of the ODE $y'(t) = f(t, y(t))$. Assume for some $\delta > 0$ that the incremental function $\widehat{\Phi}$ is well-defined on the tube

$$\{(x, u, h) \mid 0 \leq h \leq \bar{h}, a \leq t \leq b, \|u - y(t)\| < \delta\}$$

where it satisfies the uniform Lipschitz condition

$$\|\widehat{\Phi}(t, y(t), h) - \widehat{\Phi}(t, u, h)\| \leq L \|y(t) - u\|.$$

. Assume that the associated local truncation error $\widehat{\tau}$ is of the form

$$\widehat{\tau}(t, h) = e(t)h^q + r_0(t, h)h^{q+1},$$

for some continuously differentiable function $e: [a, b] \rightarrow \mathbb{R}$ and a bounded function r_0 . Moreover let Φ and $\widetilde{\Phi}$ be defined on the same tube as $\widehat{\Phi}$. Then, the associated local truncation errors τ and $\widetilde{\tau}$ fulfill

$$\begin{aligned} \tau(t, h) &= e(t) \left(\frac{h}{2}\right)^q + r(t, h)h^{q+1}, \\ \widetilde{\tau}(t, h) &= R(t, h)h^{q+1}. \end{aligned}$$

with bounded functions r and R .

Proof. The proof is optional. See below in gray. □

Proof. It holds

$$\begin{aligned} \tau(t, h) &= \frac{y(t+h) - y(t)}{h} - \Phi(t, y(t), h) \\ &= \frac{1}{2} \left\{ \frac{y(t + \frac{h}{2}) - y(t)}{\frac{h}{2}} - \widehat{\Phi}(t, y(t), \frac{h}{2}) \right\} \\ &\quad + \frac{1}{2} \left\{ \frac{y(t+h) - y(t + \frac{h}{2})}{\frac{h}{2}} - \widehat{\Phi}(t + \frac{h}{2}, y(t + \frac{h}{2}), \frac{h}{2}) \right\} \\ &\quad - \frac{1}{2} \left\{ \widehat{\Phi}\left(t + \frac{h}{2}, y(t) + \frac{h}{2} \widehat{\Phi}(t, y(t), \frac{h}{2}), \frac{h}{2}\right) - \widehat{\Phi}\left(t + \frac{h}{2}, y(t + \frac{h}{2}), \frac{h}{2}\right) \right\} \end{aligned} \tag{1.20}$$

By assumption it holds

$$\frac{y(t + \frac{h}{2}) - y(t)}{\frac{h}{2}} - \widehat{\Phi}(t, y(t), \frac{h}{2}) = \widehat{\tau}(t, \frac{h}{2}) = e(t) \left(\frac{h}{2}\right)^q + r_0(t, \frac{h}{2}) \left(\frac{h}{2}\right)^{q+1} \quad (1.21)$$

and

$$\begin{aligned} \frac{y(t+h) - y(t + \frac{h}{2})}{\frac{h}{2}} - \widehat{\Phi}(t + \frac{h}{2}, y(t + \frac{h}{2}), \frac{h}{2}) &= e(t + \frac{h}{2}) \left(\frac{h}{2}\right)^q + r_0(t + \frac{h}{2}, \frac{h}{2}) \left(\frac{h}{2}\right)^{q+1} \\ &= e(t) \left(\frac{h}{2}\right)^q + \int_0^1 e'(t + \frac{h}{2}\eta) d\eta \left(\frac{h}{2}\right)^{q+1} + r_0(t + \frac{h}{2}, \frac{h}{2}) \left(\frac{h}{2}\right)^{q+1} \\ &= e(t) \left(\frac{h}{2}\right)^q + r_1(t, \frac{h}{2}) \left(\frac{h}{2}\right)^{q+1} \end{aligned} \quad (1.22)$$

and for the third term

$$\begin{aligned} &\left\| \widehat{\Phi}\left(t + \frac{h}{2}, y(t) + \frac{h}{2}\widehat{\Phi}(t, y(t), \frac{h}{2}), \frac{h}{2}\right) - \widehat{\Phi}\left(t + \frac{h}{2}, y(t + \frac{h}{2}), \frac{h}{2}\right) \right\| \\ &\leq L \left\| y(t) + \frac{h}{2}\widehat{\Phi}(t, y(t), \frac{h}{2}) - y(t + \frac{h}{2}) \right\| \\ &= L \left\| y(t + \frac{h}{2}) - \frac{h}{2}\widehat{\tau}(t, \frac{h}{2}) - y(t + \frac{h}{2}) \right\| \\ &= L \left\| \frac{h}{2} \left(e(t) \left(\frac{h}{2}\right)^q + r_0(t, \frac{h}{2}) \left(\frac{h}{2}\right)^{q+1} \right) \right\| \end{aligned} \quad (1.23)$$

Hence,

$$\widehat{\Phi}\left(t + \frac{h}{2}, y(t) + \frac{h}{2}\widehat{\Phi}(t, y(t), \frac{h}{2}), \frac{h}{2}\right) - \widehat{\Phi}\left(t + \frac{h}{2}, y(t + \frac{h}{2}), \frac{h}{2}\right) = r_2(t, \frac{h}{2}) \left(\frac{h}{2}\right)^{q+1} \quad (1.24)$$

with a bounded function r_2 . So combining (1.20), (1.21), (1.22) and (1.24) yields

$$\frac{y(t+h) - y(t)}{h} - \Phi(t, y(t), h) = e(t) \left(\frac{h}{2}\right)^q + r(t, h) \left(\frac{h}{2}\right)^{q+1}$$

for a bounded function r depending on r_0, r_1, r_2 yielding the first claim.

Using this result we obtain

$$\begin{aligned} \frac{y(t+h) - y(t)}{h} - \widetilde{\Phi}(t, y(t), h) &= \frac{2^q}{2^q - 1} \left\{ \frac{y(t+h) - y(t)}{h} - \Phi(t, y(t), h) \right\} \\ &\quad - \frac{1}{2^q - 1} \left\{ \frac{y(t+h) - y(t)}{h} - \widehat{\Phi}(t, y(t), h) \right\} \\ &= \frac{2^q}{2^q - 1} \left\{ e(t) \left(\frac{h}{2}\right)^q + r(t, h) \left(\frac{h}{2}\right)^{q+1} \right\} \\ &\quad - \frac{1}{2^q - 1} \left\{ e(t) h^q + r_0(t, h) h^{q+1} \right\} \\ &= \frac{1}{2^q - 1} \left\{ \frac{1}{2} r(t, h) - r_0(t, h) \right\} h^{q+1} \\ &= R(t, h) h^{q+1}. \end{aligned}$$

□

Note that explicit Runge-Kutta methods for smooth enough right hand sides f satisfy the assumptions of Theorem 1.53. Consequently, originating from a Runge-Kutta scheme of order q , we can design a one-step method based on extrapolation as described before with order $q + 1$. The new scheme can be seen as an explicit Runge-Kutta method with **triple** amount of stages. This is **far from optimal**, in particular requiring many evaluations of the right hand side.

Embedded Runge-Kutta methods The idea of embedded Runge-Kutta methods is at step k to construct two iterates of the form

$$Y_{k+1} = Y_k + h \sum_{i=1}^s b_i K_i(t, Y_k, h), \quad \text{and} \quad \tilde{Y}_{k+1} = Y_k + h \sum_{i=1}^s \tilde{b}_i K_i(t, Y_k, h)$$

where only the parameters b_i and $\tilde{b}_i$ possibly differ and the Runge-Kutta scheme using parameters $(\tilde{b}_i)$ has a different order $\tilde{q}$ than the one using (b_i) with order q , typically $\tilde{q} = q + 1$ or $\tilde{q} = q - 1$. Consequently, the computation of $\tilde{Y}_{k+1}$ does **not require additional evaluations** of the right hand side f of the IVP. Such an embedded Runge-Kutta method is denoted as $\text{RK}\tilde{q}(q)$.

The embedded Runge-Kutta scheme can be summarized as

$$\begin{array}{c|cccc} 0 & 0 & \dots & \dots & 0 \\ c_2 & a_{21} & & & \vdots \\ c_3 & a_{31} & a_{32} & & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & a_{s2} & \dots & a_{s,s-1} & 0 \end{array} = \frac{\mathbf{c} \mid \mathbf{A}}{\mathbf{b}^\top \mid \tilde{\mathbf{b}}^\top}$$

Then, Theorem 1.42 can be applied to construct such a parameter sets for various schemes, assuming they exist!

Example 1.54 (Construction is not trivial) This example shows, that naive embedding may not work. Let us start with the classical Runge-Kutta scheme of order $\tilde{q} = 4$ from Example 1.36 given as

$$\begin{array}{c|cccc} 0 & 0 & \dots & \dots & 0 \\ \frac{1}{2} & \frac{1}{2} & & & \vdots \\ \frac{1}{2} & 0 & \frac{1}{2} & & \vdots \\ 1 & 0 & 0 & 1 & 0 \end{array} \quad \begin{array}{c} \vdots \\ \vdots \\ \vdots \\ \vdots \end{array}$$

We now seek for an embedded scheme by determining b_1, b_2, b_3, b_4 such that its corresponding incremental function Φ is of order $q = 3$. Application of Theorem 1.42 leads the only solution $\mathbf{b}^\top = (\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6})$, which is nothing but the classical Runge-Kutta scheme of order $q = 4$. The construction failed. $\triangle$

Surprisingly, adding a stage s may be required to actually allow for the construction of embedded pairs. To compensate for the increased number of evaluations of the right hand side f , the so called *Fehlberg-Trick* (FSAL, first same as last) requires, that the s -th stage K_s of the current iterate and the first stage K_1^* of the next iteration coincide in case of the same chosen (adaptive) stepsize h . In particular, adding the constrains

$$K_s = f(t+c_sh, t+h \sum_{j=1}^{s-1} a_{sj} K_j(t, y, h)) \stackrel{!}{=} f(t+h, y+h \sum_{j=1}^s b_j K_j(t, y, h)) = f(t+h, y+h\Phi(t, y, h)) = K_1^*. \quad (1.25)$$

Note that, in general there must not exist a solution for coefficients. But when they exist, they yield an efficient solution scheme.

Example 1.55 (Erwin Fehlberg's RK4(3), 1969) Based on classical Runge-Kutta method of order $\tilde{q} = 4$ we consider the embedded tableau

0	0		0		
$\frac{1}{2}$	$\frac{1}{2}$				
$\frac{1}{2}$	0	$\frac{1}{2}$			
1	0	0	1		
c_5	a_{51}	a_{52}	a_{53}	a_{54}	0
	b_1	b_2	b_3	b_4	b_5
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$	0

Applying Fehlberg-Trick and Theorem 1.42 leads parameters for order $q = 3$, [Exercise](#). $\triangle$

The RK4(3) is only a order 3 method, although its cost behave like a order 4 method. As an opposite idea one can use the control scheme of lower order, still obtain error estimation but having higher order scheme as the following example.

Example 1.56 (DOPRI5, $s = 6$, Dormand& Prince 1980) On of the most popular schemes is up to day is the RK4(5) scheme with an effective stage number $s = 6$ (due to Fehlberg Trick) by J.R. Dormand and P.J.Prince (1980) given as

0	0		0					
$\frac{1}{5}$	$\frac{1}{5}$							
$\frac{3}{10}$	$\frac{3}{40}$	$\frac{9}{40}$						
$\frac{4}{5}$	$\frac{44}{45}$	$-\frac{56}{15}$	$\frac{32}{9}$					
$\frac{8}{9}$	$\frac{19372}{6561}$	$-\frac{25360}{2187}$	$\frac{64448}{6561}$	$-\frac{212}{729}$				
1	$\frac{9017}{3168}$	$-\frac{355}{33}$	$\frac{46732}{5247}$	$\frac{49}{176}$	$-\frac{5103}{18656}$			
1	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	0	
	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	0	$q = 5$
	$\frac{5179}{57600}$	0	$\frac{7571}{16695}$	$\frac{393}{640}$	$-\frac{92097}{339200}$	$\frac{187}{2100}$	$\frac{1}{40}$	$\tilde{q} = 4$

It is implemented in DOPRI5 and Matlabs ode45, relies on a lower order control scheme and is of order 5, which is optimal according to Remark 1.47. $\triangle$

Remark 1.57 Other explicit one step methods in praxis are RKF4(5) with $s = 6$ (Fehlberg (1964)), RKF7(8) with $s = 13$ (Fehlberg, 1969) or DOPRI7(8) with $s = 13$ (Prince&Dormand (1981)). $\triangle$

1.2.4 Approximation of solution between grid points

Recalling Remark 1.32, a common approach to obtain approximate solution between grid points is obtained by piecewise affine linear interpolation between data pairs (t_k, Y_k) and (t_{k+1}, Y_{k+1}) . However, not high accuracy has to be expected then. However in some applications this is required.

Continuous Runge-Kutta-Methods

For this reason *continuous Runge-Kutta schemes* have been developed. They aim for high accuracy between grid points, with neglectable additional cost, i.e. without additional evaluation of the right hand side f .

The ansatz is to parametrize the nodes $\mathbf{b} \rightarrow \mathbf{b}(\theta)$ for some $\theta \in [0, 1]$ and approximate

$$y(t_k + \theta h_k) \approx Y_{k+\theta} := Y_k + h_k \sum_{i=1}^s b_i(\theta) K_i(t_k, Y_k, h_k),$$

while keeping the increments K_i untouched. Their Butcher-tableau reads

$$\begin{array}{c|cccc} 0 & 0 & \cdots & \cdots & 0 \\ c_2 & a_{21} & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ c_s & a_{s1} & \cdots & \cdots & a_{s,s-1} & 0 \\ \hline & b_1(\theta) & \cdots & \cdots & \cdots & b_s(\theta) \end{array} = \begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}(\theta)^\top \end{array}$$

The goal then, is to choose coefficients c_i, a_{ij} and b_i such that the consistency order is uniformly as high as possible for all $\theta \in [0, 1]$.

Example 1.58 The order-3 RK scheme from Heun denoted as *Heun3* and its continuous extension with consistency order 2 uniform in $\theta \in (0, 1)$ are given as

$$\begin{array}{c|ccc} 0 & 0 & \cdots & \cdots & 0 \\ \frac{1}{3} & \frac{1}{3} & \ddots & \ddots & \vdots \\ \frac{2}{3} & 0 & \frac{2}{3} & \ddots & 0 \\ \hline & \frac{1}{4} & 0 & \frac{3}{4} & \end{array} \quad \begin{array}{c|ccc} 0 & 0 & \cdots & \cdots & 0 \\ \frac{1}{3} & \frac{1}{3} & \ddots & \ddots & \vdots \\ \frac{2}{3} & 0 & \frac{2}{3} & \ddots & 0 \\ \hline & b_1(\theta) & b_2(\theta) & b_3(\theta) & \end{array} \quad b_i(\theta) = \begin{cases} \frac{3}{2}\theta^3 - \frac{9}{4}\theta^2 + \theta & i = 1, \\ 3\theta^2(1 - \theta) & i = 2, \\ \frac{3}{4}\theta^2(2\theta - 1) & i = 3. \end{cases}$$

Note that for $\theta = 1$ the continuous scheme recovers the order-3 scheme from Heun.

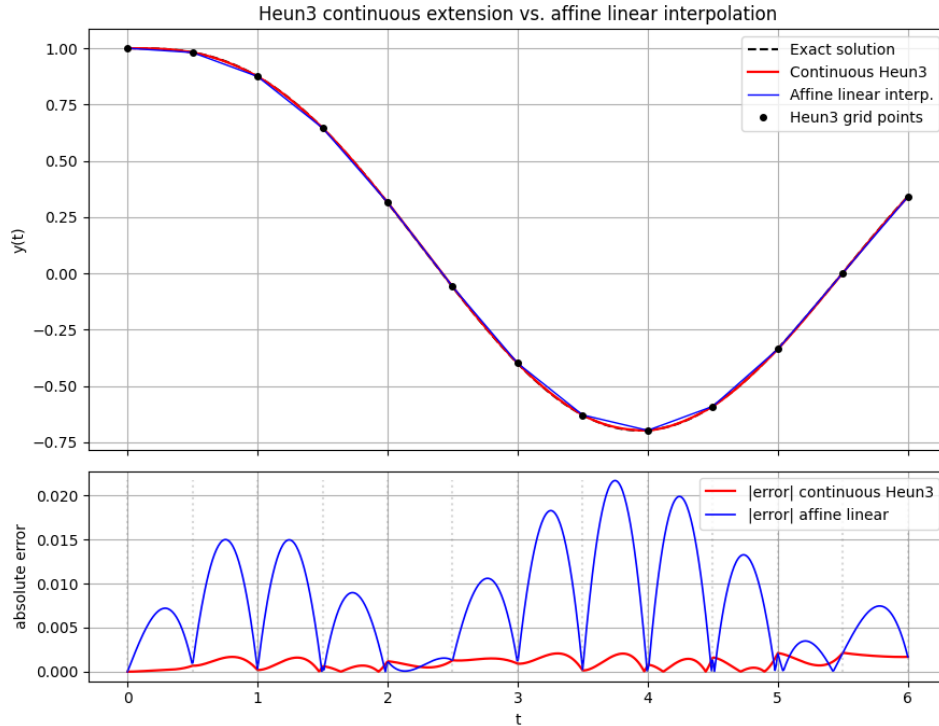


Figure 1.3: Comparison of Heun3 and continuous Heun3 for the IVP $y'(t) = \cos(t) - y(t)$ with $y(0) = 1$ with exact solution $y(t) = 0.5(\sin(t) + \cos(t)) + 0.5\exp(-t)$.

△

Example 1.59 A convenient, commonly used dense-output (continuous extension) for DOPRI5 is a quartic polynomial in $\theta \in [0, 1]$ written in the nested form (Numerical Recipes / Shampine style):

$$y(t_k + \theta h) = r_1 + \theta(r_2 + (1 - \theta)(r_3 + \theta(r_4 + (1 - \theta)r_5))),$$

with

$$\begin{aligned} r_1 &= Y_k, \\ r_2 &= Y_{k+1} - Y_k, \\ r_3 &= Y_k + hf_k - Y_{k+1}, \\ r_4 &= 2(Y_{k+1} - Y_k) - h(f_k + f_{k+1}), \\ r_5 &= d_1 hf_n + d_3 K_3 + d_4 K_4 + d_5 K_5 + d_6 K_6 + d_7 hf_{n+1}, \end{aligned}$$

where k_i are the stage increments $K_i = hf(t_k + c_i h, \cdot)$ for stages $i = 3, 4, 5, 6$ and $f_n = f(t_n, y_n)$, $f_{n+1} = f(t_{n+1}, y_{n+1})$. The numerical constants (as used in Numerical Recipes / ode45 implementations) are

$$\begin{aligned} d_1 &= -\frac{12715105075}{11282082432}, \\ d_3 &= \frac{87487479700}{32700410799}, \quad d_4 = -\frac{10690763975}{1880347072}, \\ d_5 &= \frac{701980252875}{199316789632}, \quad d_6 = -\frac{1453857185}{822651844}, \quad d_7 = \frac{69997945}{29380423}. \end{aligned}$$

Expanding the nested polynomial yields an explicit quartic polynomial in θ . This dense output

uses only the already computed stages and is the formula used in many implementations (e.g. ‘ode45’), see Shampine (1986) and discussion in Numerical Recipes. Note that this construction is of the form

$$y(t_k + \theta h) = y(t_k) + h \sum_{i=1}^{s=7} b_i(\theta) K_i$$

with $b_i(0) = 0$ and $b_i(1) = b_i$ with b_i from non-continuous DOPRI5. The choice of polynomials satisfying this is not unique, and in the construction above $b_2(\theta) \equiv 0$ leading no K_2 depends at all (cp. to definition of r_5 .)

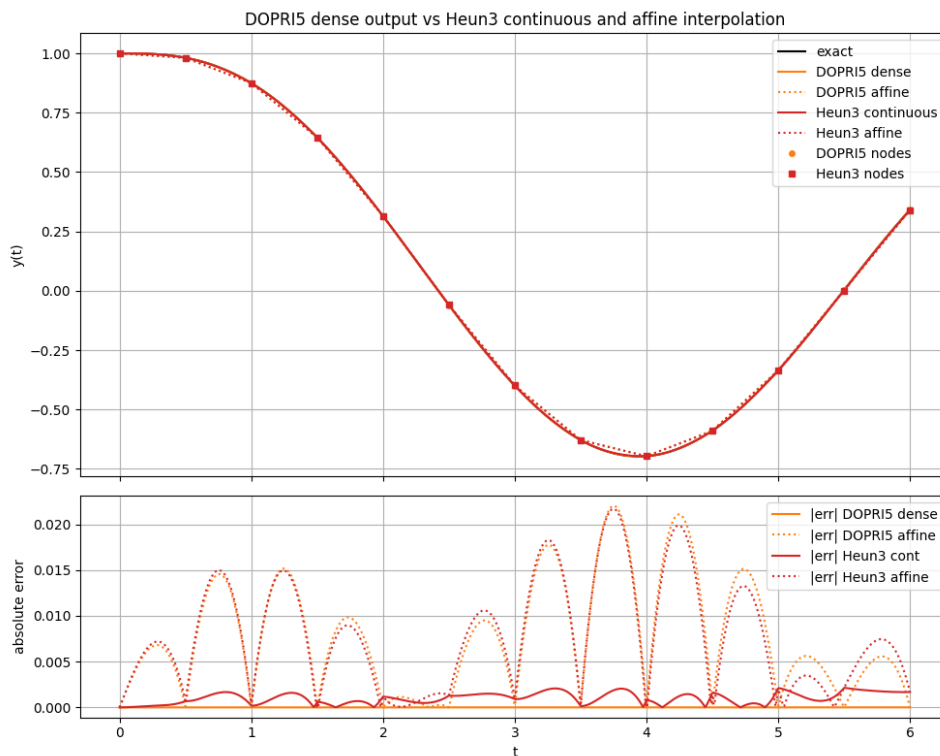


Figure 1.4: Comparison of continuous Heun3 and continuous DOPRI5 for the IVP $y'(t) = \cos(t) - y(t)$ with $y(0) = 1$ with exact solution $y(t) = 0.5(\sin(t) + \cos(t)) + 0.5 \exp(-t)$.

△

1.3 Multi-step methods

Motivation A key observation from (explicit) one-step methods such as the (explicit) Runge-Kutta methods was the fact that high order require multiple evaluations of the right hand side f in each step. Can we obtain high order and only have a few or even **one** evaluation of the right hand side per step?

An simple alternative for this is by utilizing multi-step methods. Here, the idea of using only the previous iterate Y_k to compute Y_{k+1} can be extended to the idea to utilizing $Y_k, Y_{k-1}, \dots, Y_{k-m}$ for the construction for $m \in \mathbb{N}$.

General multi-steps methods for iterates $(Y_k)_k$

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k \Phi_k(Y_{k-m+1}, \dots, Y_k, Y_{k+1}) \quad (1.26)$$

with initial given $Y_0, \dots, Y_{m-1}$. The general multi-step methods are *explicit* if

$$\Phi_k(Y_{k-m+1}, \dots, Y_k, Y_{k+1}) = \Phi_k(Y_{k-m+1}, \dots, Y_k),$$

e.g. in the k -th step does not depend on Y_{k+1} .

Definition 1.60 We call a multistep method linear, if it is of the form

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k \sum_{j=0}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j}). \quad (1.27)$$

In particular explicit linear multi-step methods with m steps is of the form

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k \sum_{j=1}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j}). \quad (1.28)$$

Remark 1.61 (Connection to One-step methods) Rearranging terms and normalize $a_{k,0} = 1$, we obtain the form

$$Y_{k+1} = \sum_{j=0}^{m-1} \tilde{a}_{k,j} Y_{k-j} + h_k \sum_{j=0}^{m-1} \tilde{b}_{k,j} f(t_{k-j}, Y_{k-j}) + h_k \tilde{b}_{k,-1} f(t_{k+1}, Y_{k+1}).$$

For $m = 1$, the multi-step method is a one step method. For $\tilde{b}_{k,-1} = 0$ the linear multi-step method is explicit and for *implicit* else. $\triangle$

Remark 1.62 (Initial values for m -step methods) A m -step method requires m initial values $Y_0, \dots, Y_{m-1}$. In praxis those can be obtained by a sequence of one-step, two-step and so one methods. Their approximation error and propagation effect needs to be accounted in rigorous analysis of multi-step methods. $\triangle$

1.3.1 Simple multi-step methods

We first discuss popular multi-step methods with simple structure of coefficients $(a_{k,j})_j$ in particular

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = Y_{k+1} - Y_k. \quad (1.29)$$

We derive a simple stability result as follows.

Theorem 1.63 Consider a grid $a = t_0 < t_1 < \dots < t_N = b$, for $k = 0, 1, \dots, N - 1$ functions $\Phi_k: (\mathbb{R}^n)^k \mapsto \mathbb{R}^n$ and for given $y_0 \in \mathbb{R}^n$ let $Y_1, \dots, Y_N$ defined by $Y_0 = y_0$ and

$$Y_{k+1} = Y_k + h_k \Phi_k(Y_0, \dots, Y_k).$$

Now for a given function $y: [a, b] \rightarrow \mathbb{R}^n$ consider the truncation errors

$$\tau_k = \frac{y(t_{k+1}) - y(t_k)}{h_k} - \Phi_k(Y_0, \dots, Y_k)$$

and assume for $k = 0, \dots, N - 1$ and $u_0, \dots, u_{N-1} \in \mathbb{R}^n$ that

$$\|\Phi_k(y(t_0), \dots, y(t_k)) - \Phi_k(u_0, \dots, u_k)\| \leq L \max_{j=0, \dots, k} \|y(t_j) - u_j\|.$$

Then, it holds for $k = 1, \dots, N$ that

$$\|y(t_k) - Y_k\| \leq e^{L(t_k - t_0)} \sum_{j=0}^{k-1} h_j \|\tau_j\|. \quad (1.30)$$

Proof. For $k = 0, 1, \dots, N - 1$ it holds

$$\begin{aligned} \|y(t_{k+1}) - Y_{k+1}\| &= \|y(t_k) + h_k \Phi_k(y(t_0), \dots, y(t_k)) + h_k \tau_k - Y_k - h_k \Phi_k(Y_0, \dots, Y_k)\| \\ &\leq \|y(t_k) - Y_k\| + h_k \|\Phi_k(y(t_0), \dots, y(t_k)) - \Phi_k(Y_0, \dots, Y_k)\| + h_k \|\tau_k\| \\ &\leq \|y(t_k) - Y_k\| + h_k L \max_{j=0, \dots, k} \|y(t_j) - Y_j\| + h_k \|\tau_k\| \end{aligned}$$

and then

$$\max_{j=0, \dots, k+1} \|y(t_j) - Y_j\| \leq (1 + h_k L) \max_{j=0, \dots, k} \|y(t_j) - Y_j\| + h_k \|\tau_k\|.$$

Then by induction over k it follows

$$\max_{j=0, \dots, k} \|y(t_j) - Y_j\| \leq e^{L(t_k - t_0)} \sum_{j=0}^{k-1} h_j \|\tau_j\|.$$

□

Remark 1.64 Theorem 1.63 is conservative and involves all previous steps from $0, \dots, k$. For one-step methods

$$\Phi_k(Y_0, \dots, Y_k) = \Phi(t_k, Y_k, h_k),$$

thus implying Theorem 1.24. For explicit multistep methods

$$\Phi_k(Y_0, \dots, Y_k) = \sum_{j=1}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j}).$$

Hence, to apply the stability Theorem 1.63 a sufficient condition for a Lipschitz-constant L requires $\sum_{j=1}^m |b_{k,j}|$ being bound independent of k and $f \in \mathcal{C}_{PL}$. $\triangle$

Now let us state some popular schemes of simple multi-step methods where the coefficients $a_{k,j}$ satisfying (1.29).

Example 1.65 (Adam-Bashforth method) Consider the approximations $f(t_{k+1-m}, Y_{k-m+1}), \dots, f(t_k, Y_k)$ for the derivatives $y'(t_{k-j}) = f(t_{k-j}, y(t_{k-j}))$ for $j = m-1, \dots, 0$. Define the associated unique Lagrange interpolation polynomial p of order m

$$p(t) = \sum_{j=1}^m f_{k+1-j} \prod_{\substack{\ell=1 \\ \ell \neq j}}^m \frac{t - t_{k+1-\ell}}{t_{k+1-j} - t_{k+1-\ell}},$$

with $f_{k+1-j} = f(t_{k+1-j}, Y_{k+1-j})$ for $j = 1, \dots, m$. Compute

$$Y_{k+1} = Y_k + \int_{t_k}^{t_{k+1}} p(t) dt \quad \text{and} \quad f_{k+1} = f(t_{k+1}, Y_{k+1}).$$

Then it holds

$$Y_{k+1} = Y_k + h_k \sum_{j=1}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j}) \quad (1.31)$$

with

$$b_{k,j} = \frac{1}{h_k} \int_{t_k}^{t_{k+1}} \prod_{\substack{\ell=1 \\ \ell \neq j}}^m \frac{t - t_{k+1-\ell}}{t_{k+1-j} - t_{k+1-\ell}} dt.$$

For $m = 1$ we obtain the **explicit Euler method**.

For $m = 2$ it holds

$$Y_{k+1} = Y_k + h_k (b_{k,1} f(t_k, Y_k) + b_{k,2} f(t_{k-1}, Y_{k-1}))$$

with

$$b_{k,1} = \frac{1}{h_k} \int_{t_k}^{t_{k+1}} \frac{t - t_{k-1}}{t_k - t_{k-1}} dt = 1 + \frac{h_k}{2h_{k-1}},$$

$$b_{k,2} = \frac{1}{h_k} \int_{t_k}^{t_{k+1}} \frac{t - t_k}{t_{k-1} - t_k} dt = -\frac{h_k}{2h_{k-1}}.$$

For constant step size $h_k \equiv h$ we obtain

$$\boxed{Y_{k+1} = Y_k + h \left(\frac{3}{2} f(t_k, Y_k) - \frac{1}{2} f(t_{k-1}, Y_{k-1}) \right)} \quad (1.32)$$

△

Remark 1.66 (Adam-Bashforth convergence) Under a local constraint of the step size, i.e. there exists $C > 0$ such that $h_k \leq Ch_{k-1}$ for all $k \in \mathbb{N}$, one can show that the truncation error

$$\begin{aligned} \tau_k &= \frac{y(t_{k+1}) - y(t_k)}{h_k} - \sum_{j=1}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j}) \\ &= \frac{y(t_{k+1}) - y(t_k)}{h_k} - \sum_{j=1}^m b_{k,j} y'(t_{k+1-j}) \end{aligned}$$

of the Adam-Bashforth scheme decays to 0 with order m for smooth enough right hand side f . Moreover one can show convergence of order m if

- $\sum_{j=1}^m |b_{k,j}|$ can be bounded independent of k (cp. Remark 1.64). A sufficient condition for this is $h_k \leq Ch_{k-1}$.
- accurate enough initial approximations $Y_0, Y_1, \dots, Y_{m-1}$.

△

Example 1.67 (Adam-Moulton scheme (implicit)) Let $m \in \mathbb{N}$ and P a polynomial of degree $m+1$. Then, the coefficients

$$b_{k,j}^* = \frac{1}{h_k} \int_{t_k}^{t_{k+1}} \prod_{\substack{\ell=0 \\ \ell \neq j}}^m \frac{t - t_{k+1-\ell}}{t_{k+1-j} - t_{k+1-\ell}} dt.$$

satisfy

$$P(t_{k+1}) = P(t_k) + h_k \sum_{j=0}^m b_{k,j}^* P'(t_{k+1-j}).$$

We compute Y_{k+1} as solution of

$$Y_{k+1} = Y_k + h_k \sum_{j=0}^m b_{k,j}^* f(t_{k+1-j}, Y_{k+1-j}).$$

For $m = 0$ one obtains the *implicit Euler method*, which has order $q = 1$.

For $m = 1$ one obtains *Crank-Nicolson Trapezoidal rule* (of order $q = 2$)

$$\boxed{Y_{k+1} = Y_k + \frac{h_k}{2} (f(t_k, Y_k) + f(t_{k+1}, Y_{k+1}))}. \quad (1.33)$$

For $m = 2$ we obtain a order 3 scheme as

$$Y_{k+1} = Y_k + h_k (b_{k,0}^* f(t_{k+1}, Y_{k+1}) + b_{k,1}^* f(t_k, Y_k) + b_{k,2}^* f(t_{k-1}, Y_{k-1}))$$

with

$$b_{k,0}^* = \frac{1}{6} \frac{3h_{k-1} + 2h_k}{h_{k-1} + h_k}, \quad b_{k,1}^* = \frac{2}{3} \frac{h_k}{h_{k-1}}, \quad b_{k,2}^* = -\frac{1}{6} \frac{h_k}{h_{k-1} + h_k}.$$

Note that $b_{k,0}^*$ and $b_{k,1}^*$ can be bounded independent of the grid. For $h_k \equiv h$ we obtain

$$\boxed{Y_{k+1} = Y_k + h \left(\frac{5}{12} f(t_{k+1}, Y_{k+1}) + \frac{2}{3} f(t_k, Y_k) - \frac{1}{12} f(t_{k-1}, Y_{k-1}) \right)} \quad (1.34)$$

△

Remark 1.68 (Consistency error of Adam-Moulton) Using Taylor argument for $(m+2)$ -times continuous differentiable solution y of (IVP) one can show that the truncation error

$$\tau_k^* := \frac{y(t_{k+1}) - y(t_k)}{h_k} - \sum_{j=0}^m b_{k,j}^* f(t_{k+1-j}, Y_{k+1-j})$$

satisfies

$$\|\tau_k^*\| \leq B \frac{(t_{k+1} - t_{k+1-m})^{m+1}}{(m+1)!}, \quad B = \left(1 + \sum_{j=0}^m |b_{k,j}^*|\right) \max_{t \in [t_{k+1-m}, t_{k+1}]} \|y^{(m+2)}(t)\|.$$

Step-size control as $h_{k-1} \leq Ch_k$ again ensures grid independent bounds on $|b_{k,j}^*|$ and it holds $|b_{k,0}^*| \leq 1$. $\triangle$

Example 1.69 (Predictor-corrector schemes) We introduce the PECE-class of Predictor-corrector schemes which is *explicit* and combines Adam-Bashforth and ideas of Adam-Moulton. Given t_k, Y_k, h_k and $f_{k+1-j} = f(t_{k+1-j}, Y_{k+1-j})$ for $j = 1, \dots, m$. We compute Y_{k+1}, f_{k+1} as follows:

(P)	$Y_{k+1}^{(0)} = Y_k + h_k \sum_{j=1}^m b_{k,j} f_{k+1-j}$	<i>prediction</i>	(1.35)
(E)	$f_{k+1}^{(0)} = f(t_{k+1}, Y_{k+1})$	<i>evaluation</i>	
(C)	$Y_{k+1} = Y_k + h_k \left(b_{k,0}^* f_{k+1}^{(0)} + \sum_{j=1}^m b_{k,j}^* f_{k+1-j} \right)$	<i>correction</i>	
(E)	$f_{k+1} = f(t_{k+1}, Y_{k+1})$	<i>evaluation</i>	

This scheme, independent of m requires 2 evaluations of the right hand side f per step. $\triangle$

Theorem 1.70 (PECE convergence of order $m+1$) Let $m \in \mathbb{N}$, y be $(m+2)$ -times continuous differentiable, $f \in \mathcal{C}_{PL}$ with Lipschitz constant L and assume step sizes such that $h_{k-1} \leq Ch_k$ for some $C > 0$. Then, PECE has convergence order $m+1$, where m is the order of convergence of the underlying Adam-Bashforth scheme.

Proof. The incremental functions Φ_k can be written as

$$\Phi_k(Y_0, \dots, Y_k) = b_{k,0}^* f(t_{k+1}, Y_k + h_k \sum_{j=1}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j})) + \sum_{j=1}^m b_{k,j}^* f(t_{k+1-j}, Y_{k+1-j})$$

and hence meets the requirements of Theorem 1.63. Then the claim follows once we showed, that the truncation error τ_k of PECE satisfies

$$\|\tau_k\| \leq \|\tau_k^{\text{AM}}\| + Lh_k \|\tau_k^{\text{AB}}\|,$$

where τ_k^{AM} and τ_k^{AB} are the truncation errors of the Adam-Moulton and the Adam-Bashforth scheme respectively. The latter follows relation

$$\begin{aligned} \tau_k &= \frac{y(t_{k+1}) - y(t_k)}{h_k} - \Phi_k(y(t_0), \dots, y(t_k)) \\ &= \tau_k^{\text{AM}} - b_{k,0}^* (f(t_{k+1}, y(t_{k+1})) + h_k \tau_k^{\text{AB}}) - f(t_{k+1}, y(t_{k+1})) \end{aligned}$$

and the Lipschitz continuity of f and the fact that $|b_{k,0}^*| \leq 1$ as stated in Remark 1.68. $\square$

1.3.2 Linear multi-step methods I - grid dependence case

So up to now we found a consistency error of order q if for all polynomials p of degree $\leq q$, it holds that

$$\frac{1}{h_k}(p(t_{k+1}) - p(t_k)) = \sum_{j=0}^m b_{k,j} p'(t_{k+1-j}), \quad (1.36)$$

including the case $b_{k,0} = 0$ for the explicit case. We saw an order $q \leq m$ and $q \leq m+1$ for Adams-Bashforth and Adams-Moulton scheme respectively. The question now is, if we can increase the order by introduction of more degrees of freedom $a_{k,j}$ for $j = 0, \dots, m$. So we turn back our view to linear multi-step methods with grid dependence that read

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k \sum_{j=0}^m b_{k,j} f(t_{k+1-j}, Y_{k+1-j}). \quad (1.37)$$

First let us fix, when such scheme is well-posed, i.e. a solution Y_{k+1} exists.

Theorem 1.71 (Well-posedness) Assume that the functions $y \mapsto \Phi_k(y_0, \dots, y_k, y)$ are L -Lipschitz for all $y_0, \dots, y_k$ and for all k it holds $Lh_k < 1$. Then, for any initial given initial values $Y_0, \dots, Y_{m-1}$, there exists a unique Y_{k+1} with

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k \Phi_k(Y_{k-m}, \dots, Y_k, Y_{k+1}), \quad k = m-1, \dots, N-1. \quad (1.38)$$

Proof. Application of Banach fixpoint Theorem to the function

$$p(y) = \frac{1}{a_{k,0}} \left(- \sum_{j=1}^m a_{k,j} Y_{k+1-j} + h_k \Phi_k(Y_{k-m}, \dots, Y_k, y) \right).$$

□

So given (1.36), we can generalize the construction of multi-step methods by the insertion of polynomials.

Theorem 1.72 (Consistence of grid dependent multi-step methods) Let y be solution of (IVP) and assume y is $(q+2)$ -times continuous differentiable. Moreover assume that for all polynomials P of degree $q+1$ it holds

$$\sum_{j=0}^m a_{k,j} P(t_{k+1-j}) = h_k \sum_{j=0}^m b_{k,j} P'(t_{k+1-j}).$$

Then, the truncation error

$$\tau_k = \frac{1}{h_k} \sum_{j=0}^m a_{k,j} y(t_{k+1-j}) - \sum_{j=0}^m b_{k,j} f(t_{k+1-j}, y(t_{k+1-j}))$$

satisfies

$$\|\tau_k\| \leq C \max_{t \in [t_{k+1-m}, t_{k+1}]} \|y^{(q+2)}(t)\| \frac{(t_{k+1} - t_{k+1-m})^{q+1}}{(q+1)!}$$

with a grid dependent constant

$$C := \sum_{j=1}^m \frac{t_{k+1} - t_{k+1-j}}{t_{k+1} - t_k} |a_{k,j}| + \sum_{j=0}^m |b_{k,j}|.$$

Proof. Insert Taylor polynomial of degree $q+1$ □

The novelty of this result is the presence of the values $|a_{k,j}|$. These may be dangerous terms, that in fact need to be controlled/bounded properly.

Note as a simple observation to obtain consistency (taking $P \equiv 1$) a general linear multi-step scheme on requires

$$\boxed{\sum_{j=0}^m a_{k,j} = 0, \quad \forall k.} \tag{1.39}$$

So the simple question is, do such schemes simply inherit convergence as for the simple schemes such as Adams-Bashforth/Moulton? The answer in general is negative, as the following example shows.

Without loss of generality in what follows with **normalize** and set

$$a_{k,0} = 1. \tag{1.40}$$

Example 1.73 (Counter example) Let $h_k \equiv h$ and consider the explicit multi-step method

$$a_2 Y_{k-1} + a_1 Y_k + a_0 Y_{k+1} = h(b_1 f(t_k, Y_k) + b_2 f(t_{k+1}, Y_{k+1})).$$

Requiring maximal polynomial exactness ($= 3$) leads the scheme

$$-5Y_{k-1} + 4Y_k + Y_{k+1} = h(4f(t_k, Y_k) + 2f(t_{k-1}, Y_{k-1})).$$

This can be verified by using the polynomial basis $\{1, \frac{t-t_k}{h}, (\frac{t-t_k}{h})^2, (\frac{t-t_k}{h})^3\}$. Now consider the IVP

$$y'(t) = -y(t), \quad t \in [0, 1], \quad y(0) = 1.$$

and use step size $h = \frac{1}{100}$ and initial values $Y_0 = 1$ and $Y_1 = e^{-h} = y(h)$. Then, compare the scheme to an explicit Euler scheme for the resulting error as in Figure 1.73. △

In spirit of Theorem 1.63 the question is now if such general linear multi-step methods are stable. As Example 1.73 illustrates, things may be more involved here.

In fact it turns out that an additional stability requirement is necessary to proof stability.

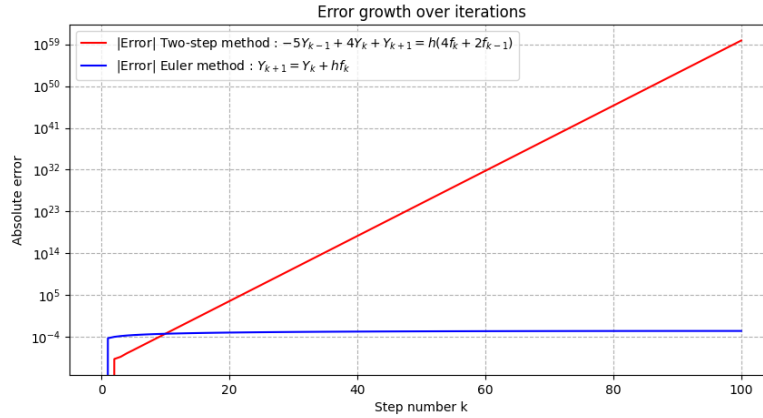


Figure 1.5: The two-step method of exactness 3 (cp. Theorem 1.72) diverges.

Definition 1.74 (Assumption (S)) An estimation by the initial values is called Assumption (S) if for $\ell = 0, \dots, N - m$ and all real numbers u_k with $k = \ell, \dots, N - 1$ that satisfy

$$\sum_{j=0}^m a_{k,j} u_{k+1-j} = 0, \quad k = \ell + m - 1, \dots, N - 1$$

it holds for some $C_0 > 0$ independent of k that

$$|u_k| \leq C_0 \max_{i=\ell, \dots, \ell+m-1} |u_i|, \quad k = \ell + m, \dots, N.$$

It will turn out that Assumption (S) is **necessary** for convergence as the next Lemma illustrates!

Lemma 1.75 Let the consistency condition $\sum_{j=0}^m a_{k,j} = 0$ from 1.39 hold. Then, Assumption (S) holds with constant C_0 if and only if for all grid points $t_\ell, \ell = 0, \dots, N - 1$ and all solutions y of the scalar ODE

$$y'(t) = 0, \quad t \in [t_\ell, t_N]$$

and all iterates Y_k for computed by the associated multi-step scheme

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = 0, \quad k = \ell + m - 1, \dots, N - 1$$

it holds

$$\max_{i=\ell, \dots, k+1} |y(t_i) - Y_i| \leq C_0 \left(\sum_{i=\ell+m-1}^k h_i |\tau_i| + \max_{i=\ell, \dots, \ell+m-1} |y(t_i) - Y_i| \right).$$

Proof. Since $y(t) \equiv \text{const}$ for $k = \ell + m - 1, \dots, N - 1$ it is

$$\tau_k = \frac{1}{h_k} \sum_{j=0}^m a_{k,j} y(t_{k+j-1}) = \frac{1}{h_k} \text{const} \sum_{j=0}^m a_{k,j} = 0$$

and

$$\sum_{j=0}^m a_{k,j} (y(t_{k+1-j}) - Y_{k+1-j}) = 0.$$

Since the initial values $y(t_k) - Y_k$, for $k = \ell, \dots, \ell + m - 1$ can be chosen arbitrary, the claims follows immediately. $\square$

Example 1.76 Recall Example 1.73. Then the sequence $u_k = (-5)^k$ is a solution of

$$-5u_{k-1} + 4u_k + u_{k+1} = 0.$$

Choosing $\ell = 0$, there is no $C_0 > 0$ such that $5^k = |u_k| \leq C_0 \max(|u_0|, |u_1|) = C_0 5$ for $k \geq 2$. $\triangle$

Example 1.77 One-step -, Adams- and Prediction-Correction schemes are of the form

$$Y_{k+1} - Y_k = h_k \Phi_k(Y_0, \dots, Y_{k+1}).$$

Assumption (S) is satisfied with $C_0 = 1$: Let $\ell = 0, \dots, N - m$ and u_k , $k = \ell, \dots, N - 1$ satisfy

$$\sum_{j=0}^m a_{k,j} u_{k+1-j} = u_{k+1} - u_k = 0, \quad \text{for } k = \ell + m - 1, \dots, N - 1.$$

That means $u_{\ell+m-1} = u_{\ell+m} = \dots = u_N$. As a consequence for any $k = \ell + m, \dots, N$ we have

$$|u_k| = |u_{\ell+m-1}| \leq \max_{i=\ell, \dots, \ell+m-1} |u_i|$$

Methods of this structure do not exhibit any internal amplification of past errors, since each subsequent term depends directly only on the difference $u_{k+1} - u_k$, which vanishes in the homogeneous case. $\triangle$

As a next goal we will show that Assumption (S) is even **sufficient** for the stability and hence with the given consistency ultimately leads to convergence.

Lemma 1.78 Assume Assumption (S) holds with constant $C_0 > 0$, then for any norm $\|\cdot\|$ on $\mathbb{R}^n$ and all vectors $Y_0, \dots, Y_N \in \mathbb{R}^n$ it holds

$$\max_{i=0, \dots, k+1} \|Y_i\| \leq C_0 \left(\sum_{i=m-1}^k \left\| \sum_{j=0}^m a_{k,j} Y_{k+1-j} \right\| + \max_{i=0, \dots, m-1} \|Y_i\| \right), \quad k = m - 1, \dots, N - 1.$$

Proof. For $i = 0, \dots, m-1$ let $\gamma_k^{(i)} \in \mathbb{R}$ for $k = 0, \dots, N$ defined as $\gamma_k^{(i)} = \delta_{ik}$ for $k = 0, \dots, m-1$ and for $k = m-1, \dots, N-1$ via

$$\sum_{j=0}^m a_{k,j} \gamma_{k+1-j}^{(i)} = 0$$

with Kronecker product δ_{ik} . Furthermore for $i = m-1, \dots, N$ define $\Gamma_k^{(i)} \in \mathbb{R}$ as $\Gamma_k^{(i)} = 0$ for $k = 0, \dots, m-1$ and for $k = m-1, \dots, N-1$ via

$$\sum_{j=0}^m a_{k,j} \Gamma_{k+1-j}^{(i)} = \partial_{ik}.$$

Then, it holds that

$$\sum_{i=0}^{m-1} |\gamma_k^{(i)}| \leq C_0, \quad k = 0, \dots, N \quad (1.41)$$

$$|\Gamma_k^{(i)}| \leq C_0, \quad k = 0, \dots, N. \quad (1.42)$$

First we show (1.41). Let $\ell \in \{0, \dots, N\}$ and write

$$\sum_{i=0}^{m-1} |\gamma_\ell^{(i)}| = \sum_{i=0}^{m-1} \sigma_i \gamma_\ell^{(i)}$$

for numbers $\sigma_i = \pm 1$ and define $\gamma_k = \sum_{i=0}^m \sigma_i \gamma_k^{(i)}$. Then, $\gamma_k = \sigma_k$ for $k = 0, \dots, m-1$ and

$$\sum_{j=0}^m a_{k,j} \gamma_{k+1-j} = 0, \quad k = m-1, \dots, N-1.$$

By Assumption (S) with $|\gamma_k| \leq C_0$ for $k = 0, \dots, N$, in particular

$$\sum_{i=0}^{m-1} |\gamma_\ell^{(i)}| = |\gamma_\ell| \leq C_0.$$

We now show (1.42). Since $\Gamma_k^{(i)} = 0$ for $k = 0, \dots, i$ and

$$\sum_{j=0}^m a_{k,j} \Gamma_{k+1-j}^{(i)} = 0, \quad \text{for } k = i+1, \dots, N-1$$

by Assumption (S) it holds (using corresponding relation $i+1 = \ell+m-1$, hence $\max_{\ell, \dots, \ell+m-1} |\Gamma_j^{(i)}| = |\Gamma_{i+1}^{(i)}|$) that

$$|\Gamma_k^{(i)}| \leq C_0 |\Gamma_{i+1}^{(i)}|, \quad k = 0, \dots, N.$$

Since $a_{i,0} = 1$ by normalization (1.40) and $\Gamma_k^{(i)} = 0$ for $k \leq i$ we have

$$\Gamma_{i+1}^{(i)} = \sum_{j=0}^m a_{i,j} \Gamma_{i+1-j}^{(i)} = 1,$$

which shows (1.42).

Now by construction for $Y_0, \dots, Y_N \in \mathbb{R}^n$ the following representation holds true

$$Y_k = \sum_{i=0}^{m-1} \gamma_k^{(i)} Y_i + \sum_{i=m-1}^{N-1} \Gamma_k^{(i)} \left(\sum_{j=0}^{m-1} a_{k,j} Y_{k+1-j} \right). \quad (1.43)$$

In order to proof (1.43) for given $Y_0, \dots, Y_N \in \mathbb{R}^n$ let

$$Z_k := \sum_{i=0}^{m-1} \gamma_k^{(i)} Y_i + \sum_{i=m-1}^{N-1} \Gamma_k^{(i)} \left(\sum_{j=0}^{m-1} a_{k,j} Y_{k+1-j} \right).$$

Then for $k = 0, \dots, m-1$, $Z_k = Y_k$ and for $k = m-1, \dots, N-1$:

$$\sum_{j=0}^{m-1} a_{k,j} Z_{k+1-j} = \sum_{j=0}^{m-1} a_{k,j} Y_{k+1-j}.$$

Consequently, $Z_k = Y_k$ for $k = 0, \dots, N$. Now combining (1.43) together with (1.41) and (1.42) we obtain

$$\max_{i=0, \dots, N} \|Y_i\| \leq C_0 \left\{ \sum_{i=m-1}^{N-1} \left\| \sum_{j=0}^m a_{k,j} Y_{k+1-j} \right\| + \max_{i=0, \dots, m-1} \|Y_i\| \right\}.$$

Since we can should this recursion as any previous step k , we conclude the claim. $\square$

Lemma 1.79 Assume Assumption (S) holds with $C_0 > 0$ and the functions $\Phi_k: (\mathbb{R}^n)^{k+2} \rightarrow \mathbb{R}^n$ for $k = m-1, \dots, N-1$ satisfy

$$\|\Phi_k(Y_0, \dots, Y_{k+1}) - \Phi_k(Z_0, \dots, Z_{k+1})\| \leq L \max_{j=0, \dots, k+1} \|Y_j - Z_j\|,$$

for some $L > 0$ and the step sizes $h_k = t_{k+1} - t_k$ fullfill $C_0 L h_k \leq \frac{1}{2}$. Then, for all $Y_0, \dots, Y_N; Z_0, \dots, Z_N \in \mathbb{R}^n$ the following stability estimate holds true:

$$\max_{i=0, \dots, N} \|Y_i - Z_i\| \leq C \left\{ \sum_{i=m-1}^{N-1} \|B_i\| + \max_{i=0, \dots, m-1} \|Y_i - Z_i\| \right\} \quad (1.44)$$

with a constant

$$C = 2C_0 e^{2C_0 L(t_N - t_0)}$$

and

$$B_k = \left\{ \sum_{j=0}^m a_{k,j} Y_{k+1-j} - h_k \Phi_k(Y_0, \dots, Y_{k+1}) \right\} - \left\{ \sum_{j=0}^m a_{k,j} Z_{k+1-j} - h_k \Phi_k(Z_0, \dots, Z_{k+1}) \right\}.$$

Proof. For $k = m - 1, \dots, N - 1$ with Lemma 1.78 it holds

$$\begin{aligned} \max_{i=0, \dots, k+1} \|Y_i - Z_i\| &\leq C_0 \left\{ \sum_{i=m-1}^k \left\| \sum_{j=0}^m a_{k,j} (Y_{k+1-j} - Z_{k+1-j}) \right\| + \max_{i=0, \dots, m-1} \|Y_i - Z_i\| \right\} \\ &= C_0 \left\{ \sum_{i=m-1}^k \|B_i + h_i \{\Phi_i(Y_0, \dots, Y_{i+1}) - \Phi_i(Z_0, \dots, Z_{i+1})\}\| + \max_{i=0, \dots, m-1} \|Y_i - Z_i\| \right\} \\ &\leq C_0 \left\{ \sum_{i=m-1}^k \|B_i\| + \max_{i=0, \dots, m-1} \|Y_i - Z_i\| \right\} + C_0 L \sum_{i=m-1}^k h_i \max_{j=0, \dots, i+1} \|Y_j - Z_j\| \end{aligned}$$

Now let $v_k := \max_{i=0, \dots, k} \|Y_i - Z_i\|$ and $K = \sum_{i=m-1}^{N-1} \|B_i\| + \max_{i=0, \dots, m-1} \|Y_i - Z_i\|$, then

$$v_{k+1} \leq C_0 L h_k v_{k+1} + C_0 L \sum_{i=m-1}^{k-1} h_i v_{i+1} + C_0 K, \quad k = m, \dots, N-1$$

$$\text{and} \quad v_m \leq C_0 L h_{m-1} v_m + C_0 K.$$

Since $C_0 L h_i \leq \frac{1}{2}$ it follows that

$$v_m \leq 2C_0 K \quad \text{and} \quad v_{k+1} \leq 2(C_0 L \sum_{i=m-1}^{k-1} h_i v_{i+1} + C_0 K).$$

Hence, by induction

$$v_{k+1} \leq e^{2C_0 L(t_k - t_{m-1})} 2C_0 K.$$

And finally the claim,

$$v_N \leq e^{2C_0 L(t_N - t_0)} 2C_0 K = CK.$$

□

Theorem 1.80 (Stability of grid-dependent multi-step methods) Let Assumption from Lemma 1.79 hold. For $k = m - 1, \dots, N - 1$ let

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k \Phi_k(Y_0, \dots, Y_k, Y_{k+1}).$$

For a function $y: [a, b] \rightarrow \mathbb{R}^n$ define

$$\tau_k = \frac{1}{h_k} \left[\sum_{j=0}^m a_{k,j} y(t_{k+1-j}) - h_k \Phi_k(y(t_0), \dots, y(t_{k+1})) \right].$$

Then, it holds that

$$\max_{i=0, \dots, N} \|y(t_i) - Y_i\| \leq C \left\{ \sum_{i=m-1}^{N-1} h_i \|\tau_i\| + \max_{i=0, \dots, m-1} \|y(t_i) - Y_i\| \right\}$$

with $C > 0$ from Lemma 1.79.

Proof. Application of Lemma 1.79 with $Z_k = y(t_k)$, since then $B_k = -h_k \tau_k$. $\square$

Consequently, we have stability and convergence of one-step-, Adams- and predictor methods under the relevant assumptions.

1.3.3 General multi-step methods II - grid independent case

So as a consequence of the previous section we saw that Assumption (S) characterizes the stability of the related linear multi-step scheme. So next we want to discuss on how to ensure such an assumption. We do so, by investigating of linear multi-step methods with coefficients $a_{k,j} = a_j$. Hence, we investigate the following setup of a general multi-step method:

$$\sum_{j=0}^m a_j Y_{k+1-j} = h_k \Phi_k(Y_0, \dots, Y_{k+1}) \quad (1.45)$$

with a_j independent of k and $a_0 = 1$ and

$$\|\Phi_k(y_0, \dots, y_{k+1}) - \Phi_k(z_0, \dots, z_{k+1})\| \leq L \max_{j=0, \dots, k+1} \|y_j - z_j\|.$$

Then, Assumption (S) can be formulated as follows. When do solutions of the difference equation

$$\sum_{j=0}^m a_j u_{k+1-j} = 0, \quad k = m-1, m, \dots, N-1 \quad (1.46)$$

satisfy

$$\max_{i=0, \dots, k} |u_i| \leq C_0 \max_{i=0, \dots, m-1} |u_i|, \quad k \geq m$$

and some $C_0 > 0$ independent of k . Note, that in this setup there is a translation invariance with respect to the initial approximations, that is not true in the grid dependent case. See the following remark.

Remark 1.81 In the case of constant coefficients a_j , the recurrence

$$\sum_{j=0}^m a_j u_{k+1-j} = 0, \quad k \geq m-1,$$

is *translation invariant*. Hence, the solution sequence $(u_k)_{k \geq 0}$ is uniquely determined by the first m initial values $u_0, \dots, u_{m-1}$, and Assumption (S) can be written in the global form

$$\max_{0 \leq i \leq k} |u_i| \leq C_0 \max_{0 \leq i \leq m-1} |u_i|, \quad k \geq m.$$

For grid-dependent coefficients $a_{k,j}$, this translation invariance is lost, and the estimate must be formulated locally for each starting index ℓ as

$$|u_k| \leq C_0 \max_{\ell \leq i \leq \ell+m-1} |u_i|, \quad k = \ell + m, \dots, N.$$

Indeed, in the constant-coefficient case the local and global formulations are equivalent: if $v_j := u_{j+\ell}$ satisfies the same constant-coefficient recurrence, applying the global estimate to v yields the local form above for u . $\triangle$

Definition 1.82 (Characteristic polynomial and null stability) The characteristic polynomial of (1.46) reads $\rho(z) = \sum_{j=0}^m a_j z^{m-j}$. It is called *null stable*, if it possesses only roots z_i with $|z_i| \leq 1$ and all roots with $|z_i| = 1$ are simple.

Lemma 1.83 Assume the characteristic polynomial $\rho(z) = \sum_{j=0}^m a_j z^{m-j}$ of the difference equation (1.46) has the factorization

$$\rho(z) = a_0(z - z_1)^{m_1} \cdots (z - z_\ell)^{m_\ell}, \quad z_i \neq z_j \text{ for } i \neq j.$$

Then, the sequences

$$u_k^{(i,\nu)} := \begin{cases} \frac{(m-1+k)!}{(m-1+k-\nu)!} z_i^{m-1+k-\nu}, & m+k-1 \geq \nu \\ 0, & \text{else} \end{cases} \quad i = 1, \dots, \ell, \nu = 0, \dots, m_i - 1.$$

form a fundamental system of the difference equation (1.46).

Proof. Let $\nu = 0, 1, 2, \dots$ and $k = 0, 1, 2, \dots$, then it holds that

$$\left(\frac{\partial}{\partial z} \right)^\nu \{ z^k \rho(z) \} = \left(\frac{\partial}{\partial z} \right)^\nu \left\{ \sum_{j=0}^m a_j z^{k+m-j} \right\} = \sum_{\substack{j=0 \\ m+k-j \geq \nu}}^m a_j \frac{(m+k-j)!}{(m+k-j-\nu)!} z^{m+k-j-\nu}.$$

Hence for $i = 1, \dots, \ell$ and $\nu = 0, \dots, m_i - 1$

$$\sum_{j=0}^m a_j u_{k+1-j}^{(i,\nu)} = \sum_{\substack{j=0 \\ k+m-j \geq \nu}}^m a_j \frac{(m+k-j)!}{(m+k-j-\nu)!} z_i^{m+k-j-\nu} = \left(\frac{\partial}{\partial z} \right)^\nu \{ z^k \rho(z) \} \Big|_{z=z_i}.$$

Since z_i is a root of $z^k \rho(z)$ with multiplicity m_i as well and $\nu < m_i$, therefore

$$\sum_{j=0}^m a_j u_{k+1-j}^{(i,\nu)} = 0$$

and hence the given sequences form a solution of the difference equations. As next, we show uniqueness (linear independence). Let

$$\sum_{i=1}^{\ell} \sum_{\nu=0}^{m_i-1} c_{i,\nu} u_k^{(i,\nu)} = 0 \quad \text{for } k = 0, 1, \dots, m-1,$$

with constants $c_{i,\nu}$. Writing these relations for the m indices $k = 0, \dots, m-1$ yields a homogeneous linear system $M\mathbf{c} = 0$, where $\mathbf{c}$ is the vector of the coefficients $c_{i,\nu}$ (there are m of them) and M is the $m \times m$ matrix with columns given by the vectors $(u_0^{(i,\nu)}, u_1^{(i,\nu)}, \dots, u_{m-1}^{(i,\nu)})^T$.

The matrix M is precisely a *confluent Vandermonde* matrix associated with the nodes $z_1, \dots, z_\ell$ and multiplicities $m_1, \dots, m_\ell$. Its determinant is the well-known nonzero product

$$\det M = C \prod_{1 \leq i < j \leq \ell} (z_j - z_i)^{m_i m_j},$$

where $C \neq 0$ is an explicit nonzero constant depending only on the multiplicities (in particular it is independent of the distinct values z_i). Since the z_i are pairwise distinct the product above is nonzero, hence $\det M \neq 0$. Therefore the only solution of $M\mathbf{c} = 0$ is $\mathbf{c} = 0$, so the sequences $\{u_k^{(i,\nu)}\}$ are linearly independent.

As there are exactly $m = \sum_i m_i$ independent solutions of an order- m linear recurrence, these sequences form a fundamental system of solutions of (1.46), which finishes the proof. $\square$

Theorem 1.84 (Assumption (S) equals null stability) The solutions of the difference equation

$$\sum_{j=0}^m a_j u_{k+1-j} = 0, \quad k = 0, 1, \dots$$

satisfy with a $C_0 > 0$ independent of k the condition

$$|u_k| \leq C_0 \max_{i=0, \dots, m-1} |u_i|$$

if and only if the related characteristic polynomial $\rho(z) = \sum_{j=0}^m a_j z^{m-j}$ is null stable.

Proof. Write the factorization

$$\rho(z) = a_0 \prod_{i=1}^{\ell} (z - z_r)^{m_r}, \quad \sum_{r=1}^{\ell} m_r = m,$$

and let the basis of solutions be

$$u_k^{(i,\nu)} = \frac{(m-1+k)!}{(m-1+k-\nu)!} z_r^{m-1+k-\nu}, \quad i = 1, \dots, \ell, \quad \nu = 0, \dots, m_i - 1,$$

(with the understanding $u_k^{(i,\nu)} = 0$ for $k < \nu$). As shown previously these m sequences are linearly independent and every solution has the unique representation

$$u_k = \sum_{r=1}^{\ell} \sum_{\nu=0}^{m_r-1} c_{i,\nu} u_k^{(i,\nu)}.$$

Null-stability $\Rightarrow$ Assumption (S). Assume the null-stability root condition holds: every root z_r satisfies $|z_r| \leq 1$ and every root on the unit circle is simple. Then each basis sequence is uniformly bounded in k : there exists a finite constant

$$S := \sum_{r=1}^{\ell} \sum_{\nu=0}^{m_r-1} \sup_{k \geq m} |u_k^{(r,\nu)}| < \infty,$$

because (i) if $|z_r| < 1$ the sequences decay exponentially and (ii) if $|z_r| = 1$ then $m_r = 1$ so only the bounded sequence z_r^k appears.

Form the $m \times m$ confluent Vandermonde matrix M whose m columns are the truncated basis vectors $(u_0^{(i,\nu)}, u_1^{(i,\nu)}, \dots, u_{m-1}^{(i,\nu)})^T$ for $i = 1, \dots, \ell$ and $\nu = 0, \dots, m_i$. Note here, that $m =$

$$\sum_{i=1}^{\ell} \sum_{\nu=0}^{m_i} 1.$$

As before $\det M \neq 0$ (the z_i are distinct), hence M is invertible. Writing the initial block $\mathbf{u}^{(0)} = (u_0, \dots, u_{m-1})^T$ and the coefficient vector $\mathbf{c} = (c_{i,\nu})$, we have $\mathbf{u}^{(0)} = M\mathbf{c}$ and therefore

$$\|\mathbf{c}\|_\infty \leq \|M^{-1}\|_\infty \|\mathbf{u}^{(0)}\|_\infty = \|M^{-1}\|_\infty \max_{0 \leq i \leq m-1} |u_i|.$$

Now for any $k \geq m$,

$$|u_k| \leq \sum_{r,\nu} |c_{r,\nu}| |u_k^{(r,\nu)}| \leq \|\mathbf{c}\|_\infty \sum_{r,\nu} \sup_{k \geq m} |u_k^{(r,\nu)}| = \|\mathbf{c}\|_\infty S.$$

Combining the two inequalities gives, for all $k \geq m$,

$$|u_k| \leq \|M^{-1}\|_\infty S \max_{0 \leq i \leq m-1} |u_i|.$$

Hence the desired uniform constant exists; take $C_0 := \|M^{-1}\|_\infty S$.

Assumption (S) $\Rightarrow$ Null-stability. Conversely, the proof that the uniform bound implies null-stability is as in the standard argument. Note that

$$|u_k^{(i,\nu)}| \leq \begin{cases} (m-1+k)^\nu |z_i|^{m-1} |z_i|^{k-\nu}, & k+m-1 \geq \nu \\ 0, & \text{else.} \end{cases}$$

Hence, if there were a root z_i with $|z_i| > 1$ or a unit root of multiplicity ≥ 2 one constructs a solution (scaled so its initial block has unit maximum) that grows unboundedly, contradicting the assumed bound. This completes the proof. $\square$

So as a consequence combining Theorem 1.80, Lemma 1.78 and Theorem 1.84, we can derive the convergence result.

So recall our framework

$$\sum_{j=0}^m a_j Y_{k+1-j} = h_k \Phi_k(Y_0, \dots, Y_{k+1}) \quad (1.47)$$

with a_j independent of k and $a_0 = 1$ and

$$\|\Phi_k(y_0, \dots, y_{k+1}) - \Phi_k(z_0, \dots, z_{k+1})\| \leq L \max_{j=0, \dots, k+1} \|y_j - z_j\|.$$

with associated characteristic polynomial $\rho(z) = \sum_{j=0}^m a_j z^{m-j}$.

Theorem 1.85 (Null-stability and stability) Assume that ρ is null-stable, then there exists a constant $C = C(\rho, L, t_N - t_0)$ such that

$$\max_{k=0, \dots, N} \|y(t_k) - Y_k\| \leq C \left\{ \sum_{j=m-1}^{N-1} h_j \|\tau_j\| + \max_{i=0, \dots, m-1} \|y(t_i) - Y_i\| \right\},$$

provided the step size is below a certain threshold depending on ρ and L in the **implicit** case.

Proof. The null-stability is equivalent in the grid-independent case to the assumption (S) with $C_0 = C_0(\rho)$ by Theorem 1.84. Hence, with $C = 2C_0 e^{2C_0 L(t_N - t_0)}$ from Lemma 1.79 ensuring that $C_0 L h_k \leq \frac{1}{2}$ (in the implicit case) the result is a direct consequence of Theorem 1.80. $\square$

Hence the null-stability is **necessary** and **sufficient** for the stability and together with the consistency for the convergence.

As for now we constructed simple linear multi-step methods and gave constructive approaches to build general multi-step methods in the grid dependent/independent case.

The question that arises next is about the limits of possible convergence orders, determined by limits of the possible maximum consistency order. Here we restrict to the equidistant grid case $h_k \equiv h$.

Theorem 1.86 (First Dahlquist barrier) Assume that the characteristic polynomial of the linear multi-step scheme

$$\sum_{j=0}^m a_j Y_{k+1-j} = h \sum_{j=0}^m b_j f(t_{k+1-j}, Y_{k+1-j})$$

is null-stable and for all polynomials P of degree $\leq q$, for $q \in \mathbb{N}$ we have

$$\sum_{j=0}^m a_j P(k+1-j) = \sum_{j=0}^m b_j P'(k+1-j).$$

Then, for the consistency q it must hold

$$q \leq \begin{cases} m+1, & m \text{ is odd,} \\ m+2 & m \text{ is even,} \\ m & \text{if } b_0 \leq 0, \text{ i.e. the method is explicit.} \end{cases}$$

Moreover, if $q = m+2$, then all roots z_i of the characteristic polynomials satisfy $|z_i| = 1$.

Proof. Rather complicated based on functional theory, e.g. see [?, section 4.2.3] □

Note that the condition

$$\sum_{j=0}^m a_j P(k+1-j) = \sum_{j=0}^m b_k P'(k+1-j),$$

is equivalent to

$$\sum_{j=0}^m a_j P(t_{k+1-j}) = h \sum_{j=0}^m b_k P'(t_{k+1-j})$$

since the equidistant grid allows to remove the factor h .

Example 1.87 The (**implicit**) Adams-Moulton schemes with m -steps have the convergence order $m+1$, that for m odd have the maximum possible order. △

Example 1.88 The Milne-Simpson scheme on an equidistant grid reads

$$Y_{k+1} - Y_{k-1} = \frac{h}{3}(f_{k-1} + 4f_k + f_{k+1}), \quad f_i = f(t_i, Y_i), \quad t_{i+1} = t_i + h.$$

It is a linear two-step scheme of order $q = 4$ and, hence reaches the maximum possible order for $m = 2$. Its characteristic polynomial is $\rho(z) = z^2 - 1$ with roots ± 1 . However, while it

sounds favorable in the application of the scheme to simple ODEs such as $y' = \lambda y$ with $\lambda \ll -1$, undesirable oscillation appear. We will investigate this in the analysis of stability regions in the next section. $\triangle$

1.4 Stiff Ordinary Differential Equations

As for now the motivation for ODE solvers was based on high accuracy while requiring a low-number of right hand side evaluations. For this we derived (explicit) Runge-Kutta one-step methods and linear explicit and implicit multi-step methods.

Question: Do these solvers perform well for any ODE?

For this consider the *model problem* for a diagonalize matrix $A \in \mathbb{R}^n$ given as

$$y'(t) = Ay(t), \quad A = T^{-1}DT, \quad D = (\lambda_1, \dots, \lambda_d), \quad \lambda_i \in \mathbb{C}. \quad (1.48)$$

Note that the set of diagonalizable matrices is dense in the set of matrices.

1.4.1 A-stability

Lemma 1.89 The solutions of the model problem 1.48 remain bounded for $t \rightarrow \infty$ if and only if $\operatorname{Re}(\lambda_i) \leq 0$ for $i = 1, \dots, n$.

Proof. If y is a solution of (1.48), then z with $z(t) = Ty(t)$ is solution of

$$z'(t) = Dz(t),$$

since $z'(t) = Ty'(t) = T Ay(t) = TT^{-1}DTy(t) = Dz(t)$. The components $z_i(t)$ hence solve

$$z'_i(t) = \lambda_i z_i(t) \quad \Rightarrow \quad z_i(t) = e^{\lambda_i t} z_i(a).$$

□

Let us recall the explicit Runge-Kutta-Scheme with s stages

$$Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k), \quad \Phi(t, Y, h) = \sum_{i=1}^s b_i K_i(t, Y, h)$$

with

$$K_1(t, Y, h) = f(t, Y), \quad K_i(t, Y, h) = f(t + c_i h, Y + h \sum_{j=1}^{i-1} a_{ij} K_j(t, Y, h)), \quad i = 2, \dots, s$$

And recall our stability function $R(z) = 1 + z\mathbf{b}^\top(\mathbf{I} - z\mathbf{A})^{-1}\mathbf{1}$ with $z = h\lambda$ derived for the scalar valued ODE $y'(t) = \lambda y$, which lead to the relationship $Y_{k+1} = R(z)Y_k$ for such an ODE. Recall, that R is a polynomial of degree s in z in the explicit case.

Theorem 1.90 If we apply the explicit Runge-Kutta scheme of stage $s \in \mathbb{N}$ to the model problem (1.48), then it holds

$$Y_{k+1} = P(h_k A)Y_k,$$

where P is a polynomial of degree s given as

$$P(x) = 1 + \sum_{i=1}^s b_i P_i(x), \quad P_1(x) = x, \quad P_i(x) = x(1 + \sum_{j=1}^{i-1} a_{ij} P_j(x)), \quad i = 2, \dots, s.$$

If $h_k \equiv h$, then

$$Y_k = T^{-1} \text{diag}(P(h\lambda_1)^k, \dots, P(h\lambda_n)^k) T Y_0.$$

Proof. The polynomial relation follows by induction, it holds $hK_i(t, Y, h) = P_i(hA)y$. In particular $P = R$. For constant step size, since

$$A^j = (T^{-1}DT)^j = T^{-1}D^jT$$

we have

$$P(hA)^k = T^{-1}P(hD)^kT.$$

□

As a consequence we have the following corollary

Corollary 1.91 Let the step size be constant with $h_k \equiv h$. Then, the approximation sequence Y_k of the explicit Runge-Kutta scheme remain bounded if and only if for all $i = 1, \dots, n$ it holds

$$\lambda_i h \in S = \{z \in \mathbb{C} \mid |P(z)| \leq 1\}$$

where S denotes the *stability region* of the considered scheme.

Example 1.92 For the explicit Euler scheme it holds

$$Y_{k+1} = Y_k + h_k A Y_k = (I + h_k A) Y_k, \quad P(z) = 1 + z.$$

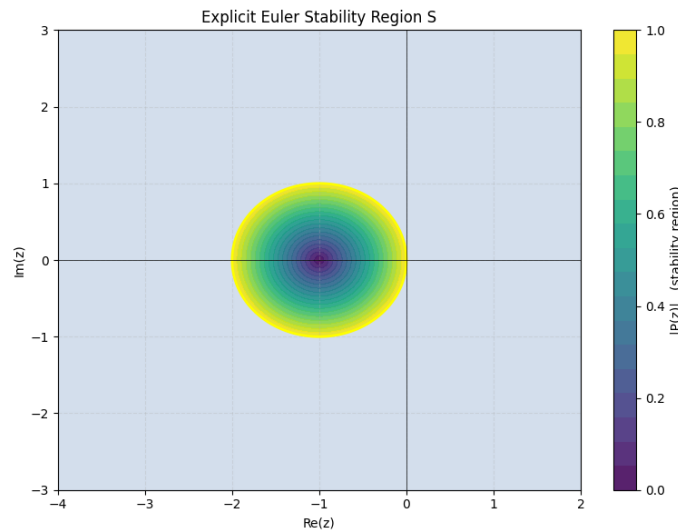


Figure 1.6: Stability region for the explicit Euler method.

If λ pure imaginary, then $h = 0$ is the only admissible step size. The scheme oscillates. $\triangle$

Example 1.93 For the classical Runge-Kutta-scheme

$$Y_{k+1} = Y_k + h\left(\frac{1}{6}K_1 + \frac{1}{3}K_2 + \frac{1}{3}K_3 + \frac{1}{6}K_4\right)$$

with

$$K_1 = AY_k, \quad K_2 = A(Y_k + \frac{1}{2}hK_1), \quad K_3 = A(Y_k + \frac{h}{2}K_2), \quad K_4 = A(Y_k + hK_3).$$

it holds

$$P(z) = R(z) = 1 + z + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \frac{1}{24}z^4 = e^z + \mathcal{O}(z^5).$$

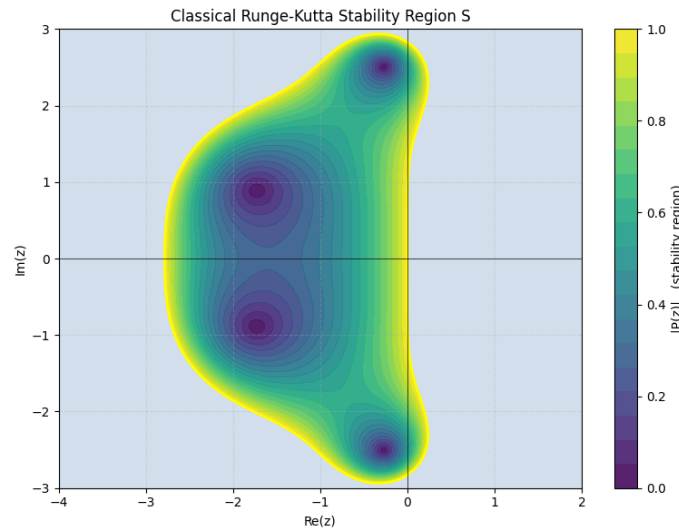


Figure 1.7: Stability region for the explicit classical Runge Kutta method.

Here, the dependence on the step size is not so critical, much better than the explicit Euler scheme, since for step sizes small enough more of the imaginary axis is contained in S . $\triangle$

Definition 1.94 An IVP discretization scheme is called A-stable if its stability region S satisfies

$$\mathbb{C}_- := \{z \in \mathbb{C} \mid \operatorname{Re} z \leq 0\} \subset S. \quad (1.49)$$

In particular this definition is motivated by the fact that, the numerical scheme should stay bounded in the case when our model problem solution remains bounded, i.e. $\operatorname{Re} \lambda_i \leq 0$.

As we can see from the examples, the stability region S is bounded for the explicit Euler method and the classical Runge-Kutta scheme. What happens for general Runge-Kutta schemes?

Theorem 1.95 The stability area S of any explicit Runge-Kutta scheme is bounded, in particular they cannot be A-stable.

Proof. Since $\sum_{i=1}^s = 1$, $P(z) = R(z) = 1 + \mu + \dots$ is not constant polynomial and hence

$$\lim_{|z| \rightarrow \infty} P(z) = +\infty.$$

□

Remark 1.96 In the exercises we had for general RK scheme $(\mathbf{A}, \mathbf{b}, \mathbf{c})$ given in the Butcher tableau

$$\begin{array}{c|c} \mathbf{c} & \mathbf{A} \\ \hline & \mathbf{b}^\top \end{array},$$

with s stages and $\mathbf{A}$ is non-singular that $R(\infty) = 0$ if one of the following conditions is satisfied:

- (i) $a_{si} = b_i$ for $i = 1, \dots, s$, or
- (ii) $a_{i1} = b_1$ for $i = 1, \dots, s$, $b_1 \neq 0$.

This does not contradict our Theorem 1.95, since in the explicit case $\mathbf{A}$ is not invertible!. $\triangle$

We introduce a concept of stiff systems in the case of linear ODEs.

Definition 1.97 The linear system of ordinary differential equations

$$y'(t) = Ay$$

is called stiff, if all eigenvalues of A have negative real part and up ordering

$$|\operatorname{Re}\lambda_1|, \dots, |\operatorname{Re}\lambda_m| \ll |\operatorname{Re}\lambda_{m+1}|, \dots, |\operatorname{Re}\lambda_n|.$$

This means the solution has contributions that are defined on different (time) scales. The largest magnitude of $|\lambda_i|$ will determine restrictions on h for the stability of the method.

Moreover, we even saw that explicit Runge-Kutta schemes are non-favorable for stiff systems and even in the case when all λ_i have non-positive real part, i.e. $\operatorname{Re}\lambda_i \leq 0$ and

$$|\lambda_1|, \dots, |\lambda_m| \ll |\lambda_{m+1}|, \dots, |\lambda_n|.$$

Let us now consider **linear multi-step methods** for the linear model problem. They become

$$\sum_{j=0}^m a_j Y_{k+1} - hA \sum_{j=0}^m Y_{k+1-j} = 0 \quad (1.50)$$

We define the set

$$S = \{\mu \in \overline{\mathbb{C}} \mid z \mapsto \rho(z) - \mu\sigma(z) \text{ is null-stable}\} \quad (1.51)$$

for $\rho(z) = \sum_{j=0}^m a_j z^{m-j}$ and $\sigma(z) = \sum_{j=0}^m b_j z^{m-j}$.

Lemma 1.98 Let $h > 0$ be fixed. Then the iterates Y_k of the linear multi-step scheme for the solutions of our linear model problem are bounded for all initial values $Y_0, \dots, Y_{m-1}$ if and only if $\lambda_i h \in S$, $i = 1, \dots, n$ for S from (1.51).

Proof. In this case for $Z_k = TY_k$ it holds

$$\sum_{j=0}^m a_j Z_{k+1-j} - hD \sum_{j=0}^m b_j Z_{k+1-j} = 0.$$

For each component, hence

$$\sum_{j=0}^m a_j (Z_{k+1-j})_i - h\lambda_i \sum_{j=0}^m b_j (Z_{k+1-j})_i = 0.$$

Monomial ansatz to solve this difference relation yields the investigation of the polynomial $P(z) = \rho(z) - h\lambda_i \sigma(z)$, similar to the characteristic polynomial case of multi-step methods. $\square$

Theorem 1.99 The boundary of S from (1.51) of the linear multi-step methods is subset of

$$R = \{\mu \in \mathbb{C} \mid \rho(z) - \mu\sigma(z) = 0 \text{ for one } z \text{ s.t. } |z| = 1\}.$$

Proof. We recall a consequence of Cauchy residuum theorem applied to polynomials. If a polynomial $P(z)$ has the decomposition $P(z) = (z - r_1)^{m_1} (z - r_2)^{m_2} \cdots (z - r_k)^{m_k} Q(z)$, where $Q(z)$ has no zeros inside $|z| < 1$, then $\frac{P'}{P}$ has simple poles at $r_1, \dots, r_k$ with residues $m_1, \dots, m_k$. By the residue theorem:

$$\frac{1}{2\pi i} \oint_{|z|=1} \frac{P'(z)}{P(z)} dz = m_1 + m_2 + \cdots + m_k.$$

That means, the term counts the number of zeros of P (counted with its multiplicities) inside the contour $|z| = 1$, which is the unit disc.

Let $\mu_0 \in \partial S$. If μ_0 such that $\rho(z) - \mu_0 \sigma(z) = 0$ for some $|z| = 1$, when $\mu_0 \in R$ by definition.

Assume now that it holds

$$\rho(z) - \mu_0 \sigma(z) \neq 0, \quad \forall z, \text{ s.t. } |z| = 1.$$

In particular this polynom has no zeros on the unit disc.

Then, due to continuity, for some $\epsilon > 0$, we have

$$\rho(z) - \mu\sigma(z) \neq 0 \quad \forall |\mu - \mu_0| < \epsilon, \forall z : |z| = 1.$$

Then, for all μ with $|\mu - \mu_0| < \epsilon$ the map

$$\mu \mapsto N(\mu) = \frac{1}{2\pi i} \oint_{|z|=1} \frac{\rho'(z) - \mu\sigma'(z)}{\rho(z) - \mu\sigma(z)} dz$$

is a continuous function, evaluating to the number zeros of the polynomial $z \mapsto \rho(z) - \mu\sigma(z)$ for $|\mu - \mu_0| < \epsilon$ and hence is constant in the interior of the unit circle, i.e. $|z| < 1$. Now there are two cases:

- $N(\mu) = m$, with m from the multi-step method. Then, all roots of $z \mapsto \rho(z) - \mu\sigma(z)$ for $|\mu - \mu_0| < \epsilon$ are in the interior of S and μ_0 is interior point of S . Contradiction.
- $N(\mu) < m$ for $|\mu - \mu_0| < \epsilon$. Then, at least one root of each polynomial $z \mapsto \rho(z) - \mu\sigma(z)$ for $|\mu - \mu_0| < \epsilon$ must be outside the unit disc and μ_0 is a inner point of $\mathbb{C} \setminus S$ and thus no boundary point of S .

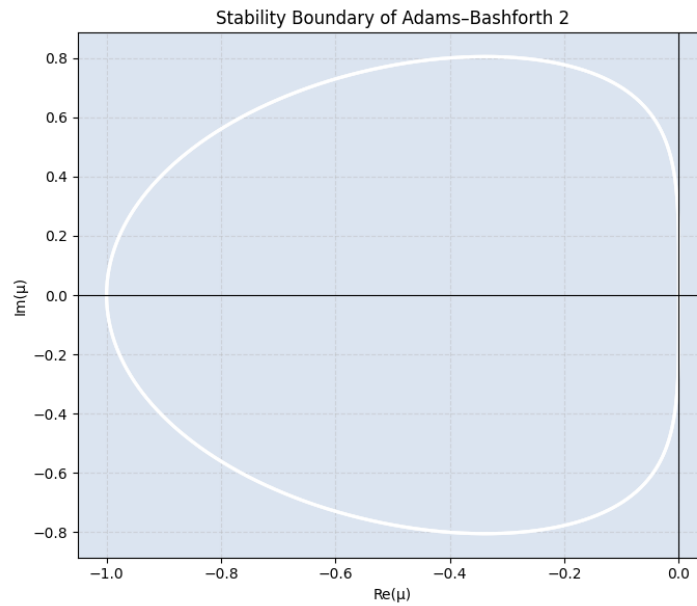


Figure 1.8: Boundary contour of R with respect to the 2-step Adam-Bashforth scheme.

Consequently, μ_0 is a boundary point of S , then there must be z with $|z| = 1$ and $\rho(z) - \mu_0\sigma(z) = 0$, which means $\mu_0 \in R$. $\square$

Conclusion: In order to understand the boundary of S , one has to draw the curve

$$\mu(\phi) = \frac{\rho(e^{i\phi})}{\sigma(e^{i\phi})}, \quad 0 \leq \phi \leq 2\pi.$$

Example 1.100 The 2-step Adam-Bashforth scheme for equidistant step size h reads

$$Y_{k+1} = Y_k + h \left(\frac{3}{2}f(t_k, Y_k) - \frac{1}{2}f(t_{k-1}, Y_{k-1}) \right)$$

Hence, for $m = 2$ we obtain $\rho(z) = z^2 - z$ and $\sigma(z) = \frac{3}{2}z - \frac{1}{2}$. The set R is boundary of a bounded domain. $\triangle$

Without proof, one even can show

Theorem 1.101 Let S be the stability region of the explicit m -step Adam-Bashforth scheme with equidistant step-size. Then, there exists a $q > 0$ such that

$$S \cap (-\infty, 0] = [-q, 0].$$

Proof. TODO ADD reference. $\square$

Consequence to ask for A-stability:

It seems we cannot avoid solving implicit systems for onestep or linear multi-step methods!

Theorem 1.102 (Stability function for general Runge-Kutta) Consider a Runge-Kutta scheme with s stages with parameters $(\mathbf{A}, \mathbf{b}, \mathbf{c})$. Then, the stability function $R: \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$ with

$$R(z) = 1 + z\mathbf{b}^\top (\mathbf{I} - z\mathbf{A})^{-1} \mathbf{1}$$

is a rational function with nominator and denominator being polynomials of degree at most s . If the scheme is explicit, then R is a polynomial.

Proof. The explicit case follows the proof in Theorem 1.46. For the general case, note that $(\mathbf{I} - z\mathbf{A})^{-1} \mathbf{1}$ can be written as the solution of $(\mathbf{I} - z\mathbf{A})\mathbf{x} = \mathbf{1}$. By Cramers rule it holds

$$x_i = \frac{\det((\mathbf{I} - z\mathbf{A})_i)}{\det(\mathbf{I} - z\mathbf{A})},$$

where $(\mathbf{I} - z\mathbf{A})_i$ is defined as the matrix $\mathbf{I} - z\mathbf{A}$, with i -th column replaced by the right hand side, i.e. $\mathbf{1}$. Consequently, the denominator is a polynomial of degree at most s and the nominator is a polynomial of degree maximum $s - 1$. The remaining multiplication with $z\mathbf{b}^\top$ yields the claim. $\square$

For the scalar case the general Runge-Kutta scheme satisfies $Y_{k+1} = R(z)Y_k$. Hence the stability region is given as follows:

Definition 1.103 The stability region for the general Runge Kutta scheme is defined as

$$S = \{z \in \overline{\mathbb{C}} : |R(z)| \leq 1\}.$$

Consequence: For a Runge-Kutta scheme to be A-stable, we necessarily need $|R(\infty)| \leq 1$: The stability function R is a rational function for **implicit** Runge-Kutta schemes. In order to be A-stable, which requires $\{z \in \mathbb{C} \mid \operatorname{Re} z \leq 0\} \subset S$, the denominator needs to have degree equal or bigger than the nominator, else it would be unbounded. In the equal degree case the limit is finite (independent of z) and needs to be not larger than 1.

Example 1.104 (Implicit Runge-Kutta with $s = 1$ / Adam-Moulton with $m = 0$) The implicit Euler-scheme

$$Y_{k+1} = Y_k + h_k f(t_{k+1}, Y_{k+1})$$

has the stability function $R(z) = \frac{1}{1-z}$. The scheme is A-stable, but we have to solve a implicit system in each iteration. $\triangle$

Example 1.105 The *implicit trapezoidal one-step scheme* reads

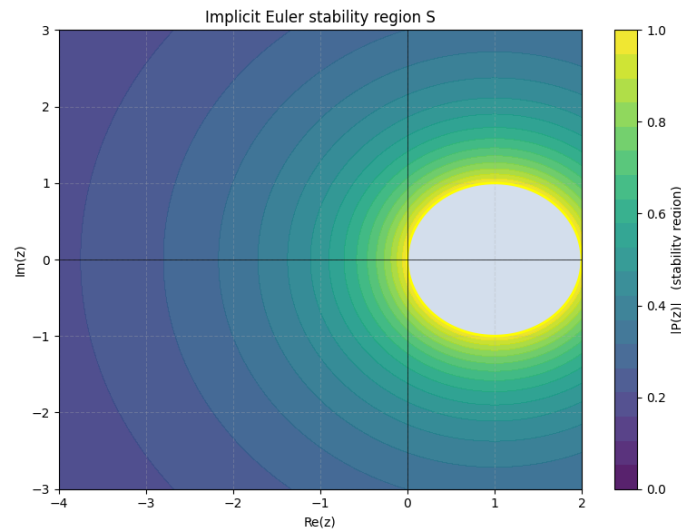
$$Y_{k+1} = Y_k + h_k \left(\frac{1}{2} f(t_k, Y_k) + \frac{1}{2} f(t_{k+1}, Y_{k+1}) \right)$$

and implicit *midpoint rule* are given as

$$Y_{k+1} = Y_k + h_k f\left(\frac{t_k + t_{k+1}}{2}, \frac{Y_k + Y_{k+1}}{2}\right)$$

For the linear model problem and $h_k \lambda_i \neq 2$ they satisfy the relation

$$Y_{k+1} = T^{-1} \operatorname{diag}(R(h_k \lambda_i)) T Y_k$$

Figure 1.9: The stability region S of the implicit Euler scheme.

with

$$R(z) = \frac{1 + \frac{1}{2}z}{1 - \frac{1}{2}z}.$$

Hence,

$$S = \{z \in \mathbb{C} \mid |R(z)| \leq 1\} = \{z \in \mathbb{C} \mid \operatorname{Re}(z) \leq 0\}.$$

Hence trapezoidal and midpoint scheme are both A -stable. $\triangle$

Remark 1.106 The implicit trapezoidal and midpoint rule both are **isometric stable**: For all $z \in \mathbb{C}$ with $\operatorname{Re}(z) = 0$ it holds $|R(z)| = 1$ since

$$R(z) = \frac{1 + \frac{1}{2}z}{1 - \frac{1}{2}z} = \sqrt{\frac{(1 + \frac{1}{2}\operatorname{Re}(z))^2 + \operatorname{Im}(z)^2}{(1 - \frac{1}{2}\operatorname{Re}(z))^2 + \operatorname{Im}(z)^2}} = 1.$$

$\triangle$

We now consider popular special cases of multi-step methods and Runge-Kutta methods

Gaussian Runge-Kutta methods

In order to obtain a maximal consistency order $q = 2s$ for **implicit** Runge-Kutta schemes, recalling Lemma 1.37, that for a one-step method of the form

$$Y_{k+1} = Y_k + h \sum_{i=1}^s b_i f(t_k + c_i h, \eta_{a_j})$$

we require that the associated quadrature rule

$$\sum_{i=1}^s b_i g(c_i) \approx \int_0^1 g(t) dt$$

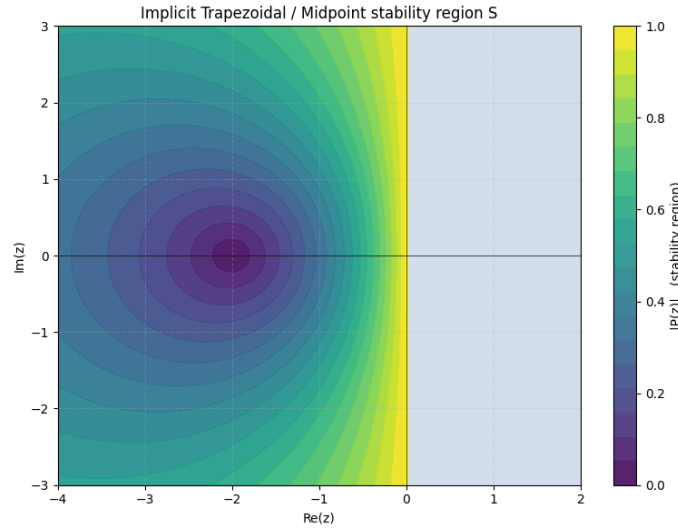


Figure 1.10: The stability region S of the implicit trapezoidal scheme (one-step method / Adam-Moulton with $m = 1$) and the midpoint scheme (one step scheme).

is exact for any polynomial of degree $2s - 1$. A quadrature rule that satisfies this exactness is the *Gauss-Legendre quadrature*.

We now derive the construction. Let ξ_i for $i = 1, \dots, s$ denote the roots of the Legendre-Polynomial $L_s: [-1, 1] \rightarrow \mathbb{R}$ of degree s , and denote w_i the associated weights of the Gauss-Quadrature formula. Then, transformation from $(-1, 1)$ to $(0, 1)$ leads to the definition

$$c_i = \frac{1}{2}(1 + \xi_i), \quad b_i = \frac{1}{2}w_i.$$

It remains to determine the coefficients a_{ij} for $j = 1, \dots, s$. One chooses them based on Newton-Cotes-Conditions as

$$a_{ij} = \int_0^{c_i} \ell_j(t) dt, \quad \ell_j(x) = \prod_{\substack{k=1 \\ k \neq i}}^s \frac{x - c_k}{c_j - c_k}.$$

These quadrature rules are at least exact for polynomials up to order $s - 1$ on $[0, c_i]$. As a consequence, this choice of coefficients a_{ij} offers maximum exactness degree for the approximation

$$h \sum_{j=1}^s a_{ij} f(t_k + c_j h, \boldsymbol{\eta}_j) \approx \sum_{t_k}^{t_k + c_j h} f(t, y(t)) dt.$$

The associated Runge-Kutta scheme is called the (**implicit**) Gauss methods.

Remark 1.107 All eigenvalues of $\mathbf{A}$ from the Gaussian methods have positive real part. [Exercise!](#) $\triangle$

Example 1.108 (Examples of Gaussian methods) For $s = 1$ we obtain (using the midpoint rule), that $b_1 = 1, c_1 = \frac{1}{2}$ and $a_{11} = \int_0^{c_1} 1 dt = \frac{1}{2}$. For $s = 1$ and $s = 2$ the Butcher-Tableaus are given as

$$\begin{array}{c|c|c|c}
\frac{1}{2} & \frac{1}{2} & \frac{1}{2} - \frac{\sqrt{3}}{6} & \frac{1}{4} \\
\hline
\frac{1}{2} & \frac{1}{2} & \frac{1}{2} + \frac{\sqrt{3}}{6} & \frac{1}{4} + \frac{\sqrt{3}}{6} \\
\hline
1 & 1 & \frac{1}{2} & \frac{1}{2}
\end{array} \quad (1.52)$$

The respective stability functions R_s for $s = 1, 2$ are given as

$$R_1(z) = \frac{1 + \frac{z}{2}}{1 - \frac{z}{2}}, \quad R_2(z) = \frac{z^2 + 6z + 12}{z^2 - 6z + 12}.$$

Calculation yields $|R_1(z)| \leq 1$ and $|R_2(z)| \leq 1$ for all $z \in \mathbb{C}_-$. [Exercise!](#) Both methods are A-stable. $\triangle$

Theorem 1.109 Gaussian methods of stage number s have consistency order $2s$.

Proof. For a proof we refer to Theorem II.7.9 in Hairer, Ernst, Syvert P. Nørset und Gerhard Wanner: Solving Ordinary Differential Equations I: Nonstiff Problems. Springer, 1993. $\square$

We saw in the previous example that Gaussian methods for $s = 1$ and $s = 2$ are A-stable. Does this hold for the general case $s \in \mathbb{N}$?

Lemma 1.110 (Collocation polynomial) Consider a Gaussian method with coefficients $(\mathbf{A}, \mathbf{b}, \mathbf{c})$ and assume $h > 0$ such that the iterates of Y_i are well-defined. Then for each t_k with $[t_k, t_{k+1}] \subset [a, b]$, there exists a vector-valued polynomial P of degree s such that

$$\begin{aligned}
p(t_k) &= Y_k \\
p(t_{k+1}) &= Y_{k+1}, \\
p'(t_k + c_i h) &= f(t_k + c_i h, p(t_k + c_i h)), \quad j = 1, \dots, s.
\end{aligned}$$

Proof. We choose p' component wise to be the interpolation polynomial of degree $s - 1$ such that

$$P'(t_k + c_i h) = f(t_k + c_i h, \boldsymbol{\eta}_i), \quad i = 1, \dots, s$$

and define P as the principal function of P' with $P(t_k) = Y_k$. Then, by design of consistent Runge-Kutta methods we have the relation

$$\begin{aligned}
\boldsymbol{\eta}_i &= Y_k + h \sum_{j=1}^s a_{ij} f(t_k + c_j h, \boldsymbol{\eta}_j) = P(t_k) + h \sum_{j=1}^s a_{ij} P'(t_k + c_j h) \\
&= P(t_k) + \int_{t_k}^{t_k + c_i h} P'(t) dt \\
&= P(t_k + c_i h)
\end{aligned}$$

for $i = 1, \dots, s$ by using that the defined a_{ij} leads to quadrature with exactness of order $s - 1$. Hence, it follows that

$$f(t_k + c_i h, \boldsymbol{\eta}_i) = f(t_k + c_i h, p(t_k + c_i h))$$

and consequently,

$$\begin{aligned} Y_{k+1} &= Y_k + h \sum_{i=1}^s b_i f(t_k + c_i h, \boldsymbol{\eta}_i) = P(t_k) + h \sum_{i=1}^s b_i P'(t_k + c_i h) = P(t_k) + \int_{t_k}^{t_k+h} P'(t) dt \\ &= P(t_{k+1}). \end{aligned}$$

□

Theorem 1.111 All Gaussian methods are A -stable.

Proof. Consider the Gaussian method of stage $s \in \mathbb{N}$ for the test equation $y'(t) = \lambda y(t)$, $y(0) = 1$ with $\lambda \in \mathbb{C}$ and $\operatorname{Re}(\lambda) \leq 0$. W.l.o.g. let $h = 1$. Since all eigenvalues of $\mathbf{A}$ have positive real part (cp. Remark 1.107), the matrix $\mathbf{I} - \lambda \mathbf{A}$ is invertible. And consequently,

$$Y_1 = R(\lambda) = 1 + \lambda \mathbf{b}^T (\mathbf{I} - \lambda \mathbf{A})^{-1} \mathbf{1},$$

is well-defined. Now let P be the collocation polynomial from Lemma 1.110 and define $Q = |P|^2 = \overline{P}P$. Then

$$|R(\lambda)|^2 = |Y_1|^2 = |P(1)|^2 = Q(1) = Q(0) + \int_0^1 Q'(t) dt = 1 + 2\operatorname{Re} \left(\int_0^1 \overline{P(t)} P'(t) dt \right).$$

Since $\overline{P}P'$ is a polynomial of degree $2s - 1$, exactness of the Gaussian-quadrature rules yields

$$|R(\lambda)|^2 = 1 + 2\operatorname{Re} \left(\sum_{i=1}^s b_i \overline{P(c_i)} P'(c_i) \right),$$

where $b_i > 0$ for all $i = 1, \dots, s$. By construction of P , it holds $P'(c_i) = \lambda P(c_i)$ and consequently

$$|R(\lambda)|^2 = 1 + 2\operatorname{Re} \left(\lambda \sum_{i=1}^s b_i |P(c_i)|^2 \right).$$

We conclude, that $|R(\lambda)| \leq 1$ if $\operatorname{Re}(\lambda) \leq 0$. □

Remark 1.112 As seen in the previous proof, we have the relation

$$|R(\lambda)|^2 = 1 + 2\operatorname{Re} \left(\lambda \sum_{i=1}^s b_i |P(c_i)|^2 \right).$$

If $\operatorname{Re}(\lambda) = 0$, we have $|R(\lambda)| = 1$. This means, all Gaussian methods are *isometric stable*, see Remark 1.106. △

BDF multi-step methods and $A(\alpha)$ -stability

We now discuss a special case of implicit multi-step methods. Recall the construction of Adam-Bashforth or Adam-Moulton schemes. They are based on interpolation polynomials through data $(t_{k+1-j}, f(t_{k+1-j}))$ as carried out for the Adam-Bashforth method in Example 1.65 for $j = 1, \dots, m$ and Adam-Moulton method in Example 1.67 for $j = 0, \dots, m$.

In contrast *Backward Difference Formula* (BDF) methods rely on approximation of the solution by an interpolation polynomial p of degree m through data points $(t_{k+1-m}, Y_{k+1-m}), \dots, (t_k, Y_k)$. Note that for a unique polynomial, one additional condition is required for p . For BDF methods, we require p to locally satisfy the ODE in t_{k+1} , i.e.

$$p'(t_{k+1}) = f(t_{k+1}, Y_{k+1}).$$

The latter condition leads to a Hermite-type interpolation and makes BDF methods **implicit**. This construction can be characterized as follows.

Theorem 1.113 For all grid $a = t_0 < \dots < t_k < \dots < t_N = b$ exists unique coefficients $a_{k,j}$, such that for all polynomials P of degree m it holds

$$\sum_{j=0}^m a_{k,j} P(t_{k+1-j}) = h_k P'(t_{k+1}).$$

These coefficients form the BDF scheme

$$\sum_{j=0}^m a_{k,j} Y_{k+1-j} = h_k f(t_{k+1}, Y_{k+1}).$$

In the equidistant case $h_k \equiv h$, it holds

$$a_{k,j} = a_j = -\ell_j(0), \quad \ell_j(t) = \prod_{\substack{\nu=0 \\ \nu \neq j}}^m \frac{\nu - t}{\nu - j}.$$

Proof. By construction it holds

$$P(t) = \sum_{j=0}^m P(t_{k+1-j}) \ell_{k,j}(t), \quad \ell_{k,j}(t) = \prod_{\substack{\nu=0 \\ \nu \neq j}}^m \frac{t - t_{k+1-\nu}}{t_{k+1-j} - t_{k+1-\nu}}.$$

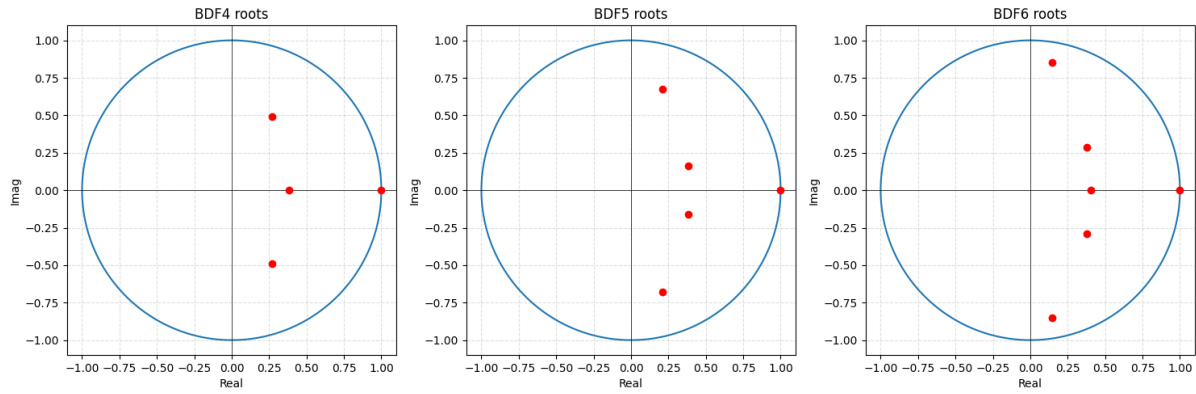
Then,

$$\sum_{j=0}^m a_{k,j} P(t_{k+1-j}) = h_k P'(t_{k+1})$$

holds true for all polynomials P of degree m if $a_{k,j} = h_k \ell'_{k,j}(t_{k+1})$. $\square$

Example 1.114 (BDF for $m = 1, 2, 3$ and $m = 4, 5, 6$) Consider equidistant grid $h_k \equiv h$. Then:

- For $m = 1$ the BDF scheme coincides with the implicit Euler scheme.

Figure 1.11: Roots of the associated characteristic polynomial of the BDF scheme for $m = 4, 5, 6$.

- For $m = 2$ and $h_k \equiv h$ we obtain

$$\frac{3}{2}Y_{k+1} - 2Y_k + \frac{1}{2}Y_{k-1} = hf(t_{k+1}, Y_{k+1}).$$

The latter scheme, of one of the most popular multi-step method for stiff problems.

- For $m = 3$ we obtain

$$\frac{11}{6}Y_{k+1} - \frac{18}{6}Y_k + \frac{9}{6}Y_{k-1} - \frac{2}{6}Y_{k-2} = hf(t_{k+1}, Y_{k+1}).$$

- $m = 4$:

$$\frac{25}{12}Y_{k+1} - 4Y_k + 3Y_{k-1} - \frac{4}{3}Y_{k-2} + \frac{1}{4}Y_{k-3} = hf(t_{k+1}, Y_{k+1}).$$

- $m = 5$:

$$\frac{137}{60}Y_{k+1} - 5Y_k + 5Y_{k-1} - \frac{10}{3}Y_{k-2} + \frac{5}{4}Y_{k-3} - \frac{1}{5}Y_{k-4} = hf(t_{k+1}, Y_{k+1}).$$

- $m = 6$:

$$\frac{147}{60}Y_{k+1} - 6Y_k + \frac{15}{2}Y_{k-1} - \frac{20}{3}Y_{k-2} + \frac{15}{4}Y_{k-3} - \frac{6}{5}Y_{k-4} + \frac{1}{6}Y_{k-5} = hf(t_{k+1}, Y_{k+1}).$$

△

Lemma 1.115 (Nullstability for BDF) Let $h_k = h$. Then BDF is null stable for $m = 1, 2, \dots, 6$.

Proof. Recall the normalization $a_0 = 1$. Then, the characteristic polynomials $\rho(z) = \sum_{j=0}^m a_j z^{m-j}$ for $m = 1, 2, 3$ for BDF are given as

m	$\rho(z)$	roots
1	$z - 1$	$z_1 = 1$
2	$z^2 - \frac{4}{3}z + \frac{1}{3}$	$z_1 = 1, z_2 = \frac{1}{3}$
3	$z^3 - \frac{18}{11}z^2 + \frac{9}{11}z - \frac{2}{11}$	$z_1 = 1, z_{2,3} = \frac{7 \pm i\sqrt{39}}{22}$

For $m = 4, 5, 6$, we refer to Figure 1.12. □

Remark 1.116 (Null stability and convergence for general BDF schemes) Consider equidistant step size $h_k \equiv h$. Then, the characteristic polynomial of BDF schemes is null-stable if and only if $m \leq 6$, see [?]. By consistency construction, they are convergent with order m (under the previous discussed assumptions). △

Theorem 1.117 (A-stability for BDF schemes) The BDF-schemes with equidistant step size are A-stable for $m = 1, 2$.

Proof. The case $m = 1$ corresponds to the implicit Euler scheme which is A-stable. For $m = 2$, by Theorem 1.99 δS is subset of

$$\left\{ \mu(\phi) \in \mathbb{C} \mid \mu(\phi) = \frac{\frac{3}{2}(e^{i\phi})^2 - 2e^{i\phi} + \frac{1}{2}}{(e^{i\phi})^2}, \phi \in [0, 2\pi] \right\}.$$

It holds that

$$\mu(\phi) = (1 - \cos \phi)^2 + 2i(1 - \cos \phi) \sin \phi.$$

Hence $\mu(\phi) = 0$ for $\phi = 0, 2\pi$ and $\operatorname{Re} \mu(\phi) > 0$ for $\phi \in (0, 2\pi)$. Now either $\mathbb{C}_- \subset S$ is subset or $\mathbb{C}_- \cap S = \{\mu = 0\}$. The associated polynomial (choosing $\mu = -1$)

$$z \mapsto \rho(z) - (-1)\sigma(z) = \frac{5}{2}z^2 - 2z + \frac{1}{2}$$

has roots $z_{1,2} = \frac{2 \pm i}{5}$ and hence is null-stable. Thus $\mu = -1$ is in S and consequently $\mathbb{C}_- \subset S$. □

Definition 1.118 ($A(\alpha)$ -stability.) A linear multi-step method with stability region S is called $A(\alpha)$ -stable with $\alpha \in (0, \frac{\pi}{2}]$ if

$$\{z \in \mathbb{C}_- \mid |\arg(z) - \pi| \leq \alpha\} \subset S.$$

Remark 1.119 Let $m = 3, 4, 5, 6$. Then, the associated m -step BDF schemes are $A(\alpha)$ stable, see also Figure 1.12:

m	1	2	3	4	5	6
α	90°	90°	86.03°	73.35°	51.84°	17.84°

△

The, following result refines the statement about A -stability.

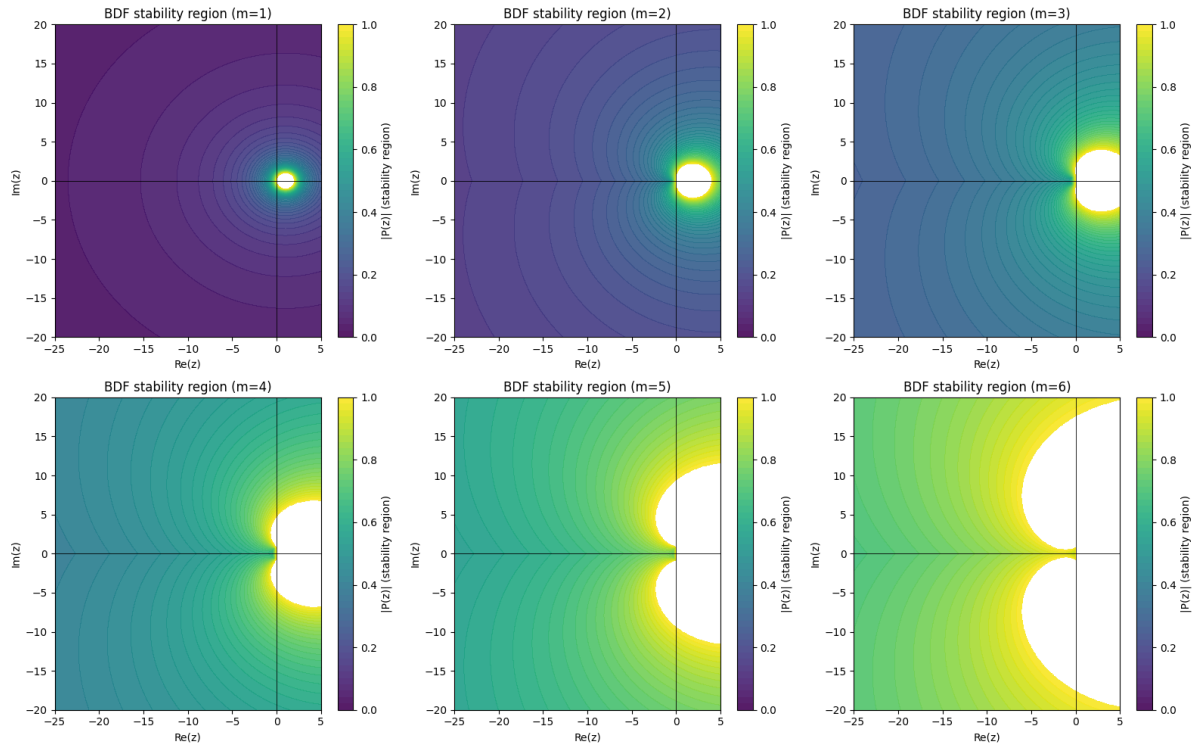


Figure 1.12: Stability region S (in dark blue) for each $m = 1, 2, 3$ (top row) and $m = 4, 5, 6$ bottom row for corresponding m -step BDF schemes.

Theorem 1.120 (Dahlquist: Second barrier 1963) An A -stable linear multi-step methods on a equidistant grid of the form

$$\sum_{j=0}^m a_j Y_{k+1-j} = h \sum_{j=0}^m b_j f(t_{k+1-j}, Y_{k+1-j})$$

has consistency order at most of **second** order.

Proof. See for example [?]. □

Example 1.121 Recall the Milne-Simpson scheme

$$Y_{k+1} - Y_{k-1} = \frac{h}{3}(f_{k-1} + 4f_k + f_{k+1}), \quad f_i = f(t_i, Y_i), \quad t_{i+1} = t_i + h.$$

from Example 1.88. The boundary of its stability region S is subset of the curve $\mu(\phi) = \frac{3 \sin \phi}{2 + \cos \phi} i$ and

$$S = \{ix \mid |x| < \sqrt{3}\}.$$

Although, consistency order is $m + 2 = 4$, they are not suitable for stiff problems. △

Effective vs. formal convergence order

In general, the classical convergence order that is derived by the expansion of the truncation error in monomials in h is not reached for stiff problems, since the appearing constants in the

estimates depend on derivatives of the right side f .

Example 1.122 Consider the IVP

$$y'(t) = \lambda y(t) - (1 + \lambda)e^{-t} = f(t, y(t)), \quad y(0) = 1$$

with solution $y(t) = e^{-t}$. In particular the solution is independent of λ . For $\lambda \rightarrow -\infty$ the problem becomes stiff. The implicit mid point rule

$$Y_{k+1} = Y_k + h_k f\left(\frac{t_k + t_{k+1}}{2}, \frac{Y_k + Y_{k+1}}{2}\right)$$

yields in this case

$$Y_{k+1} = Y_k + h_k \left(\frac{1}{2} \lambda Y_k + \frac{1}{2} \lambda Y_{k+1} - (1 + \lambda) e^{-t_k - \frac{1}{2} h_k} \right).$$

Solving for Y_{k+1} yields

$$Y_{k+1} = Y_k + h_k \Phi(t_k, Y_k, h_k), \quad \Phi(t, Y, h) = \frac{\lambda Y - (1 + \lambda) e^{-t - \frac{1}{2} h}}{1 - \frac{1}{2} h \lambda}.$$

The consistency error in the solution $y(t) = e^{-t}$ hence is

$$\tau(t, h) = \frac{y(t+h) - y(t)}{h} - \Phi(t, y(t), h) = c(h, \lambda) h e^{-t}$$

with

$$c(h, \lambda) = \frac{1}{h} \left(\frac{e^{-h} - 1}{h} - \frac{\lambda - (1 + \lambda) e^{-\frac{1}{2} h}}{1 - \frac{1}{2} h \lambda} \right).$$

It holds

$$c(h, \lambda) = - \left(\frac{1}{24} + \frac{\lambda}{8} \right) h + \mathcal{O}(h^2).$$

So the method in case as formal order $q = 2$. But

$$\lim_{\lambda \rightarrow -\infty} c(h, \lambda) = \left(\frac{1 - e^{\frac{1}{2} h}}{h} \right)^2 = \frac{1}{4} - \frac{1}{8} h + \dots$$

and consequently the order reduces from $q = 2$ to $q = 1$ as $\lambda \rightarrow -\infty$. $\triangle$

Remark 1.123 For Gaussian methods with $s \in \mathbb{N}$ stages, the formal order is $q = 2s$, The effective order reduces to $q = s$. $\triangle$

1.4.2 Computational aspects of implicit methods

Implicit Runge-Kutta schemes

Consider implicit Runge-Kutta scheme with coefficients $(\mathbf{A}, \mathbf{b}, \mathbf{c})$ such as

$$\begin{aligned}\boldsymbol{\eta}_i &= Y_k + h \sum_{j=1}^s a_{ij} f(t_k + c_i h, \boldsymbol{\eta}_j), & \sum_{j=1}^s a_{ij} &= c_i, \quad i = 1, \dots, s \\ Y_{k+1} &= Y_k + h \sum_{i=1}^s b_i f(t_k + c_i h, \boldsymbol{\eta}_i), & \sum_{j=1}^s b_j &= 1,\end{aligned}$$

where the coefficients constraints are based on the necessary conditions derived for Runge-Kutta schemes.

Let the iteration step $0 \leq k \leq N - 1$ be fixed. Define the increments K_i

$$K_i = f(t_k + c_i h, \boldsymbol{\eta}_i), \quad i = 1, \dots, s. \quad (1.53)$$

While, the increments K_i can be sequentially computed in the case of explicit Runge-Kutta schemes, they are implicitly defined for **implicit** Runge Kutta schemes. In particular, in this case we seek for K_i , that solves

$$K_i = f(t_k, c_i h, Y_k + h \sum_{j=1}^s a_{ij} K_j), \quad i = 1, \dots, s. \quad (1.54)$$

So define

$$\mathbf{K} = \begin{pmatrix} K_1 \\ \vdots \\ K_s \end{pmatrix} \in \mathbb{R}^{sn}, \quad \Psi(\mathbf{K}) = \begin{pmatrix} f(t_k + c_1 h, Y_k + h \sum_{j=1}^s a_{1j} K_j) \\ \vdots \\ f(t_k + c_s h, Y_k + h \sum_{j=1}^s a_{sj} K_j) \end{pmatrix} \in \mathbb{R}^{sn}.$$

Hence, in compact form we seek $\mathbf{K} \in \mathbb{R}^{sn}$ that solves $\mathbf{K} = \Psi(\mathbf{K})$. Once obtained, the next iterate then can be computed as

$$Y_{k+1} = Y_k + h \sum_{i=1}^s b_i K_i.$$

The solution $\mathbf{K} = \Psi(\mathbf{K})$ can be obtained by root finding algorithms. One instance is to use the NEWTON-scheme on $\Psi(\mathbf{K}) - \mathbf{K} = 0$, leading to iterates

$$\mathbf{K}^{(m+1)} = \mathbf{K}^{(m)} + \mathbf{d}^{(m)} \quad (1.55)$$

with a descent $\mathbf{d}^{(m)}$ that solves

$$\left[\mathbf{I} - \Psi'(\mathbf{K}^{(m)}) \right] \mathbf{d}^{(m)} = \Psi(\mathbf{K}^{(m)}) - \mathbf{K}^{(m)},$$

starting at $\mathbf{K}^{(0)} = (f(t_k, Y_k))_{i=1}^s \in \mathbb{R}^{sn}$. The Jacobian matrix $\Psi'(\mathbf{K})$ is given (in block matrix notation) as

$$\Psi'(\mathbf{K}) = h \begin{pmatrix} a_{11} J_1(\mathbf{K}) & \cdots & a_{1s} J_1(\mathbf{K}) \\ \vdots & \ddots & \vdots \\ a_{s1} J_s(\mathbf{K}) & \cdots & a_{ss} J_s(\mathbf{K}) \end{pmatrix}, \quad J_j(\mathbf{K}) := (\partial_y f)(t_k + c_i h, Y_k + h \sum_{j=1}^s a_{ij} K_j).$$

In praxis one can approximate the resulting matrix

$$(\partial_y f)(t_k + c_i h, Y_k + h \sum_{j=1}^s a_{ij} K_j) \approx \partial_y f(t_k, Y_k),$$

resulting in a quasi-Newton-scheme.

Remark 1.124 While Gaussian methods are favorable in terms of A-stability properties and their formal consistency order. One has to solve in general non-linear problems in each step with ns unknowns. Hence, obtaining higher order comes with a price in particular if the matrix $\mathbf{A}$ is dense. $\triangle$

Diagonal and linearly implicit Runge-Kutta methods

We saw that general (dense) implicit Runge-methods with higher order are expensive. On the other extreme explicit Runge-Kutta schemes are directly computable but are not suitable for stability discussions.

Hence a compromiss is required in between both worlds.

Definition 1.125 An **implicit** Runge-Kutta scheme with coefficient matrix $\mathbf{A}$ is called:

- Diagonally implicit Runge Kutta (DIRK) scheme, if $\mathbf{A}$ is lower triangular.
- Singly diagonally implicit Runge Kutta (SDIRK) scheme if $\mathbf{A}$ is lower triangular and $a_{ii} = a \in \mathbb{R}$ for all $i = 1, \dots, s$.

Revisiting equation (1.54), we then have to solve

$$K_i = f(t_k, c_i h, Y_k + h \sum_{j=1}^{i-1} a_{ij} K_j + a_{ii} h K_i), \quad i = 1, \dots, s.$$

sequentially for $K_1, \dots, K_s \in \mathbb{R}^n$. Then, the quasi-Newton scheme with $\mathbf{J} = (\partial_y f)(t_k, Y_k)$ for $m = 0, 1 \dots$ reads

$$(\mathbf{I} - h a_{ii} \mathbf{J})(K_i^{(m+1)} - K_i^{(m)}) = f \left(t_k + c_i h, Y_k + h \sum_{j=1}^{i-1} a_{ij} K_j + a_{ii} h K_i^{(m)} \right) - K_i^{(m)},$$

or equivalently

$$(\mathbf{I} - h a_{ii} \mathbf{J}) K_i^{(m+1)} = f \left(t_k + c_i h, Y_k + h \sum_{j=1}^{i-1} a_{ij} K_j + a_{ii} h K_i^{(m)} \right) - h a_{ii} \mathbf{J} K_i^{(m)} \quad (1.56)$$

In particular, in the single diagonally case $a_{ii} \equiv a$ and the resulting matrix $(\mathbf{I} - h a \mathbf{J})$ can be factorized a single time and used for the solution of all $i = 1, \dots, s$.

Example 1.126 For $s = 2$, there are two SDIRK schemes of consistency order $q = s + 1 = 3$ given as

$$\begin{array}{c|cc} a_{\pm} & a_{\pm} & 0 \\ 1-a & 1-2a_{\pm} & a_{\pm} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}, \quad a_{\pm} = \frac{1}{2} \pm \frac{\sqrt{3}}{6}.$$

The choice of a_+ leads to an A-stable SDIRK method. $\triangle$

Remark 1.127 (Order of Consistency for DIRK-schemes) The consistency order of DIRK-schemes with $s \in \mathbb{N}$ stages is limited by $s + 1$ or s (depending on s). $\triangle$

Remark 1.128 (Reducing the cost: Linear-implicit schemes and Rosenbrock-Wanner schemes)) Instead of performing several Newton-steps $m = 0, 1, \dots$ per iteration k one can simplify to a single Newton step. In case of sufficiently close initial value (small enough step size) and its local fast convergence one expects sufficient accurate approximation. Make the linear combination ansatz

$$K_i^{(0)} := \sum_{j=1}^{i-1} \frac{d_{ij}}{a_{ii}} K_j, \quad K_1^{(0)} = \mathbf{0},$$

for coefficients d_{ij} to be chosen. Then, a single Newton-step leads the scheme denoted as **linearly implicit Runge-Kutta methods** for $i = 1, \dots, s$ as

$$(\mathbf{I} - h a_{ii} \mathbf{J}) K_i^{(m+1)} = f \left(t_k + c_i h, Y_k + h \sum_{j=1}^{i-1} (a_{ij} + d_{ij}) K_j + a_{ii} h K_i^{(m)} \right) - h \mathbf{J} \sum_{j=1}^{i-1} d_{ij} K_j^{(m)},$$

$$Y_{k+1} = Y_k + h \sum_{j=1}^s b_j K_j.$$

If $\mathbf{A}$ is non-singular, then the stability function of the original diagonal implicit scheme with (quasi)-Newton step and the linear-implicit Runge Kutta method coincide independent of the choice of coefficients d_{ij} . Hence, stability properties inherit from the original diagonal implicit scheme. However the choice of coefficients d_{ij} still has influence on the possible order of the linear implicit method. In fact linear implicit method may have lower order than the original diagonal implicit method. In case $a_{ii} \equiv a$ we arrive at an popular instance of a Rosenbrock-Wanner scheme. $\triangle$

Implicit Linear Multistep Methods

We briefly discuss the computational structure arising in implicit linear multistep schemes, with a particular focus on Backward Differentiation Formulae (BDF). A q -step linear multistep method is given by

$$\sum_{i=0}^m a_i Y_{k+1-i} = h \sum_{i=0}^m b_i f(t_{k+1-i}, Y_{k+1-i}), \quad a_0 = 0, b_0 \neq 0 \quad (1.57)$$

where the coefficients $(\alpha_i)_{i=0}^q, (\beta_i)_{i=0}^q$ are determined by the consistency and order conditions.

Rearranging (1.57) yields the nonlinear equation

$$\Psi(Y_{k+1}) := Y_{k+1} - h b_0 f(t_{k+1}, Y_{k+1}) - R_k = 0, \quad (1.58)$$

where all terms involving past values are known,

$$R_k := - \sum_{i=1}^m a_i Y_{k+1-i} + h \sum_{i=1}^m b_i f(t_{k+1-i}, Y_{k+1-i}). \quad (1.59)$$

Thus, in contrast to implicit Runge-Kutta schemes which require solving for all stage values simultaneously, implicit linear multi-step methods require the solution of a *single* n -dimensional nonlinear equation at each time step.

Newton iteration. Newton's method applied to (1.58) produces iterates

$$Y_{k+1}^{(m+1)} = Y_{k+1}^{(m)} + d^{(m)}, \quad (1.60)$$

where the Newton correction $d^{(m)}$ solves

$$\left[I - hb_0(\partial_y f)(t_{k+1}, Y_{k+1}^{(m)}) \right] d^{(m)} = -\Psi(Y_{k+1}^{(m)}). \quad (1.61)$$

In practice one again may use quasi-Newton updates or freeze the Jacobian across several steps to reduce computational cost.

1.5 Beyond A-stability: An overview

This section is devoted to some extended discussion about stability concepts in the world of ODE solvers. We may summarize them as follows,

A scheme is stable in a certain sense, if it mimics properties of certain ODEs.

We recall the property of a scheme being *A-stable* if for the scalar valued test equation

$$y'(t) = \lambda y(t), \quad y(0) = 1, \quad \lambda \in \mathbb{C}$$

and the produced iterates Y_k obtained using an arbitrary uniform step-size $h > 0$, it holds that

$$|Y_{k+1}| \leq |Y_k|, \quad k = 0, 1, \dots,$$

whenever $\operatorname{Re}(\lambda) \leq 0$. We saw, that A-stability was closely related to properties of the stability function R .

Example 1.129 The solution $y(t) = e^{\lambda t}$ has the following behavior

$$\lim_{t \rightarrow \infty} |y(t)| = \begin{cases} \infty, & \operatorname{Re} \lambda > 0, \\ 1, & \operatorname{Re} \lambda = 0, \\ 0, & \operatorname{Re} \lambda < 0. \end{cases}$$

△

1.5.1 Refined A-stability concepts

Isometry

Definition 1.130 A Runge-Kutta scheme with stability function R is called *isometric stable* if $|R(z)| = 1$ for all $z \in \mathbb{C}$ with $\operatorname{Re}(z) = 0$.

Examples of isometric stable schemes involve Gaussian methods of s stages.

Strongly A-stable

Definition 1.131 An A-stable method with stability function R is called *strongly A-stable* if in addition $|R(\infty)| < 1$.

L-stability

The concept of L -stability refers to the strong damping for stiff modes.

Example 1.132 Consider the IVP $y'(t) = My(t)$ with

$$M = \begin{pmatrix} -(10^4 + 1) & 1 \\ 0 & -1 \end{pmatrix}, \quad y(0) = \begin{pmatrix} 10 \\ 1 \end{pmatrix}.$$

Then, a spectral decomposition $M = V\Lambda V^{-1}$ is given by

$$\Lambda = \begin{pmatrix} -(10^4 + 1) & 0 \\ 0 & -1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 10^{-4} \\ 0 & 1 \end{pmatrix}, \quad V^{-1} = \begin{pmatrix} 1 & -10^{-4} \\ 0 & 1 \end{pmatrix}.$$

Consequently, the exact solution is given as

$$y(t) = e^{Mt}y_0 = Ve^{\Lambda t}V^{-1}y_0 = e^{-(10^4+1)t}v_1 + e^{-t}v_2,$$

with $v_1 = (9.9999, 0)^\top$ and $v_2 = (10^{-4}, 1)^\top$. Now applying the first order implicit Euler leads

$$Y_k = (1 + (10^4 + 1)h)^{-i}v_1 + (1 + h)^{-i}v_2.$$

and the second order implicit mid-point rule (Gauss method $s = 1$) leads

$$Y_k = \left(\frac{1 - \frac{1}{2}(10^4 + 1)h}{1 + \frac{1}{2}(10^4 + 1)h} \right)^i v_1 + \left(\frac{1 - \frac{1}{2}h}{1 + \frac{1}{2}h} \right)^i v_2.$$

The exact solution y satisfies

$$y(1) \approx \begin{pmatrix} -3.678 \cdot 10^{-5} \\ 0.368 \end{pmatrix}.$$

Now choose $h = \frac{1}{10}$ and $h = \frac{1}{100}$, leads the following approximate value at $t = 1$, i.e. $k = 10$ or $k = 100$ respectively:

h	implicit Euler	implicit midpoint
$\frac{1}{10}$	$\approx (-3.855 \cdot 10^{-5}, 0.385)^\top$	$\approx (9.608, 0.368)^\top$
$\frac{1}{100}$	$\approx (-3.697 \cdot 10^{-5}, 0.370)^\top$	$\approx (0.183, 0.368)^\top$

△

As the Example 1.132 shows, although the mid point rule is of higher order than the implicit Euler scheme and both schemes are A-stable, the latter property seems to be not sufficient to obtain good approximations with moderate step-sizes.

Definition 1.133 An A-stable Runge-Kutta scheme with stability function R is called L -stable if $\lim_{|z| \rightarrow \infty} R(z) = R(\infty) = 0$.

Recalling Remark 1.96, we saw that an A -stable Runge-Kutta scheme was L -stable if in addition

$$a_{si} = b_i, \quad \text{for } i = 1, \dots, s.$$

Runge-Kutta methods with such property are called *stiffly accurate*. In particular, L -stable schemes are strictly A -stable.

Example 1.134 (Radau-IIA schemes) A special implicit Runge-Kutta class with s stages is based on Radau-quadrature with optimal exactness $2s - 2$ under the constraint $c_s = 1$ (right node constraint). Let c_i and b_i then be defined by the s -step Radau-Legendre quadrature $\{(b_i, c_i)_i\}$ on $[0, 1]$ given as

$$\int_0^1 g(t) dt \approx \sum_{i=1}^s b_i g(c_i).$$

Choose coefficients in $\mathbf{A}$ again via Newton-Cotes conditions (integration of Lagrange basis L_j) similar to the construction of the Gaussian methods. Then, due to exactness of the Radau-quadrature of order $2s - 2$

$$a_{sj} = \int_0^{c_s} L_j(t) dt = \int_0^1 L_j(t) dt = \sum_{i=1}^s b_i L_j(c_i) = b_j.$$

Consequently, Radau-IIA schemes are stiffly accurate and L -stable. $\triangle$

1.5.2 Concepts for dissipative systems

Is a property of ODE solvers to mimic behavior of dissipative systems. Recall an ODE system with right hand side f is called dissipative if

$$\langle y - z, f(t, y) - f(t, z) \rangle \leq 0 \forall t, y, z.$$

Now let y_1 and y_2 be two solutions of $y' = f(t, y)$. Then,

$$\|y_1(t) - y_2(t)\| \leq \|y_1(a) - y_2(a)\|,$$

and moreover

$$\|y_1(t_{k+1}) - y_2(t_{k+1})\| \leq \|y_1(t_k) - y_2(t_k)\|.$$

So, when does a numerical scheme mimics such behaviour? We refer to the original work of stability concepts in this direction, see [?].

Algebraic stability

Definition 1.135 A Runge-Kutta scheme with $s \in \mathbb{N}$ stages with coefficients $\mathbf{A} = [a_{ij}]_{ij}$, $\mathbf{b} = [b_i]$, $\mathbf{c}$ is called algebraic stable if the matrix $M \in \mathbb{R}^{s,s}$ given as

$$[M]_{ij} = b_i a_{ij} + b_j a_{ji} - b_i b_j$$

is positive semi-definite and $b_i \geq 0$ for all $i = 1, \dots, s$.

The, concept of algebraic stability is relevant when comparing two iterations Y_{k+1} and Z_{k+1} , where the positive semi-definiteness of the quadratic form

$$\xi^T M \xi \forall \xi \in \mathbb{R}^s$$

to ensure contraction.

BN-stability

Let us explicitly write a numerical scheme as $Y_{k+1} = \Psi_h(Y_k)$. Note, this notation includes implicit methods.

Definition 1.136 A numerical scheme is called *BN-stable*, if applied to dissipative systems, then it is contractive with

$$\|\Psi_h(Y) - \Psi_h(Z)\| \leq \|Y - Z\|, \quad \forall Y, Z, h > 0$$

In particular B-stability means that the numerical method never increases the distances between two trajectories, when the underlying ODE would not. This is in particular relevant for dissipative Systems, where solution converge to steady states, perturbations shrink over time and no oscillatory or unstable growth behavior is present.

Theorem 1.137 (Burrage&Butcher : algebraic stable $\Rightarrow$ BN-stable) Consider a dissipative system $y' = f(t, y)$ and assume a Runge-Kutta scheme is algebraic stable, then it is B-stable.

For other sufficient conditions of BN-stability, see e.g. [?].

AN-stability

Definition 1.138 A Runge-Kutta scheme is called *AN-stable* if independent on the step-size all produced iterates $(Y_k)_k$ for the complex scalar valued ODE

$$y'(t) = \lambda(t)y(t), \quad \text{Re}(\lambda(t)) \leq 0$$

satisfy

$$|Y_{k+1}| \leq |Y_k|.$$

Theorem 1.139 (Burrage&Butcher : AN-stable $\Rightarrow$ algebraic stable) Let $c_i \in [0, 1]$ of an AN-stable Runge-Kutta scheme be distinct. Then, the RK scheme must be algebraic-stable.

So if c_i are distinct, then algebraic stability, AN-stability and BN-stability are equivalent properties. In the other case algebraic stability implies BN-stability, which itself implies AN-stability.

2. Geometric Numerical Integration

This chapter is devoted to the introduction to a important problem class of ordinary differential equations for which their solution(s) y satisfy certain geometrical properties. Such properties may include cases of *invariance* or *preservation* or differential equations that are defined on manifolds such as Stiefel, Grassmann or manifolds of low-rank matrices (tensors). It is then a goal of numerical methods so not only mimic such solution properties but to exactly satisfy them. The latter can become highly relevant, when one wishes to approximate long time behaviour.

The introductory presentation at hand is based on [?] and we refer to the detailed discussions therein for the interested reader.

Let us state a canonical geometrical property, the so called *first integral*. For this consider a general ODE of the form $y' = f(y)$.

Definition 2.1 (first integral) A non-constant function $I(y)$ is called a *first integral* if

$$I'(y)f(y) = 0 \quad \text{for all } y.$$

This implies that every solution y satisfies $I(y(t)) = I(y_0) \equiv c \in \mathbb{R}$. Synonymously with first integral the terms *invariant* or *conserved quantity* or *constant of motion* are also used.

In particular if $I(y)$ is a first integral, then the ODE's flow preserves the level sets

$$\mathcal{M}_c = \{y \in \mathbb{R}^n \mid I(y) = c\},$$

i.e. the solution trajectory remains on these sets, e.g. submanifolds.

2.1 Numerical methods for Hamiltonian Systems

For the sake of this course, we will focus on Hamiltonian systems and with it on symplectic integration methods. In particular we focus on special form of ODEs $y' = f(y)$ of the form $y = (p, q) \in \mathbb{R}^{2d}$ and

$$p' = -H_q(p, q), \quad q' = H_p(p, q) \quad (2.1)$$

for some function $H(p_1, \dots, p_n, q_1, \dots, q_n)$, the *Hamiltonian* representing the total energy. Here q_i are the (generalized) position coordinates and p_i the momenta for $i = 1, \dots, n$ for a number of degrees of freedom $n \in \mathbb{N}$. Here H_p and H_q are the vectors of partial derivatives.

2.1.1 Hamiltonian systems

Hamiltonian systems form one of the most important class of ordinary differential equations in the context of geometric numerical integration. A important related property of these system

is the so called *symplecticity* of the flow.

We start with the equations of motions in possible historical order.

Example 2.2 (Newton (1687)) Let $x \in \mathbb{R}^d$ be the position (in cartesian coordinates) of a particule with mass $m > 0$. Let $F(x)$ be a force, e.g. $F(x) = -\nabla V$ for some potential $V(x)$. Then, the movement of mass points under forces is described by **Newtonian equation of motion**:

$$mx''(t) = F(x).$$

△

Example 2.3 (Lagrange (1788)) Let $q = (q_1, \dots, q_n) \in \mathbb{R}^n$ denote $n \in \mathbb{N}$ degrees of freedom or *generalized coordinates*. Then denote the **kinetic energy** by $T(q, q')$ and the **potential energy** by $U(q)$. Then, Lagrange's equations of motion are given as

$$\frac{\partial}{\partial t} \frac{\partial T}{\partial q'_k} - \frac{\partial T}{\partial q_k} = -\frac{\partial U}{\partial q_k}, \quad k = 1, \dots, n. \quad (2.2)$$

Define the **Lagrange function** $L = T - U$. Then, (2.2) is equivalent to

$$\frac{\partial}{\partial t} \nabla_{q'} L(q, q') = \nabla_q L(q, q'). \quad (2.3)$$

A popular class of kinetic energy is of the form $T(q, q') = \frac{1}{2} q'^\top M(q) q'$, where $M(q)$ is symmetric and positive definite. △

Example 2.4 (Newtons eq. of motions $\subset$ Lagrange eq. of motions) Lagrange's equations of motion generalized Newtons formulations. Consider $q \in \mathbb{R}^3$ as the position of a mass point with mass m and define the kinetic energy $T(q, q') = \frac{m}{2} \|q'\|_2^2$ and $U(q) = F(q)$ be the potential energy. Then, Lagrange equation becomes Newton's equation of motion. △

Example 2.5 (Pendulum) See exercise. △

Then, Hamilton derived *Hamilton's equations* in 1834, while being extended and stressed by Jacobi. The generalized view point of Hamilton's equation as in the words of Jacobi (1842), made all the long history of research of Newton and Lagrange "to be considered as an introduction". In fact Hamilton simplified the structure of Lagrange's equation and turned them into a form with remarkable symmetry. For this he introduced *Poisson*' variables $p_k, k = 1, \dots, n$ also known as **conjugate momenta**:

$$p_k = \frac{\partial L}{\partial q'_k}(q, q') \quad (2.4)$$

and the **Hamiltonian** H given as

$$H(p, q) := p^\top q' - L(q, q') \quad (2.5)$$

For the latter to make sense, we assume that for every q , the conjugate momenta define a diffeomorphism between q' and p . This transformation is called *Legendre transform*.

Theorem 2.6 The Lagrange equation (2.3) is equivalent to the **Hamilton's equations**:

$$p'_k = -\frac{\partial H}{\partial q_k}(p, q), \quad q'_k = \frac{\partial H}{\partial p_k}(p, q), \quad k = 1, \dots, d. \quad (2.6)$$

Proof. Note that using the conjugate momenta, Lagrange equation is equivalent to $p' = \nabla_q L(q, q')$. The equivalence follows by the identities using the definition of the Hamiltonian H and the conjugate momenta:

$$\begin{aligned}\frac{\partial H}{\partial p} &= (q')^\top + p^\top \frac{\partial q'}{\partial p} - \frac{\partial L}{\partial q'} \frac{\partial q'}{\partial p} = (q')^\top, \\ \frac{\partial H}{\partial q} &= p^\top \frac{\partial q'}{\partial q} - \frac{\partial L}{\partial q} - \frac{\partial L}{\partial q'} \frac{\partial q'}{\partial q} = -\frac{\partial L}{\partial q}.\end{aligned}$$

□

In short: we obtain the same differential equation written in different coordinates: (q, q') vs (q, p) .

Assume the kinetic energy T is quadratic: $T = \frac{1}{2}(q')^\top M(q)q'$, with $M(q)$ symmetric positive definite. For fixed q it holds $p = M(q)q'$, hence the existence of the Legendre transform is given. Hence, replace q' by $M(q)^{-1}p$ in the definition of $H(p, q)$, we obtain

$$H(p, q) = p^\top M(q)^{-1}p - L(q, M(q)^{-1}p) = \frac{1}{2}p^\top M(q)^{-1}p + U(q).$$

Hence the Hamiltonian $H = T + U$ describes the *total energy* of the mechanical system.

Example 2.7 (Newton revisited) Newton's equation of motion uses the Legendre transform

$$p_k = \frac{\partial L}{\partial q'_k}(q, q') = \frac{\partial T}{\partial q'_k}(q, q') = m q'_k.$$

The Hamiltonian H is given as

$$H(p, q) = p^\top q' - L(q, q') = \frac{1}{2m}p^\top p + U(q).$$

Hence, the Hamiltonian equations of motions read

$$\begin{aligned}p' &= -\nabla_q H(p, q) = -\nabla_q U(q), \\ q' &= \nabla_p H(p, q) = \frac{1}{m}p.\end{aligned}$$

△

We conclude this introduction with other example of Hamiltonian systems.

Example 2.8 (N -body systems) Given the total energy or Hamiltonian

$$H(p, q) = \tilde{T}(p) + U(q)$$

with the kinetic energy

$$\tilde{T}(p) = \frac{1}{2} \sum_{i=1}^N \frac{1}{m_i} \|p_i\|_2^2 = \frac{1}{2} \sum_{i=1}^N m_i \|q'_i\|_2^2 = T(q'), \quad p_i, q_i \in \mathbb{R}^3, p_i = m_i q'_i$$

and the potential energy

$$U(q) = \sum_{i=1}^N \sum_{j=1}^{i-1} V_{ij}(\|q_i - q_j\|_2),$$

for given potentials V_{ij} . N -body systems involve

- The outer solar system: $q = (q^{(1)}, \dots, q^{(5)}) \in \mathbb{R}^{3 \cdot 5}$ positions of Jupiter, Saturn, Uranus, Neptune, Pluto relative to the sun and $p^{(i)} = m_i q^{(i)}$ refers to the impulse for $m_i > 0$. Let the sun have position $q^{(0)} = (0, 0, 0)^\top$ and momentum $p^{(0)} = (0, 0, 0)^\top$. Then, for g denoting the gravitational constant, H is given as

$$H(p, q) = \frac{1}{2} \sum_{i=1}^5 \frac{1}{m_i} \|p^{(i)}\|^2 - g \sum_{i=1}^5 \sum_{j=0}^{i-1} \frac{m_i m_j}{\|q^{(i)} - q^{(j)}\|}.$$

- Motions of atoms within a molecule (without quantum effects) with application to **molecular dynamics**.
- For $N = 2$, one obtains the Kepler problem (two-body problem), due to Kepler (1609) the describes planet movement in elliptic orbits.

△

We conclude this section with the connection of Hamiltonian systems and a first integral, recalling that H may encode the total energy.

Lemma 2.9 (Conservation of energy) Let $p(t), q(t) \in \mathbb{R}^n$ be solution of the Hamiltonian system

$$p' = -\nabla_q H(p, q), \quad q' = \nabla_p H(p, q)$$

with an arbitrary smooth Hamiltonian $H: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Then,

$$H(p(t), q(t)) \equiv \text{const}, \quad \forall t.$$

In particular H is an invariant or *first integral* of the Hamiltonian system.

Proof. In order $H(p(t), q(t))$ to be constant, its time derivative must vanish. It holds

$$\frac{\partial}{\partial t} H(p(t), q(t)) = (\nabla_p H(p, q))^\top p' + (\nabla_q H(p, q))^\top q' = (q')^\top p' - (p')^\top q' = 0.$$

□

So, it is a **goal** to find numerical methods such that iterations $Y_k = (P_k, Q_k) \approx (p(t_k), q(t_k)) = y(t_k)$ satisfy

$$H(P_k, Q_k) \approx H(P_0, Q_0) = H(y(0)),$$

in particular for long time horizons.

2.1.2 Symplectic transformations

As we saw in the last section, hamiltonian systems have its Hamiltonian $H(p, q)$ as its first integral. This section is devoted to another important property, the *symplecticity* of the corresponding ODE flow.

For this, one studies two-dimension parallelograms lying in $\mathbb{R}^{2n}$. We suppose that the parallelogram P is spanned by two vectors

$$\xi = \begin{pmatrix} \xi^p \\ \xi^q \end{pmatrix}, \quad \eta = \begin{pmatrix} \eta^p \\ \eta^q \end{pmatrix}$$

that is

$$P = \{\alpha\xi + \beta\eta \mid \alpha, \beta \in [0, 1]\}.$$

In the case $n = 1$ we consider the *oriented area*

$$\det \begin{pmatrix} \xi^p & \eta^p \\ \xi^q & \eta^q \end{pmatrix} = \xi^p \eta^q - \xi^q \eta^p.$$

In the higher dimensional case $n > 1$, we replace this by the sum of the oriented areas of the projects of P onto the coordinates planes $(p_i, q_i) \in \mathbb{R}^2$, i.e. by defining the bilinear map (a two-form) $\omega: \mathbb{R}^{2n \times 2n} \rightarrow \mathbb{R}$:

$$\omega(\xi, \eta) := \sum_{i=1}^n \det \begin{pmatrix} \xi_i^p & \eta_i^p \\ \xi_i^q & \eta_i^q \end{pmatrix} = \sum_{i=1}^n \xi_i^p \eta_i^q - \xi_i^q \eta_i^p. \quad (2.7)$$

In matrix notation this map has the form

$$\omega(\xi, \eta) = \xi^\top J \eta, \quad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \quad (2.8)$$

where I_n is the identity matrix of dimension n .

Definition 2.10 A linear mapping $A: \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ is called *symplectic* if

$$A^\top J A = J$$

or equivalently if $\omega(A\xi, A\eta) = \omega(\xi, \eta)$ for all $\xi, \eta \in \mathbb{R}^{2d}$.

Example 2.11 (Area preservation) In the case $n = 1$, where $\omega(\xi, \eta)$ represents the area of the parallelogram P , symplecticity of a linear mapping A is therefore the *area preservation* of A . $\triangle$

We now investigate the non-linear case.

Definition 2.12 Let $U \subset \mathbb{R}^{2d}$ be an open set. A differentiable map $\mathcal{F}: U \rightarrow \mathbb{R}^{2d}$ is called *symplectic* if its Jacobian matrix $\mathcal{F}'(p, q)$ is everywhere symplectic, i.e.

$$\mathcal{F}'(p, q)^\top J \mathcal{F}'(p, q) = J \quad \Leftrightarrow \quad \omega(\mathcal{F}'(p, q)\xi, \mathcal{F}'(p, q)\eta) = \omega(\xi, \eta).$$

Let us revisit the Hamiltonian system from (2.1). Using J from (2.8) and $y = (p, q)^\top$, we equivalently can introduce the *Hamiltonian IVP*

$$y'(t) = J^{-1} \nabla H(y(t)), \quad y(0) = y_0 \in \mathbb{R}^{2n}.$$

Definition 2.13 (Flow of an ODE) We denote $\mathcal{F}_t: U_t \subset U \rightarrow \mathbb{R}^{2n}$ the flow of a Hamiltonian system, that advances the solution by time t , such as $\mathcal{F}_t(y_0) = y(t)$, $\mathcal{F}_0(y_0) = y(0) = y_0$ and

$$\frac{\partial}{\partial t} \mathcal{F}_t(y_0) = y'(t) = J^{-1} \nabla H(\mathcal{F}_t(y_0)).$$

We now can obtain the following result

Theorem 2.14 (Poincaré 1899) Let $H \in \mathcal{C}^2(U; \mathbb{R})$, $U \subset \mathbb{R}^{2n}$ open. Then, for each fixed t , the flow $\mathcal{F}_t$ is a symplectic transformation on U_t .

Proof. Let $\nabla^2 H(y) \in \mathbb{R}^{2n, 2n}$ denote the Hessian of $H(y)$ and let $\mathcal{G}(t) := \frac{\partial \mathcal{F}_t}{\partial y_0}(y_0) \in \mathbb{R}^{2n, 2n}$. Then,

$$\mathcal{G}' = \frac{\partial}{\partial y_0} \frac{\partial}{\partial t} \mathcal{F}_t(y_0) = \frac{\partial}{\partial y_0} J^{-1} \nabla H(\mathcal{F}_t(y_0)) = J^{-1} \nabla^2 H(\mathcal{F}_t(y_0)) \frac{\partial \mathcal{F}_t(y_0)}{\partial y_0}.$$

In particular $\mathcal{G}$ solves the variational equation

$$\mathcal{G}' = J^{-1} \nabla^2 H(\mathcal{F}_t(y_0)) \mathcal{G}.$$

Hence, since $\nabla^2 H(p, q)$ is symmetric, $J^\top = -J$ and $J^{-\top} J = -I$ we obtain for any $t \geq 0$

$$\begin{aligned} \frac{\partial}{\partial t} \left(\left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top J \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right) \right) &= \left(\frac{\partial}{\partial t} \frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top J \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right) + \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top J \left(\frac{\partial}{\partial t} \frac{\partial \mathcal{F}_t}{\partial y_0} \right) \\ &= \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top \nabla^2 H(\mathcal{F}_t(y_0)) J^{-\top} J \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right) + \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top \nabla^2 H(\mathcal{F}_t(y_0)) \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right) \\ &= 0. \end{aligned}$$

Consequently, Since for $t = 0$, it holds

$$\left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top J \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right) = I J I = J$$

We have $\mathcal{G}(t)^\top J \mathcal{G}(t) \equiv \text{const} = J$, as long as the solution remains in the domain of definition of H . $\square$

Surprisingly also the converse almost holds true. We call a differential equation $y' = f(y)$ *locally Hamiltonian* if for every $y_0 \in U$, there exists a neighborhood where $f(y) = J^{-1} \nabla H(y)$ for some function H .

Lemma 2.15 (Integrability Lemma (EULER, LAGRANGE)) Let $D \subset \mathbb{R}^d$ be open and $f: D \rightarrow \mathbb{R}^d$ be continuously differentiable and assume that Jacobian $f'(y)$ is symmetric for all $y \in D$. Then, for every $y_0 \in D$, there exists $\epsilon > 0$ and a function $H(y)$ such that

$$f(y) = \nabla H(y), \quad y \in B_\epsilon(y_0) \subset D.$$

If $D = \mathbb{R}^{2n}$ or D is star-shaped, then H is globally defined.

Proof. Assume $y_0 = 0$ and consider $\epsilon > 0$ such that $B_\epsilon(y_0) \subset D$. Define for a constant $C \in \mathbb{R}$

$$H(y) = \int_0^1 y^\top f(ty) dt + C, \quad y \in B_\epsilon$$

Now differentiating H , with respect to y_k leads for $f = (f_1, \dots, f_d)$

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 f_k(ty) + y^\top \frac{\partial f}{\partial y_k}(ty) t \, dt = \int_0^1 \frac{\partial}{\partial t} (t f_k(ty)) \, dt = f_k(y),$$

where we used the relationship implied by the symmetry assumption

$$y^\top \frac{\partial f}{\partial y_k}(ty) = \sum_{i=1}^n y_i \frac{\partial f_i}{\partial y_k}(ty) = \sum_{i=1}^n y_i \frac{\partial f_k}{\partial y_i}(ty) = y^\top \nabla f_k(ty) = \frac{\partial}{\partial t} f_k(ty).$$

The construction of $H(y)$ via straight path $\{ty \mid t \in [0, 1]\}$ shows global existence if D is star-shaped, in particular the full EUCLIDEAN space. $\square$

Remark 2.16 The statement holds true if D is simply connected, but fails if D is not simply connected. This is a differential geometry consequence connected to Poincaré Lemma: If differential 1-forms are closed and the underlying domain is simply connected, then the differential form is exact. $\triangle$

Theorem 2.17 Let $f \in \mathcal{C}^1(U; \mathbb{R}^{2n})$. Then $y' = f(y)$ is locally Hamiltonian if and only if its flow $\mathcal{F}_t$ is symplectic for all $y \in U$ and for all sufficiently small t . If $U = \mathbb{R}^{2n}$, or U is star-shaped or or simply connected, then H is globally defined.

Proof. " $\Leftarrow$ ": Theorem 2.14. For the " $\Rightarrow$ " case assume that the flow $\mathcal{F}_t$ is symplectic. We now (locally) seek for a function H such that $f(y) = -J^{-1}H(y)$.

Since $\mathcal{G}(t) := \frac{\partial \mathcal{F}_t}{\partial y_0}(y_0) \in \mathbb{R}^{2n, 2n}$ solves the variational equation

$$\frac{\partial}{\partial t} \mathcal{G} = \nabla f(\mathcal{F}_t(y_0))^\top \mathcal{G},$$

we obtain by symplecticity of the flow $\mathcal{F}_t$, that

$$\mathcal{G}(t) [\nabla f(\mathcal{F}_t(y_0))J + J\nabla^\top f(\mathcal{F}_t(y_0))] \mathcal{G}(t) = \frac{\partial}{\partial t} \left(\left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top J \left(\frac{\partial \mathcal{F}_t}{\partial y_0} \right)^\top \right) = \frac{\partial}{\partial t} J = 0.$$

Hence, since $J = -J^\top$, and let $t = 0$ it follows that $J\nabla f(y_0)$ is a symmetric matrix for all y_0 by noting that $\mathcal{G}(0) = I$, the identity. By the Integrability Lemma 2.15 it follows that $Jf(y)$ is gradient of a function $H(y)$. $\square$

Remark 2.18 For a diffeomorphic transformation $\mathcal{F}$, with $z = \mathcal{F}(y)$ the following holds :

$$\begin{aligned} y(t) \text{ solves a Hamiltonian system} &\Rightarrow z(t) = \mathcal{F}(y(t)) \text{ solves a Hamiltonian system} \\ &\Leftrightarrow \\ &\mathcal{F} \text{ is symplectic.} \end{aligned}$$

$\triangle$

Remark 2.19 (Construction of first integrals) In (1918) NOETHER provided a constructive proof for the construction of first integrals for Hamiltonian systems. Let $H(p, q)$ and $L(q, q')$ denote

the Hamiltonian and the Lagrangian function and let $\{g_s : s \in \mathbb{R}\}$ denote a one-parameter group of (local diffeomorphic) transformations on $\mathbb{R}^n$, such that

$$g_0(q) = q, \quad g_s \circ g_r = g_{s+r}, \quad \forall r, s \in \mathbb{R}, \quad L\left(g_s(q), \frac{\partial}{\partial t} g_s(q)\right) = L(q, q'), \quad \forall s, \forall q, q'.$$

Define the vector field

$$v(q) = \left. \frac{\partial}{\partial s} g_s(q) \right|_{s=0}.$$

Then, $I(p, q) := p^\top v(q)$ is a first integral of the Hamiltonian system. $\triangle$

Remark 2.20 (Deviation from first integrals) Consider a smooth function $F : \mathbb{R}^{2n} \rightarrow \mathbb{R}$. Let y solve the Hamiltonian system $y' = J^{-1} \nabla H(y)$. Then, $F(y)$ solves

$$\frac{\partial}{\partial t} F(y(t)) = \{F, H\}(y(t)),$$

where $\{\cdot, \cdot\}$ denotes the *Poisson brackets* defined as $\{F, H\} = \nabla F^\top J^{-1} \nabla H$. In particular if F is a first integral, then $\{F, H\} = 0$. $\triangle$

2.2 Simple Symplectic methods

We saw that symplecticity is a characteristic property of Hamiltonian systems. Hence, we seek for numerical methods that share this property.

Definition 2.21 A numerical one-step method is called *symplectic* if the one-step map

$$Y_1 = \Psi_h(Y_0)$$

is symplectic whenever the method is applied to a smooth Hamiltonian system.

Example 2.22 (Symplectic Euler methods) There are two versions of so called *symplectic Euler methods*

- *Version I:* (implicit in p , explicit in q)

$$\begin{aligned} P_{k+1} &= P_n - hH_q(P_{k+1}, Q_n) \\ Q_{k+1} &= Q_n + hH_p(P_{k+1}, Q_n) \end{aligned}$$

- *Version II:* (explicit in p , implicit in q)

$$\begin{aligned} P_{k+1} &= P_n - hH_q(P_k, Q_{k+1}) \\ Q_{k+1} &= Q_n + hH_p(P_k, Q_{k+1}) \end{aligned}$$

$\triangle$

Theorem 2.23 (de Vogelaere 1956) Assume $h > 0$ such that the scheme Version I and Version II are well-defined. Then, the symplectic Euler methods are of order 1 and in fact symplectic.

Proof. We exemplarily proof proof for Version I only. First define

$$\begin{aligned} F_1(P_{k+1}, Q_{k+1}, P_n, Q_n) &:= P_{k+1} - P_n + hH_q(P_{k+1}, Q_n) \\ F_2(P_{k+1}, Q_{k+1}, P_n, Q_n) &:= Q_{k+1} - Q_n - hH_p(P_{k+1}, Q_n) \end{aligned}$$

Then the first version is giving as $F_1(\dots) = 0$ and $F_2(\dots) = 0$. Note that F_1, F_2 do not depend on Q_{k+1} within partial derivatives of H . Hence differentiation with respect to (P_k, Q_k) of the system yields for $F = F_1, F_2$ that

$$\frac{\partial F}{\partial(P_{k+1}, Q_{k+1})} \frac{\partial(P_{k+1}, Q_{k+1})}{\partial(P_n, Q_n)} + \frac{\partial F}{\partial(P_n, Q_n)} = 0,$$

which is equivalent to

$$\frac{\partial F}{\partial(P_{k+1}, Q_{k+1})} \frac{\partial(P_{k+1}, Q_{k+1})}{\partial(P_n, Q_n)} = -\frac{\partial F}{\partial(P_n, Q_n)}.$$

Now, inserting for $F = F_1, F_2$ it holds that

$$\begin{aligned} \frac{\partial F_1}{\partial P_{k+1}} &= I + hH_{qp}(P_{k+1}, Q_n)^\top = I + hH_{qp}^\top. \\ \frac{\partial F_1}{\partial P_{k+1}} &= \mathbf{0}. \\ \frac{\partial F_2}{\partial P_{k+1}} &= -hH_{pp}(P_{k+1}, Q_n) \\ \frac{\partial F_2}{\partial Q_{k+1}} &= I. \end{aligned}$$

Hence,

$$\frac{\partial(F_1, F_2)}{\partial(P_{k+1}, Q_{k+1})} = \begin{pmatrix} I + hH_{qp}^\top & 0 \\ -hH_{pp} & I \end{pmatrix}.$$

Analogously

$$-\frac{\partial(F_1, F_2)}{\partial(P_k, Q_k)} = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{pq}^\top \end{pmatrix}$$

Summarizing we obtain

$$\underbrace{\begin{pmatrix} I + hH_{qp}^\top & 0 \\ -hH_{pp} & I \end{pmatrix}}_{=:A} \left(\frac{\partial(P_{k+1}, Q_{k+1})}{\partial(P_n, Q_n)} \right) = \underbrace{\begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{pq}^\top \end{pmatrix}}_{=:B} \quad (2.9)$$

where H_{qp}, H_{pp}, H_{qq} are partial derivatives evaluated at (P_{k+1}, Q_n) . Note that for h small enough, such that the scheme is well-defined $I + hH_{qp}^\top$ needs to be invertible (implicit function theorem), hence A is invertible. In order to show symplectic property, we investigate

$$\left(\frac{\partial(P_{k+1}, Q_{k+1})}{\partial(P_n, Q_n)} \right)^\top J \left(\frac{\partial(P_{k+1}, Q_{k+1})}{\partial(P_n, Q_n)} \right) = (A^{-1}B)^\top J (A^{-1}B)$$

Since for block lower-triangular matrices it holds with A_{11} invertible that

$$\begin{pmatrix} A_{11} & 0 \\ A_{21} & I \end{pmatrix}^{-1} = \begin{pmatrix} A_{11}^{-1} & 0 \\ -A_{21}A_{11}^{-1} & I \end{pmatrix}^{-1}$$

it follows

$$A^{-1} = \begin{pmatrix} (I + hH_{qp}^\top)^{-1} & 0 \\ hH_{pp} (I + hH_{qp}^\top)^{-1} & I \end{pmatrix}.$$

Finally, using intermediate step of matrix vector multiplication such as

$$A^{-\top}JA = \begin{pmatrix} 0 & (I + hH_{qp}^\top)^{-T} \\ -(I + hH_{qp}^\top)^{-1} & 0 \end{pmatrix}$$

and symmetry properties $H_{pq}^\top = H_{qp}$, etc leads to

$$B^\top A^{-\top}JAB = J.$$

□

Remark 2.24 (Explicit version for separable systems.) Assume $H(p, q) = T(p) + U(q)$ is separable. Then both symplectic Euler schemes are explicit. $\triangle$

Lemma 2.25 The composition of symplectic numerical schemes is again symplectic.

Proof. Consider two symplectic one-step maps Ψ_h^1 and Ψ_h^2 . And consider the numerical scheme

$$Y_1 = \Psi_h(Y_0) := \Psi_h^2 \circ \Psi_h^1(Y_0).$$

Then, the result follow from chain rule and subsequent use of the symplectic property of Ψ_h^1 and Ψ_h^2 , such that

$$\left(\frac{\partial \Psi_h}{\partial y_0} \right)^\top J \left(\frac{\partial \Psi_h}{\partial y_0} \right) = J.$$

□

Example 2.26 (Störmer-Verlet schemes / leap-frog) Again there are two version.

Version I:

$$\begin{aligned} P_{k+1/2} &= P_k - \frac{h}{2} H_q(P_{k+1/2}, Q_k), \\ Q_{k+1} &= Q_k + \frac{h}{2} (H_p(P_{k+1/2}, Q_k) + H_p(P_{k+1/2}, Q_{k+1})) \\ P_{k+1} &= P_{k+1/2} - \frac{h}{2} H_q(P_{k+1/2}, Q_{k+1}). \end{aligned} \tag{2.10}$$

Let $H(p, q) = T(p) + U(q)$ be separabel. Then, the **explicit** Störmer-Verlet scheme I reads:

$$\begin{aligned} P_{k+1/2} &= P_k - \frac{h}{2} U'(Q_k), \\ Q_{k+1} &= Q_k + h T'(P_{k+1/2}), \\ P_{k+1} &= P_{k+1/2} - \frac{h}{2} U'(Q_{k+1}). \end{aligned}$$

Version II:

$$\begin{aligned}
 Q_{k+1/2} &= Q_k + \frac{h}{2} H_q(P_k, Q_{k+1/2}), \\
 P_{k+1} &= P_k - \frac{h}{2} (H_p(P_k, Q_{k+1/2}) + H_p(P_{k+1}, Q_{k+1/2})) \\
 Q_{k+1} &= Q_{k+1/2} - \frac{h}{2} H_q(P_{k+1}, Q_{k+1/2}).
 \end{aligned} \tag{2.11}$$

Positions and momenta are staggered by half a time-step, given it the popular name of **leap-frog**.
 $\triangle$

Theorem 2.27 The Störmer-Verlet scheme 2.10 and 2.11 is a symplectic method of order 2.

Proof. The scheme is a composition of two symplectic Euler methods and by Lemma 2.25 symplectic. E.g. for Version I we can equivalently write

$$\begin{aligned}
 P_{k+1/2} &= P_k - \frac{h}{2} H_q(P_{k+1/2}, Q_k), \\
 Q_{k+1/2} &= Q_k + \frac{h}{2} H_p(P_{k+1/2}, Q_k), \\
 Q_{k+1} &= Q_{k+1/2} + \frac{h}{2} H_p(P_{k+1/2}, Q_{k+1}), \\
 P_{k+1} &= P_{k+1/2} - \frac{h}{2} H_q(P_{k+1/2}, Q_{k+1}).
 \end{aligned}$$

So its a composition of half step of symplectic Euler Version I and a half step of symplectic Euler Version II. Order 2 follows by analyzing the local error as usual (note symmetry of the scheme). $\square$

Example 2.28 We consider

$$q = (q_1, q_2) \in \mathbb{R}^2, \quad p = (p_1, p_2) \in \mathbb{R}^2.$$

$$H(p, q) = \frac{1}{2} \|p\|^2 + \frac{1}{2} (q_1^2 + q_2^2) + q_1^2 q_2 - \frac{1}{3} q_2^3,$$

describing a unit-mass particle in the plane, confined by a harmonic trap and subject to nonlinear, asymmetric anharmonic forces. We have

$$\nabla_p H(p, q) = p, \quad \nabla_q H(p, q) = \begin{pmatrix} q_1 + 2q_1 q_2 \\ q_2 + q_1^2 - q_2^2 \end{pmatrix}.$$

So the Hamiltonian dynamical system reads

$$\begin{aligned}
 \dot{q}_1 &= p_1, \\
 \dot{q}_2 &= p_2, \\
 \dot{p}_1 &= -(q_1 + 2q_1 q_2), \\
 \dot{p}_2 &= -(q_2 + q_1^2 - q_2^2).
 \end{aligned}$$

Applying the symplectic Euler schemes and the Strörmer-Verlet I scheme for a time horizon of $[a, b] = (0, 100)$ with step size $h = 0.1$ leads the result in Figure 2.1 and in comparison with the explicit Euler scheme for $h = 0.01$ in Figure 2.2. On the right of theses Figures we see the

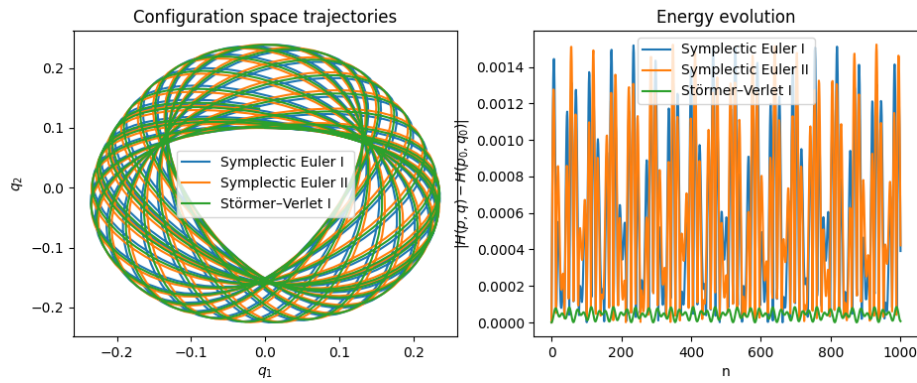


Figure 2.1: Various methods applied to the Hamiltonian system from Example 2.28

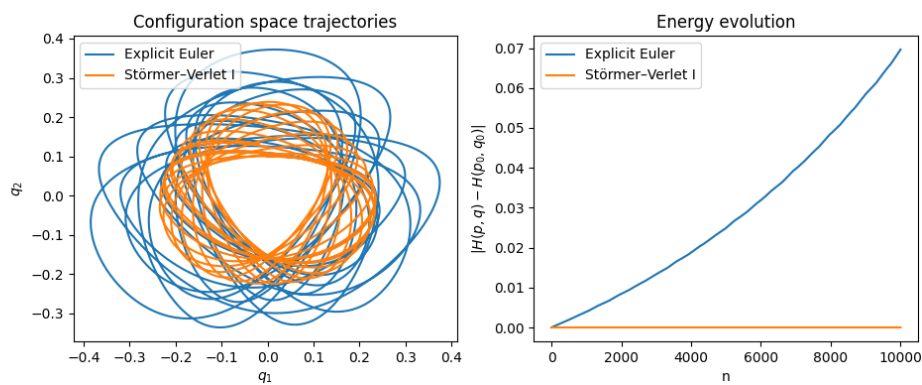


Figure 2.2: Explicit Euler method in comparison applied to the Hamiltonian system from Example 2.28 including energy loss when applying the non-symplectic Euler-scheme.

deviation from the first integral property of the Hamiltonian. As we can see in the Explicit Euler case, the energy is lost in the system, while other methods oscillates around the constant value of the Hamiltonian along the trajectory.

△